Affirmed Networks Value-Added Services Administration Guide

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Preface

This guide describes an overview of the Affirmed Networks Value Added Services (VAS) running on the Mobile Content Cloud. This guide also describes the Command Line Interface (CLI) commands and objects used to configure, control, and monitor those services.

For information on accessing the Command Line Interface (CLI), see the *Affirmed Networks CLI Reference Guide*. For information on statistics, see *Acuitas Performance Statistics*.

Note: *The Affirmed Networks Mobile Content Cloud*™ (MCC) *is the mobile services software solution that operates on virtual platforms such as blade systems.*

When deploying VAS, use the following VM provisioning guidelines:

- Use one of the following building block strategies:
  - specialized (preferred)
  - advanced-content-separation (small deployments only)
  - integrated (small deployments only)
- Provision a sufficient number of ISM/ASM VMs as required for your network deployment
  - use CCM, DCM and ASM personalities for specialized cluster building block strategy
  - use CPM and ASM personalities for advanced-content-separation cluster building block strategy
  - use CPM and ISM for integrated-services cluster.
- Designate a dedicated CPU to each ASM by setting the processor affinity
- Set the number of ASM/ISM virtual cores to the number of CPU cores minus 10%, reserving the remainder for the VM hypervisor. For example:
  - 18 on CPU with 20 virtual cores
  - 22 on CPU with 24 virtual cores
  - When deploying CPU intensive functionality, the minimum number of virtual cores is 18
- Provision memory for ASMs/ISMs VMs running Value-Added Services:
  - Recommend of 110GB for deployments
  - Minimum of 32GB PoC trials with limited number (up to a few hundred) of subscribers

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide), installation staff, and other qualified service personnel who want to learn about the Affirmed Networks solution and associated features and functions. The audience should also be familiar with Long Term Evolution (LTE), and 3G or 4G networks. It may also be helpful to have a basic understanding of the HTTP protocol, TCP/IP stack, and the use of MMS services.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

This guide contains the following chapters and appendices:

Chapter/Appendix	Describes
Chapter 1, Content Services Common Constructs	Common tasks that you should perform to enable content services functions.
Chapter 2, Multi-protocol Proxy	HTTP proxy and services such as IP flow, request policy, response policy, TCP splicing, and header enrichment.
Chapter 3, TCP Optimization	Overview of TCP optimization and how to configure a TCP optimization profile.
Chapter 4, Configuring a DNS Proxy and Caching Service	DNS proxy and services such as DNS policy and reverse cache.
Chapter 5, Configuring Content Filtering	Content filtering and services such as content categorization, blacklist file, whitelist file, and safe search.
Chapter 6, Configuring a Content Caching Proxy	HTTP caching proxy and services such as content caching and cache manager.
Chapter 7, HTTP Server	HTTP server and services such as entitlement.
Chapter 8, MMS Services	MMS services and wireless transport layer security and SMPT proxy services.
Chapter 9, Workflow Parameters	Workflow objects and parameters supported on the MCC.

Features in this Release

This release includes the following new features:

Note: *For a complete list of the features supported in this release, see the software release notes. In addition to the information in this guide, see the additional Affirmed Networks documentation listed in the Affirmed Networks Documentation Library. For a comprehensive list of terms and acronyms used in Affirmed Networks documentation, see Affirmed Networks Terms and Acronyms.*

Feature	Description
This release adds support for Google QUIC version Q046.	MCC supports Google QUIC versions Q039, Q043, Q044, and Q046.
Proxy is TTL transparent See Configuring a DNS Proxy and Caching Service (page 4-1).	Previously, for upstream traffic from the UE to the origin server, IP packets that *traversed* the proxy had their TTL values reset to 63 for IPv4, and their hop limit values reset to 254 for IPv6. However, IP packets that *bypassed* the proxy had their TTL and hop limit values correctly decremented by one. With this feature, traffic traversing the proxy also has the TTL/hop limit value decremented. Specifically, for any HTTP, HTTPS, and QUIC flows originating from the UE for IPv4 and IPv6, the TTL or hop limit value is decremented by one, respectively. This applies to transparent IP and non-transparent IP traffic traversing the proxy, and to traffic bypassing the proxy, regardless of whether the User Space IP stack or the Linux IP stack is used. ***Note:*** *OSU – During an upgrade, traffic traversing VM nodes that have not been upgraded will retain the existing behavior.*

<table>
<tr><th>Feature</th><th>Description</th></tr>
<tr><td>Added consent-id to Secure HTTP Identify Protocol (SHIP) services

See Configuring Secure HTTP Identify Protocol Services (page 7-7) and Consent ID (page 7-13).</td><td>This feature enables closer coordination between Content Providers (CP) and end users who have given permission for that coordination. For example, if a CP requests to share subscriber information, this feature will verify if the user has consented to share their information with a given CP. If consent was granted, the data is shared; if not, access to subscriber information is rejected.

This feature receives data from the PCRF. A new string named ‘ConsentID’ is added to the PCRF. That string is relayed to the Proxy for processing.

If this feature is invoked and the content provider’s configured consent-id matches a VSA string in the metadata for that user, then MCC processes the request. Otherwise, MCC returns an error code.

This feature adds a new content provider and content status to data records and also adds the SERVER PLUGIN FAILURES NO CONTENT statistic:

an3000-mcm-slot7cpu1# show services http-server statistics cluster-level server-plugin

This feature adds the following new show subscriber pdn-session parameter:

an3000-mcm-slot7cpu1# show subscriber pdn-session 239616

Session VSA ConsentID <Consent ID String>

This feature adds the following new show subscriber proxy-session parameter:

an3000-mcm-slot7cpu1# show subscriber proxy-session detail 239616

Session VSA - ConsentID <Consent ID String>

This feature enables all proxy metadata across MCC to store up to 1022 bytes of data.

Configure the consent-id as follows:

an3000-mcm-slot7cpu1(config)# services http-server ship content-provider <CP-name>

consent-id Consent ID for the content provider - string

This feature filters SHIP requests as follows:

■ If the configured content provider consent-id is ‘*’ (wildcard) (default), then the SHIP request is processed, regardless of the user strings supplied. The SHIP request could still fail due to other errors.

■ If the configured content provider consent-id is supplied and matches a VSA string provided for the user (not case sensitive), then the SHIP request is processed. The SHIP request could still fail due to other errors.

■ If the configured content provider consent-id is supplied but does not match a VSA string supplied for the user, then the SHIP request fails with the failure return code (1004).</td></tr>
</table>

Feature	Description

Added consent-id to Secure HTTP Identify Protocol (SHIP) services
(*continued*)
	In the HTTP Transaction data records (HTDRs), added the following field to the HTTPTransactionRecord:
	SHIPInformation Data Record:
	`string Content Provider /`
	`uint64 AppID / not used`
	`ConsentStatus Consent Status /`
	`ConsentStatus` contains the following enumerations:
	0 = Consent Not Available
	1 = Consent Not Matched
	2 = Consent Matched
	`ShipStatus Ship Status /`
	`ShipStatus` contains the following enumerations:
	0 = SHIP success
	1 = Content Provider Not Found
	2 = Info Not Available
	3 = Consent Validation Failed
	4 = SHIP Internal Server Error

Feature	Description

Factor additional latency for distance for congestion detection
 	
See Configuring Multi-protocol Proxy Services (page 2-3) and Configuring Multi-protocol Proxy Server Connections (page 2-33).
	This feature adds a configuration option for Round-trip Time (RTT) offsets when RTT is used in the congestion detection's throughput formula. RTT offsets can be configured in data centers for both *nearby* and *distant* subscribers to avoid misidentifying *distant* subscribers as congested.
	
	This feature reports the length of time that the client connection remains in each congestion level in the HTDR/EDR and also reevaluates policies for flows where the congestion levels change.
	
	This feature adds the following new HTDR/EDR fields in tcpClientConnectionInfo:
	
	■ CongestionLevelNoneTime
	■ CongestionLevelMildTime
	■ CongestionLevelModerateTime
	■ CongestionLevelSevereTime
	
	To configure RTT offsets:
	
	`an3000-mcm-slot7cpu1(config)#` **`services multi-protocol-proxy rtt-location-offset-profile RTT-PROFILE`**
	
	`Possible completions:`
	
	`compromise-rtt-offset RTT location offset for subscribers without a known TAC`
	
	`location-offset RTT location offset for subscribers with a known TAC`
	
	`an3000-mcm-slot7cpu1(config)#` **`services multi-protocol-proxy rtt-location-offset-profile RTT-PROFILE location-offset RTT-LOCATION1`**
	
	`an3000-mcm-slot7cpu1(config-location-offset-RTT-LOCATION1)# ?`
	
	`Possible completions:`
	
	`rtt-location-offset   RTT location offset(ms)`
	
	`tac-high-byte         high byte of TAC`
	
	`an3000-mcm-slot7cpu1(config)#` **`services multi-protocol-proxy instance USTCP tcp-measurement`**
	
	`Possible completions:`
	
	`admin-state Enable/Disable TCP Measurement`
	
	`congestion-level Congestion Level for applying Optimizations based on Client Side TCP measurement`
	
	`rtt-location-offset-profile   RTT Location Offset Profile for this instance`
	
	`sampling-interval Client Side TCP measurement Sampling Interval (1/ms)`

Feature	Description
Added shaping enhancement for configurable chunk sizes See Configuring a Multi-protocol Proxy ABR Optimization Profile (page 2-8) and Configuring a Multi-protocol Proxy Traffic Control Profile (page 2-47).	This feature provides a configurable option to control the data chunk size of rate-limited packets for the HTTP, HTTPS, and UDP protocols. The data size is controlled by the time interval at which rate-limited packets are sent to the UE for a given bitrate. For example: If the configured bitrate is 1.5Mbps and the configured send-interval is 50ms (0.05sec), MCC sends the following in each interval: chunksize = (1.5Mb / 8) * 0.05 = 9375 bytes A new **`send-interval`** field is added to the **`abr-optimization-profile`**. For example: **`services multi-protocol-proxy abr-optimization-profile AOP admin-state enabled`** **`send-interval`** – Configure rate limiting send interval (in ms) (min – 40ms, max – 500ms; in multiples of 10 ms; default – 100ms) A new **`send-interval`** field is added to the **`traffic-control-profile.`** For example: **`services multi-protocol-proxy traffic-control-profile ACP rate-limiting`** **`send-interval`** – Configure rate limiting send interval (in ms) (min – 40ms, max – 500ms; in multiples of 10 ms; default – 100ms) of 10ms; default – 100ms)

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	**Times New Roman bold font**	Use **configure_ha** to configure... Enter **y**. To continue, press **Enter**.
Selecting a menu item	**Menu** => **Option**	Select **Security** => **Group Management**.
Variable parameters for which you supply values	***<courier bold italic blue font>***	**configure_dr convert slave --masterhost <host IP>**
User input	**Courier bold blue font**	**cd /opt/Affirmed/NMS/bin**
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	**[ ]**	**cluster [id] node [id] reboot**
Optional keywords and arguments in curly brackets	**{ }**	***{cdr-file-name}***
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.	|	Is this correct? [y|n]
Keywords representing a single element in angle brackets	**< >**	***<host name>***
Asterisk after a CLI object indicates a required object.	*****	**name***
Variables, book and chapter titles, file path names, new terms, and emphasized text	*Times New Roman italic font*	See the *Acuitas User's Guide*.

Tip: Helpful suggestions.

Note: *Additional information that may apply to the subject text.*

CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

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Chapter 1

Content Services Common Constructs

This section describes the following common tasks that you should perform to enable content services functions.

Externally Activated Services and Service Rules

MCC content services supports:

- external activation (from PCRF) of proxy policies, specifically for applying HTTP header enrichment
- selection of the desired proxy service instance from the proxy service instance in the CRBN and the data-profiles

Any proxy-based service can be externally activated from the PCRF. The PCRF can use either the CRBN or CRN list or a combination of both lists to activate proxy service instances and specific service rules. When multiple proxy service instances are activated, MCC uses the instance with the highest priority.

MCC evaluates the policies configured under proxy-based service instances. If a policy is associated with an externally activated service-rule, it is applied only if it is activated by the PCRF.

Guidelines	■ A maximum of 64 service-rule IDs can be sent from Fastpath to the proxy. ■ The gateway validates the CRN received from the PCRF against existing services (such as QoS, metering, and charging) configured under data-profile. If the CRN is not associated to any of the services, but is configured in MCC, then the gateway sends the CRN to FastPath which in turn passes the same CRN to content/http-proxy. Then, content validates the CRN/SR against the configured http-proxy instances. If the CRN is not associated with any instance, content discards the service-rule or takes no action. However, MCC does not issue any rejection notification to the PCRF about this misconfiguration. The Operator must set the proper configuration. ■ In order to trigger externally-activated PCRF rules, you must enable **`service-activation`** and set it to either **`always-on`** or **`external-activation.`** See the *Affirmed Networks Gateways Administration Guide* for details on the service-construct service-rule service-activation setting.

Configuring a List of URLs

This section describes how to configure a list of URLs, import prefixes, import from a local file, and import from a server. The url-list database is built after a scheduled or manual download completes.

Access	config
Limitations	None
Guidelines	■ The file can be imported from the server or local file. If you make changes to the file, you must import the file from the CLI using url-list import to rebuild the database for the new file to take effect. ■ The file should have a Fully Qualified Domain Name (FQDN) for each line for HTTP request URL prefix or exact-match. ■ The protocol schema is ignored and anything before "//" and including "//" is skipped. Only the URL following "//" is used. ■ If the line contains a star "*", the entry is added in the database using the string before "*" for URL prefix-match. Otherwise, the entry is added in the database for URL exact-match. Prefix match is only supported for the URI part of the URL. Domain prefix match is not supported.

Use the following steps to configure a list of URLs:

1. Create the list of URLs.

```
an3000-mcm-slot7cpu1(config)#service-construct url-list <name>
```

2. Specify whether imported URLs should be treated as URL prefixes. Terminator "*****" is added at the end of each line in the imported file if it doesn't already exist.

```
an3000-mcm-slot7cpu1(config)#service-construct url-list
import-as-url-prefixes
```

3. Configure a filename for importing this list from a local file.

```
an3000-mcm-slot7cpu1(config)#service-construct url-list
import-from-local-file
```

The url-list database is built after an import action. Operators can upload the local file through Acuitas or using **filecommand** to scp the file to */public/content/policylists/url/userlists/* on the MCC. There is no scheduled automatic import for this mode. When the file on the disk changes, you must execute **service-construct url-list** ***<name>*** **import** to rebuild the database for the new file to take effect.

The file contains a list of URLs and each URL is a string in the following format:

http[s]://<domain-name>[:port]/[<url-path>][*]

The *port*, *url-path*, and terminating /* are optional. If /* is used, the URL is treated as a URL prefix rather than the exact-match URL.

4 Configure a rule for importing this list from a file server.

```
an3000-mcm-slot7cpu1(config)#service-construct url-list
import-from-server
```

This object references a file to be imported (see service-construct diffserv-profile) from a file server and configures a rule for importing the file. This object is referred to by the service-construct http-rule-group only and used as rules to match HTTP request URL.

The file contains a list of URLs and each URL is a string in the following format:

http[s]://<domain-name>[:port]/[<url-path>][*]

The *port*, *url-path*, and terminating /* are optional. If /* is used, the URL is treated as a URL prefix rather than the exact-match URL.

Configuring a Device Database

This section describes how to configure the device database. The device-database configuration is a singleton object and supports one instance of the device-database at any time. Also see Initiating a Device Database File Download (page 1-7).

Access	config
Limitations	None
Guidelines	■ The device-database must be in CSV format. ■ The first line in the CSV file must contain the field names since MCC uses this information for parsing. The files are stored in MCC. ■ The header information is case sensitive. ■ MCC supports a maximum of 300,000 database entries. ■ If the database is modified, the existing session's values are not changed. ■ The database supports a single level of lookup.

This feature enables Operators to use device characteristics to configure data services such as data filters used to allow/block traffic for a particular class of devices based on the device-type parameter. This feature also enables Operators to configure differentiated rate-limiting bit rates based on device-type or device screen size.

This feature supports the following device-database characteristics: TAC (Type Allocation Code), device-type, device-make (or brand), device-model, screen-width, and screen-height. The TAC information is extracted from the first 8 digits of the 15/16 digit IMEI. The device-type field indicates the type of device that corresponds to a particular TAC. Options include: handheld, feature-phone, smart-phone, modem, dongle, tablet, vehicle, connected-device, router, module, ebook, and femto-app.

The device-database is in CSV format and must be previously imported into MCC or downloaded from a remote server. See Initiating a Device Database File Download in the *Affirmed Networks Gateway Administration Guide*. MCC obtains the database attributes during session establishment. The device-database contains a mapping of the TAC of a device to its specific parameters. The database is either the GSMA device map or a modified version of the GSMA device map and may also contain a subset of GSMA device-map fields.

The service-construct session-rule-group rule objects match the following device-database parameters: TAC, device-type (exact match), make (exact match), model (exact match), screen-width, and screen-height.

The content-filter instance accepts a service-rule to which a session-rule-group is associated.

Use the following steps to configure a device database:

1. Enter a name for the device database.

```
an3000-mcm-slot7cpu1(config)#service-construct device-database <name>
```

2. Import the device-database now. Options include:

- **noalarm** – Do not raise any alarms when this import fails.
- **nolog** – Do not issue error logs when this import fails.

```
an3000-mcm-slot7cpu1(config)#service-construct device-database <name>
import noalarm|nolog
```

3. Configure strings for csv headings. The csv-field-names are case sensitive. Options include:

- **device-make** – Configure a string to indicate the name of the device-make (brand) field.
- **device-model** – Configure a string to indicate the name of the device-model field.
- **device-type** – Configure a string to indicate the name of the device-type field.
- **screen-height** – Configure a string to indicate the name of the screen-height field.
- **screen-width** – Configure a string to indicate the name of the screen-width field.
- **tac*** – Configure a string to indicate the name of the TAC field. This field is mandatory as it is the key to the database.

```
an3000-mcm-slot7cpu1(config)#service-construct device-database <name>
csv-field-names
device-make|device-model|device-type|screen-height|screen-width|tac
```

4. Configure a filename for importing this list from a local file.

- **filename** – Configure the particular list’s filename, the file should be in */public/content/policylists/<list-type>/*

```
an3000-mcm-slot7cpu1(config)#service-construct device-database <name>
import-from-local-file filename
```

5. Configure a rule for importing this list from a file server.

- **filename** – Configure the server filename, <string> For example, *folder/file-$DATE{yyyymmdd}.txt*
- **file-download-profile** – Configure the file download profile.

```
an3000-mcm-slot7cpu1(config)#service-construct device-database <name>
import-from-server filename|file-download-profile
```

Initiating a Device Database File Download

The file download process can occur either locally or from a remote server, but not both simultaneously. Only one device-database can be configured at a time. If the database is modified or removed, all history is cleared. In order to add a different device-database, the first device-database must be deleted.

Access	config
Limitations	None

Remote Server

1. Add a file download profile and specify the retry intervals and the schedule at which the file is downloaded.
2. Specify the full path on the server where the file resides.
3. In the file download profile, specify the server address, user name, and password.
4. Download the file from the remote server.
   `service-construct device-database <database-config-name> import`

Local Server

1. Copy the file to */public/content/policylists/devicedb*
2. Execute the following import command to trigger the import device-database now.
   `service-construct device-database <database-config-name> import`
3. Indicate **noalarm** or ***nolog*** to prevent an alarm or log from being raised when this import fails.

Configuring an Encryption Profile

This section describes how to configure an encryption profile.

Enable/disable the encryption-profile and enter the public-key-filename. A security department provides the Public Key. The configured encryption-profile is applied to the http-proxy instance.

Customer data may be considered confidential and should not be shared. The system provides the ability to encrypt the Mobile Subscriber Integrated Services Digital Network-Number (MSISDN). The MSISDN number uniquely identifies a subscription in a GSM or UMTS mobile network. There are two types of encryption; static and dynamic.

Access	config
Limitations	None

Use the following steps to configure an encryption profile:

1 Create the encryption profile and Anonymous Customer Reference (ACR) Encryption profile name.

```
an3000-mcm-slot7cpu1(config)#services multi-protocol-proxy
encryption-profile <name>
```

2 Set the admin state to enabled (or disabled).

```
an3000-mcm-slot7cpu1(config)#services multi-protocol-proxy
encryption-profile <name> admin-state enabled
```

3 Create the name of the public key file in */public/content/publickeys/*.

```
an3000-mcm-slot7cpu1(config)#services multi-protocol-proxy
encryption-profile <name> public-key-filename
```

4 Configure the type of public key file, either PEM, DER, or X509.

```
an3000-mcm-slot7cpu1(config)#services multi-protocol-proxy
encryption-profile <name> public-key-filename
```

Downloading Files Containing Policy Definitions

This section describes how to define common parameters for downloading files containing policy definitions, such as files defining a list of URLs from file servers. Enable or disable automatic-download and assign a name to this file-download-profile. The profile is referred to by multiple import-from-server configuration objects. See Configuring a List of URLs (page 1-3)

Access	config
Limitations	None

Use the following steps to configure a file download profile:

1 Name the file download profile.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name>
```

2 Enable or disable automatic download. See Configuring a List of URLs (page 1-3) for information on manually importing the policy list.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> automatic-download enabled|disabled
```

- **enabled** – Operators must configure an automatic-download schedule (days-of-week and time-of-day) or download-interval. In this mode, both manual and automatic download are allowed. The download-interval takes precedence and is therefore used for automatic download.
- **disabled** – Operators must download the policy list manually. This is called “on demand only” mode. In this mode, the days-of-week/time-of-day and download-interval objects are configurable but not mandatory.

3 Configure the file download days of week, Mon|Tue|Wed|Thu|Fri|Sat|Sun.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> days-of-week Mon
```

4 Configure a download interval in minutes for periodical download. When the download-interval is configured, it takes priority over the time-of-day and days-of-week download schedule. Operators use the download-interval as an alternative way to schedule automatic downloads in minute intervals. A value of 0 indicates no periodical download.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> download-interval
```

5 Configure a file download server network context.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> network-context <NC-name>
```

6 Configure a file download password.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> password <password>
```

7 Configure the number of old downloads to retain, <0-30>. The system can retain a maximum of 30 old downloads.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> retain-count 20
```

8 Configure the file download retry count <0-10>.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> retry-count 5
```

9 Configure the file download retry interval in minutes <1-120>.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> retry-interval 10
```

10 Configure the file download server IP address.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> server-address <ipaddress>
```

11 Configure the file download time, hh:mm '<0-23>:<0-59>'.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> time-of-day 02:15
```

12 Configure a file download username.

```
an3000-mcm-slot7cpu1(config)# service-construct file-download-profile
<name> username <username>
```

Chapter 2

Multi-protocol Proxy

This section describes the multi-protocol proxy service.

Access	config
Limitations	If http-proxy is used in conjunction with a gateway on the same network-context, a separate loopback(s) must be provisioned for http-proxy and gateway interfaces.

HTTP Overview

The system is designed to support the HTTP proxy function in forward intercepting mode, or reverse, offering advantages in redirecting client requests to the right node and supporting additional revenue-generating features for the Operator. Forward proxy mode is supported using the existing APN configuration on the mobile GW.

Forward explicit mode proxy enables Operators to perform the following functions based on the subscriber's profile and configuration:

- Parental control
- Dedicate proxy nodes for different domains/URLs
- Request/Response modification

The system hosts proxy and cache functions in each Content Services Module (CSM) and Advanced Services Module (ASM). Each CSM can be viewed as an independent caching proxy with its own storage. The Subscriber Services Module (SSM30) module hosts the load distribution function that terminates the TCP connections and distributes the user sessions among all caches (CSM).

HTTP proxy supports upgrading a plain text HTTP/1.1 connection to a different protocol (such as, WebSocket, HTTP/1.1 over TLS, HTTP/2.0, HTTP/2.0 over TLS).

When switching to another protocol is detected:

- http-proxy forwards initial "101 switching protocols" response to client
- http-proxy forwards any following server message to client
- TCP proxy splices (relays) other protocol messages between client and server

The MCC retrieves the following VSAs from the PCRF via the Gx interface and distribute these VSAs to the http-proxy. The MCC uses these VSAs to customize http-proxy services for a particular subscriber.

- *VsaEntitlement* – Indicates the type of service to which the subscriber is entitled.
- *VsaPaymentType* – Indicates subscriber payment type. Possible values include:
  - – PO – Indicates a postpaid subscriber.
  - – PR – Indicates a prepaid subscriber.
  - – SL – Indicates smart limits plan.
- *VsaSubId* – Indicates the subscriber ID.
- *VsaHttpPolicy* – Customize HTTP Proxy services. Possible values include:
  - – NOPRXY – HTTP Proxy is disabled for the subscriber.
  - – NOGZIP – Gzip compression is disabled for the subscriber.
  - – NOTFD – No Toll Free Data (TFD).
  - – NOVIDEO – Video pacing is bypassed for the subscriber.
- *VsaResellerId* – Indicates the subscriber's reseller ID.
- *VsaMiDial* – Indicates whether the subscriber opted out from Long Term Mobile ID (LTMI) header enrichment or opted for static trusted ID in LTMI header enrichment.

Configuring Multi-protocol Proxy Services

This section describes how to configure multi-protocol proxy services.

Access	config
Limitations	None
Guidelines	See Congestion Control Settings (page 3-13) for guidelines on the supported congestion-control settings when user-space-networking is enabled. **user-space-networking** ■ The following ip-protocol tcp configuration options are not supported if user-space-networking is enabled: rmem-min, wmem-min, frto, tcp-early-retrans, tcp-mem-high-threshold, tcp-mem-low-threshold, tcp-mem-pressure-threshold. See Configuring Global TCP Parameters (page 3-4). ■ See Congestion Control Algorithms (page 3-9) for the supported settings when user-space-networking is enabled or disabled. – The following congestion-control settings are not supported if user-space-networking is enabled: bic, highspeed, hybla, illinois, lp, scalable, veno, westwood, yeah. – The following congestion-control settings are not available and default to cubic if user-space-networking is disabled: bic, cdg, chd, dctcp, hd, and newreno.

Use the following steps to configure multi-protocol proxy services:

1 Enable/disable the multi-protocol-proxy service.

> ***Note:*** *Enable the admin-state prior to enabling the content-cache admin state. The admin state cannot be disabled if the content-cache and content-filter admin state are enabled. See Chapter 6, Configuring a Content Caching Proxy.*

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
admin-state enabled
```

2 Configure the ABR rate limiting optimization profile. See Configuring a Multi-protocol Proxy ABR Optimization Profile (page 2-8).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
abr-optimization-profile
```

3 Configure HTTP proxy back-end forwarding. The admin state is disabled by default.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
back-end-forwarding enabled
```

a Configure the maximum number of idle connections per back-end proxy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
max-idle-connections 2048
```

4 Configure the client connection HTTP request wait time in milliseconds. A value of 0 (zero) indicates forever. Configure a value between 0 .. 15000(0) milliseconds. Reverse hand-off is implemented and if the HTTP request is found, it is handed back to the HTTP proxy and continues to process HTTP traffic.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
client-connections request-wait-time 0
```

5 Configure client idle timeout in seconds. The default is 60 seconds.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
client-idle-timeout 60
```

6 Set the global compression level. Options include: default-compression, best-speed, and best-compression.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
compress-level default-compression
```

7 Enable/disable the encryption-profile and enter the public-key-filename. A security department provides the Public Key. The configured encryption-profile is applied to the http-proxy instance. See Configuring an Encryption Profile (page 1-8) for details.

Customer data may be considered confidential and should not be shared. The system provides the ability to encrypt the Mobile Subscriber Integrated Services Digital Network-Number (MSISDN). The MSISDN number uniquely identifies a subscription in a GSM or UMTS mobile network. There are two types of encryption; static and dynamic.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
encryption-profile
```

8 Configure the IP address and IP port for the explicit proxy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
explicit-proxy-profile
```

9 Configure a fraud detection profile. See Configuring a Multi-protocol Proxy Fraud Detection Profile (page 2-10).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
fraud-detection-profile
```

10 Configure an HTTP request policy. See Configuring a Multi-protocol Proxy HTTP Request Policy (page 2-12).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy
```

11 Configure an HTTP response policy. See Configuring a Multi-protocol Proxy HTTP Response Policy (page 2-17).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy
```

12 Configure an HTTP transaction data record profile. See Configuring a Multi-protocol Proxy HTTP Transaction Record Profile (page 2-19).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-transaction-record-profile
```

13 Configure an HTTPS profile. See Configuring a Multi-protocol Proxy HTTPS Profile (page 2-20).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
https-profile
```

14 Configure a multi-protocol proxy service instance. See Configuring a Multi-protocol Proxy Instance (page 2-21).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
```

15 Configure an IP flow. See Configuring a Multi-protocol Proxy HTTP Request Policy (page 2-12).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
```

16 Configure Long Term Mobile ID. See Configuring Multi-protocol Proxy LTMI (page 2-29).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
```

17 Configure minification of Javascrip/CSS/HTML. See Configuring Multi-protocol Proxy Minification (page 2-31)

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy minify
```

18 Configure optimization bypass CPU overload control. See Configuring Multi-protocol Proxy Optimization Overload Control (page 2-31).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
optimization-overload-control
```

19 The HTTP proxy preemptive DNS feature is used to modify URLs in the response before sending it to client by inserting a local loopback IP address with token. See Configuring Multi-protocol Proxy Preemptive DNS (page 2-32).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
preemptive-dns
```

20 Enable proxy-down-bypass (workflow acts as though proxy services are not configured for this session if the serving proxy instance is down or all proxy instances are down). If disabled, connection requests are dropped until proxies are available.

On the ISM, set to enabled or disabled as follows:

- *enabled* – If the session's proxy instance is selected (indicating that the session has passed at least 1 flow) and is brought down, then a new flow is proxy-bypassed. After the serving proxy instance is brought back, the new flow continues to pass through the previously selected serving proxy instance.
- *disabled* – The proxy tries to select another available proxy instance from the same slot.

- *Regardless of enabled or disabled* – If the session's proxy instance is not selected (the first flow), then MCC selects a valid proxy instance for the first flow (if one is available).
  - *enabled* – If no proxy instances are available for that ISM slot, MCC bypasses the flow.
  - *disabled* – If no proxy instances are available for that ISM slot, MCC drops the flow.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
proxy-down-bypass enabled
```

21 Configure an RTT location offset profile and location offset parameters when Round-trip Time (RTT) is used in the congestion detection's throughput formula. See Configuring Round-trip Time Offsets (page 3-18).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
rtt-location-offset-profile
```

22 Configure HTTP proxy server connection parameters. See Configuring Multi-protocol Proxy Server Connections (page 2-33).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
server-connections
```

23 Configure a static DNS map.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
static-dns-map
```

24 Display statistics.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy statistics
```

25 Configure a group of steering destinations. See Configuring Multi-protocol Proxy Steering Group (page 2-34).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
steering-group
```

26 Configure subscriber profile table parameters. Configure the maximum number of subscribers allowed (requires slot reboot). 100 .. 6000000(100).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
subscriber-profile-table size 100
```

27 Configure TCP Splicing parameters. See Configuring Multi-protocol Proxy TCP Splicing (page 2-35).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
tcp-splicing
```

28 Configure TCP Splicing Policy. See Configuring a Multi-protocol Proxy TCP Splicing Policy (page 2-35).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
tcp-splicing-policy
```

29 Configure HTTP Toll Free Data profile. See Configuring a Multi-protocol Proxy Toll Free Profile (page 2-40).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
toll-free-profile
```

30 Configure profile for handling of splicing traffic. See Configuring a Multi-protocol Proxy Traffic Control Profile (page 2-47).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile
```

31 Configure UDP proxy parameters. See Configuring Multi-protocol Proxy UDP (page 2-50).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp
```

32 Configure a UDP policy. See Configuring a Multi-protocol Proxy UDP Policy (page 2-51).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp-policy
```

33 Affirmed Networks recommends setting user-space-networking to **enabled** for optimal ISM node throughput. Disabling this setting may impact ISM node throughput. The user-space-networking (TCP and UDP) object supports integration for all HTTP/HTTPS/TCP/UDP proxy functionality. This feature is supported for IP transparent and non-IP transparent proxy connections. This feature is supported only on the ISM and supports an increase in per-ISM VM proxy throughput up to 15 Gbps on either ESXi or KVM Openstack.

Note: *Performance may be impacted by many factors, including CPU core count, applied functionality, gateway type (GiGW versus SAEGW), and network call model.*

Also see Congestion Control Settings (page 3-13) for guidelines on the supported congestion-control settings when user-space-networking is enabled or disabled.

Note: *This configuration requires a cluster reboot to take effect. MCC issues the anClusterRebootRequired alarm, since the core-layout or vcpu information has changed. Perform the cluster reboot after you receive this alarm.*

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
user-space-networking enabled
```

34 Configure video pacing. See Configuring Multi-protocol Proxy Video Pacing (page 2-51).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
video-pacing
```

35 Configure HTTP proxy web image optimization. See Configuring Multi-protocol Proxy Web Image Optimization (page 2-52).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization
```

Configuring a Multi-protocol Proxy ABR Optimization Profile

This section describes how to configure the ABR optimization profile.

MCC displays the rate-limiting bitrate-unit and rate-limiting burst-size commands only when you change the bitrate or burst-size. Configure the abr rate-limiting bitrate and burst-size and estimate optimization byte savings.

Access	config
Limitations	None

Use the following steps to configure the multi-protocol proxy ABR optimization profile:

1. Create the ABR profile and enable the admin status.

   ```
   an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
   abr-optimization-profile <name> enabled
   ```

2. Specify the rate limiting bitrate.

   ```
   an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
   abr-optimization-profile <name> birate
   ```

3. Specify the bitrate unit. Options include: bps, kbps, mbps, and gbps.

   ```
   an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
   abr-optimization-profile <name> bitrate-unit kbps
   ```

4. Specify the rate-limiting burst size.

   ```
   an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
   abr-optimization-profile <name> burst-size
   ```

5. Specify the burst size unit. Options include: bytes, kbytes, mbytes, and gbytes.

   When NOABR is present in the acwhttppolicy Vendor-Specific Attribute (VSA) received from the PCRF, ABR video rate limiting and encrypted video rate limiting is bypassed. This mechanism is used to allow a subscriber to opt-out from video rate limiting.

   ```
   an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
   abr-optimization-profile <name> burst-size-unit kbytes
   ```

6 Set the data chunk size of the rate-limited packets for the HTTP, HTTPS, and UDP protocols. Configure the rate limiting send-interval in milliseconds in multiples of 10 (min -40ms, max -500ms, default -100ms). The data size is controlled by the time interval at which rate-limited packets are sent to the UE for a given bitrate. For example, if the configured bitrate is 1.5Mbps and the configured send-interval is 50ms (0.05 seconds), MCC sends the following in each interval:

chunksize = (1.5 Mb/8)*0.05=9375 bytes

Also see Configuring a Multi-protocol Proxy Traffic Control Profile (page 2-47) to configure the send-interval for the traffic control profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
abr-optimization-profile <name> send-interval -100ms
```

7 Set the savings estimation to enabled or disabled. This feature adds support to estimate optimization byte savings attributed to ABR pacing and display the savings in logs, statistics, and HTDRs. This feature enables Operators to log performance statistics and provide an estimate of savings obtained by enforcing rate limiting on content.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
abr-optimization-profile <name> savings-estimation enabled
```

a Specify the sampling interval for savings estimation.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
abr-optimization-profile <name> savings-estimation enabled
sampling-interval
```

b Specify the sampling interval unit min/hr.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
abr-optimization-profile <name> savings-estimation enabled
sampling-interval-unit
```

8 Set the threshold limit for statistic calculations on secondary segments. Statistic calculations are not performed on segments below the configured threshold.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
abr-optimization-profile <name> enabled secondary-segment-threshold
```

a Specify the secondary segment threshold unit bytes. Options include: gbytes, kbytes, and mbytes.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
abr-optimization-profile <name> enabled secondary-segment-threshold
secondary-segment-threshold-unit-bytes gbytes
```

b Specify the rate limiting bitrate (bits per second) for secondary segments. Options include: bps, gbps, kbps, and mbps.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
abr-optimization-profile <name> enabled secondary-segment-threshold
secondary-segment-threshold-unit-bytes gbytes
secondary-segment-bitrate-unit bps
```

Configuring a Multi-protocol Proxy Fraud Detection Profile

This section describes how to configure a fraud detection profile.

Access	config
Limitations	None

The fraud detection profile feature adds support for HTTP-proxy fraud detection, QoS, and charging of fraud HTTP traffic. This feature enables the gateway to detect and take a configured action on subscriber devices attempting data connections through fraudulent use of a web proxy. For example, the system can detect fraudulent users utilizing rogue HTTP proxies to spoof HTTP URLs for free rated sites, such as those the Carrier assigns for browsing promotions and partnerships.

HTTP proxy performs DNS validation and a header consistency check to determine if a request is destined for a legitimate domain or destined for a rogue proxy. If the destination IP address is not within the defined range or domain returned by the DNS lookup, the system determines if a free browsing proxy is in use and enforces a specified action (allow the traffic and charge for it and/or reduce its QoS to a specified level). Workflow (page 9-1) implements QoS and charging using a specified fraud qos flow and rating group.

Use the following steps to configure the fraud detection profile:

1 Configure a name for the fraud detection profile and enable the admin state.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
fraud-detection-profile <name> admin-state enabled
```

2 Configure the action (allow) to take when fraud is detected. Options include:

- **allow** – Allow the HTTP request when a fraud is detected
- **block** – Block the HTTP request when a fraud is detected

a Configure a fraud detection block notification message (string)

```
block-notification-message <string>
```

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
fraud-detection-profile <name> action allow
```

3 Apply a service rule to the fraud detection profile. The service-rule specifies which requests to subject to the fraud check. For example, to apply the fraud check only to free rated websites for browsing promotions and partnerships, configure the service-rule with the list of free-rated domains. If the fraud detection service-rule is not configured, fraud checks are performed for all HTTP requests.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
fraud-detection-profile <name> service-rule <name>
```

4 Specify a whitelist for legitimate proxies.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
fraud-detection-profile <name> whitelist
```

5 Set the following additional parameters:

a Set **services quality-of-service qos-policy qci-dscp-map fraud-qos** to **true** to designate a particular qos-flow for handling of fraudulent activity within this qos-policy.

b Set s**ervices charging billing-plan rating-group fraud-charging** to **true** to designate a particular rating-group for charging of fraudulent activity within this billing-plan.

Configuring a Multi-protocol Proxy HTTP Request Policy

This section describes how to configure the HTTP request policy.

Access	config
Limitations	None
Guidelines	For DNS resolution guidelines, see Configuration Guidelines (page 2-16).

The HTTP request policy provides a notification to the server for cache access. The content cache may store HTTP video objects that can be cached only by private caches (that is, web browsers) to improve video caching performance. Caching and serving private-cacheable videos may cause the system to incur inaccurate video object hit accounting at the origin servers. The forced-cache-access-notification feature allows the HTTP proxy to notify the origin servers of the private-cacheable video objects access by sending an HTTP HEAD request to the origin server for each client access.

After the client connection is classified for an http-ip-flow, the http-proxy receives the connection. If the request is an HTTP request, MCC parses the headers and starts the evaluation for the modules. MCC selects a matching request-policy (if configured).

Use the following steps to configure HTTP request policies for this flow. This HTTP request policy is enabled by default:

1 Configure an HTTP request policy name.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name>
```

2 Set bypass-http-proxy to enabled or disabled (default) to indicate to the request-policy whether to bypass the HTTP proxy if this request policy is used. This field defaults to disabled, indicating that the proxy processes the transaction.

This object enables the Operator to bypass the HTTP proxy process for certain domains or URIs, by configuring a request policy specific to those domains/URL.If enabled, the transaction is sent to the traffic-control function which then takes over the client connection and completes the transaction by interacting with the origin server by opening a server connection. See Bypass HTTP Proxy (page 2-15).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> bypass-http-proxy disabled
```

3 Force on-the-fly HTTP compression for all client requests.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> compress-force
```

4 Enable on-the-fly HTTP compression for supported client requests. To use this feature, you must enable compress-force.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> compress-inline
```

5 Configure the maximum response size in bytes required for compression to take effect.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> compress-max-size
```

6 Configure the minimum response size in bytes required for compression to take effect.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> compress-min-size
```

7 Configure a name for the content-insertion-profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> content-insertion-profile <name>
```

8 Configure the DNS resolution to be used for lookup destination. See DNS Resolution (page 2-16) and Configuration Guidelines (page 2-16) for additional details.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> dns-resolution
```

9 Set to enabled (or disabled) to forced cache object access notification (to Origin Server).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> forced-cache-access-notification enabled
```

This feature provides a notification to the server for cache access. The content cache may store HTTP video objects that can be cached only by private caches (that is, web browsers) to improve video caching performance. Caching and serving private-cacheable videos may cause the system to incur inaccurate video object hit accounting at the origin servers. This feature allows the HTTP proxy to notify the origin servers of the private-cacheable video objects access by sending an HTTP HEAD request to the origin server for each client access. See Chapter 6, Configuring a Content Caching Proxy.

10 Configure a header enrichment profile to use.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> header-enrichment-profile
```

11 Select the transport protocol used for server connection. Options include http, https, and ue-http-protocol (default). Use the same protocol as the ue-connection-protocol.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> http-protocol
```

12 Configure the loopback-ip-list for server connections.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> loopback-ip-list
```

13 Enable or disable minify CSS.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> minify-css enabled
```

14 Enable or disable minify HTML.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> minify-html enabled
```

15 Enable or disable minify JavaScript.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> minify-javascript enabled
```

16 Specify the maximum response size in bytes allowed for minification to take effect.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> minify-max-size
```

17 Specify the minimum response size in bytes.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> minify-min-size
```

18 Configure the network-context used for server connections; when configured this parameter overrides network-context selected by higher level.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> network-context-override
```

19 Enable or disable preemptive DNS.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> preemptive-dns enabled
```

20 Enter a unique priority of the http-policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> priority
```

21 Configure a TCP optimization profile for server connections.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> server-tcp-optimization-profile
```

22 Enter the name of a service-rule associated with this http-policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> service-rule
```

23 Associate the steering-group to the http-proxy http-request-policy. Configure the IP address of the node to steer the request. When the steering-gw is applied to the http-request-policy, the http-proxy load balances traffic between the gateways.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> steering-group
```

24 Specify the maximum data to store before forwarding request to server, in KB. When the http-request-policy is applied, the server connection is not opened until the received request buffer size reaches the value of the store-forward-buffer-limit or entire request is received. If this value is set to 0 KB (default), the request is immediately forwarded.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> store-forward-buffer-limit 0
```

25 Configure a set of policies used to control the behavior of the TCP connections that match this HTTP IP flow. The matched traffic is forwarded to the TCP splicing component and spliced to the server.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> tcp-splicing-policy
```

26 Specify the url-rewrite-profile to use for HTTP requests.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> url-rewrite-profile
```

27 Configure the Web image optimization to use for the HTTP requests. See Chapter 2, Configuring Multi-protocol Proxy Web Image Optimization.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <name> web-image-optimization
```

Bypass HTTP Proxy

This feature enables the Operator to bypass the http-proxy process for certain domains or URIs, by configuring a request-policy specific to those domains/URL.When transactions encounter this request-policy, they are tcp-spliced rather than processed by the proxy. See Configuring Multi-protocol Proxy Server Connections (page 2-33).

The http-proxy operates in tcp-proxy mode for all protocols based on:

- Configurable URL
- Configurable user-agent string
- Configurable URL-list

Select enable or disable to indicate to the request-policy whether to bypass proxy if this request-policy is used. This object defaults to disable, indicating that the proxy processes the transaction. If set to enable, the transaction is sent to the traffic-control function which then takes over the client connection and completes the transaction by interacting with the origin server by opening a server connection.

After the client connection is classified for an http-ip-flow, the http-proxy receives the connection. If the request is an HTTP request, MCC parses the headers and starts the evaluation for the modules. MCC selects a matching request-policy (if configured).

DNS Resolution

The HTTP proxy performs DNS lookup to determine the IP address of the domain in the URL in which the user visits. The DNS server IP addresses are configured on each network-context.

The DNS resolver is a library that is used to conduct asynchronous DNS lookup. The dns-resolution is configured in the http-request-policy that is associated with the http-proxy instance through the http-ip-flow.

When multiple DNS servers exist in the network-context, the http-proxy sends a query to the DNS servers in a rotating manner. In other words, the first query is sent to the first DNS server, the second query is sent to the second DNS server, and so on. The http-proxy caches the DNS result for up to two minutes.

If dns-resolution is configured, the http-proxy uses the DNS resolver to retrieve the origin server IP address from the configured dns-server(s), and selects one IP address to open the server connection.

Configuration Guidelines

The following guidelines apply to this feature:

If BEF is in use:

Only the back-end-proxy invokes DNS lookup since the front-end-proxy opens a connection to the back-end-proxy.

If Host is already an IP address:

DNS lookup is not necessary. DNS lookup is performed for the explicit Proxy since the destination IP address is one of the configured loopback IP addresses.

If http-request-policy has a steering-gw configured:

Its ip-address is used to open a server connection and as a result DNS lookup is not involved.

The network-context used to open the server connection is based on the configured server-connection in the http-request-policy. If the server-connection is not configured, MCC uses the giVpn from serviceMetaData. If after that attempt, the server-connection is still not set, MCC uses 0. Depending on the network-context, MCC selects a proper family IP address from the query response used to open the server connection.

The MCC responds to the configured loopback-ip in the network-context and takes the following action for the HTTP server connection:

loopback-ip in network-context	Action
IPv4 *and* IPv6 loopback addresses	MCC requests any server address through dns-lookup and uses ip-family-preference to determine which address is used.
IPv4 loopback address only	MCC uses the first IPv4 IP address in the response
IPv6 loopback address only	MCC uses the first IPv6 IP address in the response

If no proper family IP address is specified in the loopback-ip in the network-context, MCC uses the first IP address in the response. If a failure occurs or no IP address is specified in the loopback-ip in the network-context, MCC rejects the http-request.

Configuring a Multi-protocol Proxy HTTP Response Policy

This section describes how to configure the HTTP response policy.

Access	config
Limitations	None

Note: *If both rate-limiting-bitrate and rate-limiting-burst-size are configured, you must configure one or the other prior to performing an upgrade.*

Use the following steps to configure the HTTP response policy:

1 Configure a name for the HTTP response policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <name>
```

2 Configure ABR optimization profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <name> abr-optimization-profile
```

3 Configure the maximum data (KB) to store when processing a response from the origin server. Response latency added by the HTTP proxy increases for responses using this option. The added latency depends on the response size and the client downlink speed.

When the http-response-policy is applied, HTTP proxy uses the configured buffering-limit to determine the biggest response data size to be stored while sending the response to client. When the limit is reached, HTTP proxy stops reading from the server socket until space is freed after some data is sent to the client. The default is 256 KB.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <name> buffering-limit
```

4 Configure the header-enrichment-profile to use.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <name> header-enrichment-profile
```

5 Configure a unique priority for the http-policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <name> priority
```

6 Configure the rate limiting bit rate in bps.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <name> rate-limiting-bitrate
```

7 Configure the rate limiting burst size in bytes.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <name> rate-limiting-burst-size
```

8 Enter the name of a service-rule associated with this http-response-policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <name> service-rule
```

9 Enter the video pacing burst time in seconds. If video pacing is enabled and the burst time is set to 0, then video pacing savings is not calculated even though video pacing is properly applied to each transaction. This issue affects only video pacing savings statistics.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <name> video-pacing-burst-time
```

10 Enter the video pacing scale factor in 1/100.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <name> video-pacing-factor
```

Configuring a Multi-protocol Proxy HTTP Transaction Record Profile

This section describes how to configure the HTTP transaction data record profile.

Access	config
Limitations	None

The http-transaction-record-profile provides information about each HTTP transaction that traverses the NE. The data is represented in a comma-separated values format (csv) and written as a string of csv values.

The Operator must specify the http-transaction-record-profile in an http-proxy instance tied to the workflow (see page 9-1) for it to take effect.

For an MMS transaction, MCC forces a URL rewrite on every request. The rewritten URL could be the same as the original URL as MCC replaces the host in the original request with the configured host value. Therefore, the rewritten URL for the transaction data record for MMS transactions could be the same as the original URL

Use the following steps to configure a multi-protocol proxy HTTP transaction record profile.

1 Configure a name for the HTTP transaction record profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-transaction-record-profile <name>
```

2 Select the type of format supported.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-transaction-record-profile <name> encode-format csv
```

3 File name for streaming HTDR record locally.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-transaction-record-profile <name> file-name <name>
```

4 File type for streaming HTDR record locally. Options include: : corefiles, datarecord, eventlog, ext-storage, psm, ramdisk-storage, storage, tech-support, tosdb-content

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-transaction-record-profile <name> file-type datarecord
```

5 Enter the length to which the URL field in the record should be truncated.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-transaction-record-profile <name> record-url-length
```

6 Enter the HTTP transaction record template.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-transaction-record-profile <name> transaction-template
```

Configuring a Multi-protocol Proxy HTTPS Profile

This section describes how to configure the HTTPS profile.

Access	config
Limitations	None

The HTTPS profile enables HTTP proxy support for HTTPS. HTTP Secure (HTTPS) is a variant of the HTTP protocol which uses secure TLS, rather than TCP connections, to communicate with clients and/or servers.

HTTPS termination supports multiple sites with different certificates to be terminated for the same virtual IP address.

When terminating TLS connections, HTTP proxy supports Server Name Indication (SNI) based selection of the certificate and key pair:

- For a request with SNI:
  - If SNI matches a configured site, use the certificate and key for that site.
  - If SNI does not match any configured site, use the certificate and key pair for the default site (if configured).
  - Otherwise, reject the TLS connection.
- For a request without SNI:
  - Use the certificate and key pair for the default site (if configured).
  - Otherwise, use the certificate and key pair of the first configured site.

When originating TLS connections, HTTP proxy sends the request Host field value as SNI for the server connection.

Affirmed Network’s HTTP proxy supports the following HTTPS features:

- HTTPS termination of client TLS connections
- HTTPS origination of TLS connections to HTTPS servers
- HTTPS pass-through

If HTTP proxy is used to terminate a clients' TLS connections to HTTPS servers, the HTTPS profile must include the TLS Termination Policy for the server. The terminated HTTP proxy decrypts and processes user HTTPS requests and applies enabled content features. If the http-proxy is used to originate HTTPS server connections, server authentication is performed using a list of trusted CA certificates which must be configured in the HTTPS profile

HTTPS traffic, for which no tls-termination-policy is configured, bypasses HTTP proxy logic and is passed-through between the client and server. Because the HTTPS pass-through traffic is not processed by the HTTP proxy, content features are not applied to HTTP requests on these connections. However, connection-level features such as TCP optimization are supported.

Use the following steps to configure the HTTPS profile.

1 Enter a name for the HTTPS profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
https-profile <name>
```

2 Enable the admin state.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
https-profile <name> admin-state enabled
```

3 Configure the HTTPS origination CA certificate list.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
https-profile <name> ca-certificate-list
```

4 Configure the HTTPS termination policies for the HTTPS profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
https-profile <name> tls-termination-policy
```

Configuring a Multi-protocol Proxy Instance

This section describes how to configure the HTTP proxy service instance. The Affirmed Networks' proxy uses an IPv4/IPv6 connection to the remote content site (when available) for all traffic with the exception of NAT64 traffic. When DNS lookup is in use, Operators can specify the preferred server ip-family-preference.

Access	config
Limitations	None

Use the following steps to configure the HTTP proxy service instance:

1 Enter a name for the HTTP proxy instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name>
```

2 Enter the ABR rate limiting optimization profile for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> abr-optimization-profile <name>
```

3 Configure the ABR opting mode that allows Operators to bypass QUIC rate-limiting for selected subscribers. Options include:

- **`opt-out`** – ABR video rate limiting is applied unless the NOABR string is received in the acwhttppolicy Vendor-Specific Attribute (VSA) from the PCRF Gx interface.
- **`opt-in`** – ABR video rate limiting is not applied unless the ABR string is received in the acwhttppolicy Vendor-Specific Attribute (VSA) from the PCRF Gx interface.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> abr-opting-mode opt-in
```

4 Enable (or disable) the HTTP proxy instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> admin-state enabled
```

5 Enter the Anonymous Customer Reference (ACR) Encryption profile applicable for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> encryption-profile <name>
```

6 Enter the explicit proxy list applicable for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> explicity-proxy-profile <name>
```

7 Enter the fraud detection profile applicable for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> fraud-detection-profile <name>
```

8 Enter the HTTP transaction record profile for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> http-transaction-record-profile <name>
```

9 Enter the HTTP profile for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> https-profile <name>
```

10 Enter the IP flows applicable for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> ip-flow <name>
```

11 Enter the LDAP profile applicable for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> ldap-profile <name>
```

12 Enter the MMSC group applicable for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> mmsc-group <name>
```

13 Configure server connections. See Configuring Multi-protocol Proxy Server Connections (page 2-33).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> server-connections
```

14 Enter the SMSC group applicable for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> smsc-group <name>
```

15 Configure TCP measurement. This object supports the reporting of congestion detection by actively monitoring the performance of all incoming TCP connections to the HTTP/HTTPS/TCP proxy. In addition, tcp-measurement supports TCP performance statistics for every connection made to the proxy for all user sessions.

TCP congestion-triggered optimization enables Operators to configure optimization policies for users experiencing congestion. Optimization is based on a subscriber's measured downstream throughput. Operators trigger optimization policies based on a subscriber's access network congestion level: mild, moderate, or severe. This feature enables Operators to apply differentiated optimization polices based on current network conditions.

Operators configure a service-rule with a session-rule-group and a congestion-level rule construct. For example, if an Operator sets the severe congestion level threshold to 500 kbps, all subscribers with an average 1-min downstream throughput less than this value are considered severely congested.

MCC calculates the saturated downlink throughput based on Maximum Segment Size (MSS), Congestion Window (CWND), and Round Trip Time (RTT).

See Chapter 3, TCP Optimization.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> tcp-measurment
```

Options include:

- **`admin-state`** – enabled/disabled. The tcp-measurement object is disabled by default. If the Operator enables tcp-measurement, the MCC calculates the TCP performance parameters for the sampling frequency set in the CLI for both uplink and downlink connections.
- **`congestion-level`** – Enables Operators to configure optimization policies for users experiencing congestion. See Configuring TCP Congestion-Triggered Optimization (page 3-15).
  - **`mild`** – Mild congestion-level throughput in kbps
  - **`moderate`** – Moderate congestion-level throughput in kbps
  - **`severe`** – Severe congestion-level throughput in kbps
- **`sampling-interval`** – The value of the sampling interval is indicated in milliseconds with a range between 250-5000 milliseconds.

16 Configure a TCP optimization profile for client connections.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> tcp-optimization-service
```

17 Enter the TCP traffic control profile for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> tcp-traffic-control-profile <name>
```

18 Configure a toll free profile for the instance. See Configuring an Encryption Profile (page 2-54).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> toll-free-profile <name>
```

19 Enter the UA profile applicable for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> ua-profile <name>
```

20 Enter the quic control profile for the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> upd-traffic-control-profile <name>
```

Configuring a Multi-protocol Proxy Server Connections

This section describes how to configure the instance server connections.

Access	config
Limitations	None

The Affirmed Networks proxy uses an IPv4/IPv6 connection to the remote content site (when available) for all traffic with the exception of NAT64 traffic. When DNS lookup is in use, Operators can specify the preferred server ip-family-preference.

Use the following steps to configure the instance server connections:

1 Configure the IP family preference.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> server-connections ip-family-preference
```

In networks running a dual-stack system (both IPv4 and IPv6 loopbacks), Operators can configure the ip-family-preference as follows:

- `client-ip-family` – HTTP proxy requests both IPv4 and IPv6 server addresses and uses the client's ip-family address unless it is unavailable.
- `IPv4` – HTTP proxy requests both IPv4 and IPv6 server addresses and uses IPv4 address unless it is unavailable.
- `IPv6` – HTTP proxy requests both IPv4 and IPv6 server addresses and uses IPv6 address unless it is unavailable.

2 Configure the loopback IP list for server connections.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> server-connections loopback-ip-list ipv4
```

3 Configure the loopback selection algorithm. Options include: round-robin and session-sticky.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> server-connections loopback-selection-algorithm round-robin
```

4 Enter the network-context used for server connections; when configured, this parameter overrides the network-context selected by the gateway subsystem.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> server-connections network-context-override
```

5 Configure a TCP optimization profile for server connections. See Configuring a Multi-protocol Proxy Traffic Control Profile (page 2-47).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> server-connections tcp-optimization-profile
```

Configuring a Multi-protocol Proxy IP Flow

This section describes how to configure the IP flow.

Access	config
Limitations	None

Use the following steps to configure one or more IP flows which should be processed by the HTTP proxy:

1 Enter a name for the ip-flow.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name>
```

2 Set the ip-flow to enabled (or disabled).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> enabled
```

3 Set the application-detection to enabled (or disabled). This feature enables Operators to use Deep Packet Inspection (DPI) in multi-protocol-proxy policies. The DPI function can detect hundreds of applications based on traffic signatures.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> application-detection enabled
```

When an application is detected, the information is passed to the multi-protocol proxy. Based on the DPI input provided, the proxy reevaluates policies, performs matching against configured protocol-application-rule-group rules, and selects a new policy. See Application Detection (page 2-27).

4 Configure an http-request-policy. See Configuring a Multi-protocol Proxy HTTP Request Policy (page 2-12).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> http-request-policy
```

5 Configure an http-response-policy. See Configuring a Multi-protocol Proxy HTTP Response Policy (page 2-17).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> http-response-policy
```

6 Set ip-transparency to enabled (or disabled). When this option is enabled, HTTP requests forwarded by the http-proxy to the origin servers contain the UE's original IP address as the source TCP/IP address in the IP packet. See IP Transparency (page 2-28).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> ip-transparency enabled
```

7 Enter a unique priority for the http-ip-flow.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> priority 5
```

8 Configure a service protocol (http/https/tcp/udp) for this ip-flow. See the note in step 10. Set to **tcp** to take advantage of optimized fast-open connection establishment.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> protocol tcp
```

9 Set proxy-bypass to enabled or (disabled). This feature enables Operators to bypass proxy for a specific IP address or domain.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> proxy-bypass disabled
```

Affirmed Networks recommends having a small number of packet-filters to bypass; the maximum is 256.

This feature applies only to proxy bypass external server (s) configured through ip-flow. This feature does not apply to the explicit proxy where a loopback is configured. Configure the bypassed ip-flow object with a higher priority than the non-bypassed ip-flow object. The proxy-bypass object defaults to disabled.

10 Enter the name of the service rule associated with this ip-flow.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> service-rule <name>
```

The service-rule associated with an ip-flow cannot contain a protocol-application-rule-group, if the protocol is configured as http (see step 8). The protocol-application-rule-group applies to TCP and UDP traffic only and is not supported for HTTP.

11 Enter the name of the TCP splicing policy for the ip-flow.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> tcp-splicing-policy <name>
```

12 Enter the name of the upd-policy (QUIC policy) for the ip-flow.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ip-flow
<name> upd-policy <name>
```

Application Detection

MCC supports the capability to use Deep Packet Inspection in multi-protocol-proxy policies. The DPI function can detect hundreds of applications based on traffic signatures.

When an application is detected, the information is passed to Multi-Protocol-Proxy. Based on the DPI input provided, Proxy reevaluates policies, performs matching against configured protocol-application-rule-group rules, and selects a new policy.

This capability can be used to apply video rate limiting to HTTPS and QUIC connections for video streaming web sites supported by DPI. For example, with a protocol-application-rule-group configured for Netflix video, a configured tcp-splicing-policy can be associated with that protocol-application-rule-group. Based on the application received, proxy re-evaluates policies and performs rule-matching to select a new tcp-splicing-policy/udp-policy. A new traffic-control-profile is selected and the profile parameters are enforced on the flow.

Use the following steps to configure application detection:

1. Configure application-detection for TCP, HTTPS, or UDP (QUIC) proxy IP flows:

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-ip-flow <name> admin-state enabled priority <value> protocol
https ip-transparency disabled application-detection enabled
service-rule <name> tcp-splicing-policy <name>
```

2. In the associated service-rule, specify a protocol-application-rule-group that defines the application received by DPI detection:

```
an3000-mcm-slot7cpu1(config)# service-construct service-rule <name>
admin-state enabled priority <value> service-data-flow-id <value>
protocol-application-rule-group <name> tcp-filter all
service-activation always-on
install-default-bearer-packet-filters-on-ue disabled
```

3. Specify the same service-rule in the tcp-splicing-policy/udp-policy:

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
tcp-splicing-policy <name> priority <value> traffic-control-profile
<name> service-rule <name>
```

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp-policy
<name> priority <value> traffic-control-profile <name>
congestion-control head-drop service-rule <name>
```

Note: *Affirmed Networks recommends that the service-rule containing a protocol-application-rule-group have a higher Priority than other service-rules (containing domain-rule-group, session-rule-group, for example).*
Note: *The service-rule associated with an ip-flow cannot contain a protocol-application-rule-group, if the protocol is configured as http. The protocol-application-rule-group applies to TCP and UDP traffic only and is not supported for HTTP.*
Note: *Order of precedence for applying rate limiting to a flow:*

- *multi-protocol-proxy instance abr-opting-mode*
- *max-rate in the Toll Free Data sponsored website manifest*
- *protocol-application-rule-group*

IP Transparency

When this option is enabled, HTTP requests forwarded by the http-proxy to origin servers contain the UE's original IP address as the source TCP/IP address in the IP packet. IP transparent proxy is an HTTP proxy mode of operation where HTTP client IP addresses (UE addresses) are used as a source when originating network-side connections to Web origin servers.

HTTP proxy, using the client addresses, hides its own network identity and becomes IP transparent to Internet origin servers. When IP transparency is used, origin servers are able to determine the identity of Web clients based on the source IP addresses of the HTTP/HTTPS connections.

In non-IP Transparent mode, Affirmed Network's HTTP proxy uses MCC IP loopback addresses when originating network-side connections.

Configuring Multi-protocol Proxy LTMI

This section describes how to configure the Long Term Mobile ID (LTMI) objects to control the key import (download) from the key server using HTTPS.

Access	config
Limitations	None

Also see Configuring an Encryption Profile (page 1-8). The encryption-profile is used in HTTP Header Enrichment (HHE) to determine the algorithm and keys to be used for header encryption.

Use the following steps to configure LTMI:

1 Configure the key import interval (minutes). A value of 0 disables this setting. Using HTTPS, the MCC periodically downloads encryption keys from a key server. The keys can be manually imported using **`import ltmi-key`** or automatically imported from the key server by configuring the key-import-interval.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
key-import-interval 20
```

2 Configure the key-server-network-context. The key server network context should consist with loopback-ip-list. LTMI download chooses the loopback from the loopback-ip-list if configured.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
key-server-network-context
```

3 Configure the key Server password for basic authentication.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
key-server-password <password>
```

4 Configure the key server URL, including the IP address or hostname, port, and URI.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
key-server-url
```

5 Configure the key server user name for basic authentication.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
key-server-username <username>
```

6 Configure the loopback IP list for the key server. The key-server-network-context should consist with loopback-ip-list.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
loopback-ip-list <loopiplist>
```

The following parameters control if and how the LTMI ID is generated. Operators can opt out of the LTMI feature using the opt-out-string and static-trusted-id-string as follows:

1. Set the id-gen-admin-state to enabled (or disabled); disabled will disable LTMI ID generation.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
id-gen-admin-state enabled
```

2. Configure the opt-out-string. If the miDial string exactly matches the opt-out-string, then the subscriber completely opts out of LTMI.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
opt-out-string <string>
```

3. Configure the static-trusted-id-string. If the miDial string ends with the static-trusted-id-string, then the static trusted ID is used instead of the dynamic trusted ID. For example, if the static-trusted-id-string is configured as "#" and the miDail string is "xyz#" (ending with "#"), then the static-trusted-id is used instead of the dynamic trusted ID.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
static-trusted-id-string <string>
```

4. Configure the trusted-id-divisor as an integer of minutes for trusted ID rotation interval. For example, a 6 month rotation would be 259,200 minutes.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
trusted-id-divisor 259200
```

5. Configure the external-id-divisor as an integer of minutes for external ID rotation interval. For example, a 24 hour rotation would be 1440 minutes.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
external-id-divisor 1440
```

6. Configure the network-carrier-code used in dynamic mobile IDs to identify the carrier.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy ltmi
network-carrier-code <NCC>
```

Configuring Multi-protocol Proxy Minification

This section describes how to configure minification to remove unnecessary characters (such as blank characters and new line characters) from source code, thereby reducing the amount of data that needs to be transferred.

Access	config
Limitations	None

MCC supports minification on the following source code for content optimization:

- JavaScript
- HTML
- CSS

Use the following steps to configure minification:

1 Set the admin-state to enabled (or disabled) for Javascript, CSS, or HTML. See Configuring a Multi-protocol Proxy Server Connections (page 2-24) for additional details.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy minify
admin-state enabled
```

Configuring Multi-protocol Proxy Optimization Overload Control

This section describes how to configure optimization bypass CPU overload control. Also see zone gateway threshold in *Affirmed Networks Gateway Administration Guide*.

Access	config
Limitations	None

Set the optimization-overload-control admin-state to enabled.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
optimization-overload-control admin-state enabled
```

Configuring Multi-protocol Proxy Preemptive DNS

This section describes how to configure preemptive DNS. The HTTP proxy preemptive DNS feature is used to modify URLs in the response before sending it to the client by inserting a local loopback IP address with a token.

Access	config
Limitations	None

This object enables the preemptive-dns features and replaces the domain request with the IP address of one of the Affirmed GI local loopback addresses and configured DNS token. This feature reduces the cost per transaction and that cost is amortized across all proxies instead of one proxy. See Configuring a Multi-protocol Proxy Server Connections (page 2-24) for additional details.

Use the following steps to configure preemptive DNS:

1 Set the preemptive-dns admin-state to enabled.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
preemptive-dns admin-state enabled
```

2 Configure the IPv4 loopback to be used to replace URLs in HTTP response.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
preemptive-dns loopback-ipv4
```

3 Configure the IPv6 loopback to be used to replace URLs in HTTP response.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
preemptive-dns loopback-ipv6
```

4 Configure the network-context to be used to select loopback-ip.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
preemptive-dns network-context
```

5 Configure a string token to be used to replace URLs in the HTTP response.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
preemptive-dns token <string>
```

Configuring Multi-protocol Proxy Server Connections

This section describes how to configure multi-protocol-proxy server connection parameters.

Access	config
Limitations	None

The server connection pool feature increases performance and decreases the client page load times since the HTTP proxy reuses previously established connections to http servers. If a client closes a connection with http-proxy, the system makes the associated server connection available for reuse by inserting it into the idle server connection pool.

Use the following steps to configure the HTTP proxy server connections:

1. Turn on (off) HTTP server fast connection. This feature supports improved page load times for HTTP connections for TCP splicing and HTTP proxy in IP transparency mode. When Fast Open is configured, the MCC proxy opens a server connection during the client connection handshake, which reduces the HTTP transaction time by approximately one Round Trip Time (RTT) between the UE and MCC proxy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
server-connections http-fast-open
```

2. Configure server connection idle timeout in seconds (30-600). The default is 60 seconds.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
server-connections idle-timeout 60
```

3. Configure the maximum number of idle connections per server. This value defaults to 64 connections. A value of 0 disables this feature.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
server-connections max-server-idle 64
```

4. Configure the total maximum number of idle connections per server. This value defaults to 131072 connections. A value of 0 disables this feature.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
server-connections max-total-idle 131072
```

Configuring Multi-protocol Proxy Steering Group

This section describes how to configure a group of steering destinations. The http-proxy steering-group is enhanced to allow peers with the same IP address to be reachable through different network-contexts.

Access	config
Limitations	None

HTTP proxy supports Layer 7 load balancing. This feature enables Operators to configure a steering-group (ip-address, port). Based on the http-proxy http-request-policy rules, the request is steered to the server using a round-robin algorithm.

Use the following steps to configure a group of steering gateways and configure the network-context to use for the steering-gateway-group. Also see **Configuring a Multi-protocol Proxy HTTP Request Policy (page 2-12).**

1 Configure a name for the steering group.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
steering-group <name>
```

2 Configure a health check profile for the steering group. See the *Affirmed Networks Gateway Administration Guide* to configure the health-check profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
steering-group <name> health-check-profile <name>
```

3 Configure a steering destination. Priorities are configurable. When a higher priority destination goes down, MCC switches traffic to a lower priority destination. MCC generates an event/alarm when it detects that a steering destination went down and clears the event/alarm when the steering destination comes back online.

Options include:

- **host-name** – Host Name (FQDN) of the steering destination.
- **ip-address** – IP address for steering destination.
- **name*** – HTTP steering destination Name (string).
- **port** – Port for steering destination port (default value is 80).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
steering-group <name> steering-destination
```

Configuring Multi-protocol Proxy TCP Splicing

This section describes how to configure TCP splicing parameters.

Access	config
Limitations	None

Use the following steps to configure tcp splicing parameters:

1 Configure the idle-timeout in seconds. The following command is a debug command and requires you to enable debug.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
tcp-splicing idle-timeout 180
```

2 Turn the server-connection-fast-open on or off. This feature supports improved page load times for HTTP and HTTPS connections for TCP splicing and HTTP proxy in IP transparency mode. When Fast Open is configured, the MCC proxy opens a server connection during the client connection handshake, which reduces the HTTP/HTTPS transaction time by approximately one Round Trip Time (RTT) between the UE and MCC proxy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
tcp-splicing server-connection-fast-open
```

Configuring a Multi-protocol Proxy TCP Splicing Policy

This section describes how to configure the TCP splicing policy. Also see Chapter 3, TCP Optimization.

Access	config
Limitations	None.
Guidelines	See Configuration Guidelines (page 2-37).

TCP Splicing Overview

The TCP proxy capability is currently realized using the http-proxy tcp-splicing function. When HTTP proxy is configured with a set of TCP protocol IP flows, it performs TCP splicing (data forwarding) between client and server TCP connections.

The TCP splicing policy (TCPSP) is used to apply non-default actions to a set of TCP spliced connections in the http-proxy. The TCPSP enable Operators to apply differentiated treatment to TCP connections based on service-rule selections, for example:

- allows rate limiting of connections
- block / allow a specific set of TCP connections
- change the core-network context
- select source loopback IP address(es)
- steer connections to a server domain

HTTP proxy supports forwarding of non-HTTP traffic received on port 80. The TCP splicing component of HTTP proxy is responsible for handling of non-HTTP traffic. The component maintains a pair of client-side and server-side connections and forwards data between these connections.

TCP splicing provides the ability to enable TCP optimization for TCP-based protocols such as SFTP, FTP and HTTPS. TCP splicing stitches the client and server TCP connections, enabling the Operator to configure the TCP parameters to be used for the client connections separately.

MCC can be configured as a TCP proxy and support the following:

- Terminate a TCP connection from the client and apply a set of TCP optimization parameters.
- Originate a TCP connection to the server and apply a set of TCP optimization parameters.
- Forward TCP traffic between the client and server TCP connections without parsing or disturbing the packets.
- Teardown the TCP connections when the peer closes the connections or idle exceeded timeout

Currently, the following non-HTTP traffic flows are handled by TCP splicing:

- Traffic on HTTP connection which has been upgraded to another protocol, for example web sockets
- Traffic received on HTTP connection but not parsable as HTTP

- Traffic over HTTP tunnel established with HTTP CONNECT method
- Traffic received on http-ip-flow configured with tcp protocol
- HTTPS traffic for which no certificates and keys are configured
- Traffic received on http-ip-flow after configured request-wait-time expired. In this scenario, the connection is handed off to the TCP splicing component after a configured request-wait-time expired.

  This is needed for protocols such as SMTP where the server sends the initial message in the exchange. Occasionally, due to a timing issue with a slow web browser not sending an HTTP request in time, a legitimate HTTP connection is handed off to TCP splicing. TCP splicing examines incoming traffic on the connection and if it detects HTTP, it hands the connection back to the HTTP proxy - this is sometimes referred to as *reverse handoff*. With reverse handoff, all HTTP proxy functionality is applied to the connection, including HTTP header enrichment and optimization features.

Configuration Guidelines

The following table describes the conditions when tcp-splicing is triggered and when a tcp-splicing-policy is applied triggering a DNS lookup.

Table 2-1 DNS Lookup

Configured http-ip-flow protocol	When tcp-splicing occurs	tcp-spicing-policy
tcp	always	applicable
https	no https credential	applicable
http	not http request	applicable
http	http request timeout	applicable
http	http connection upgrade	not applicable
	http connect	not applicable

Example Call Flow

Operators may apply a tcp-splicing-policy to provide differentiated treatment of subscriber data, such as free of charge service, based on the destination network domain. In this case, domain-based policy rules must be configured using a tcp-spicing-policy. The following example call flow describes a use case allowing a UE to access freedomain.com free of charge.

1 The UE issues a DNS lookup for *freedomain.com*.

2 The DNS proxy on the SSM returns a local loopback ip-address [2001::AFFD:1] configured in dns static-map.

3 The UE establishes a TCP connection with the destination address [2001::AFFD:1] on port 443 to MCC.

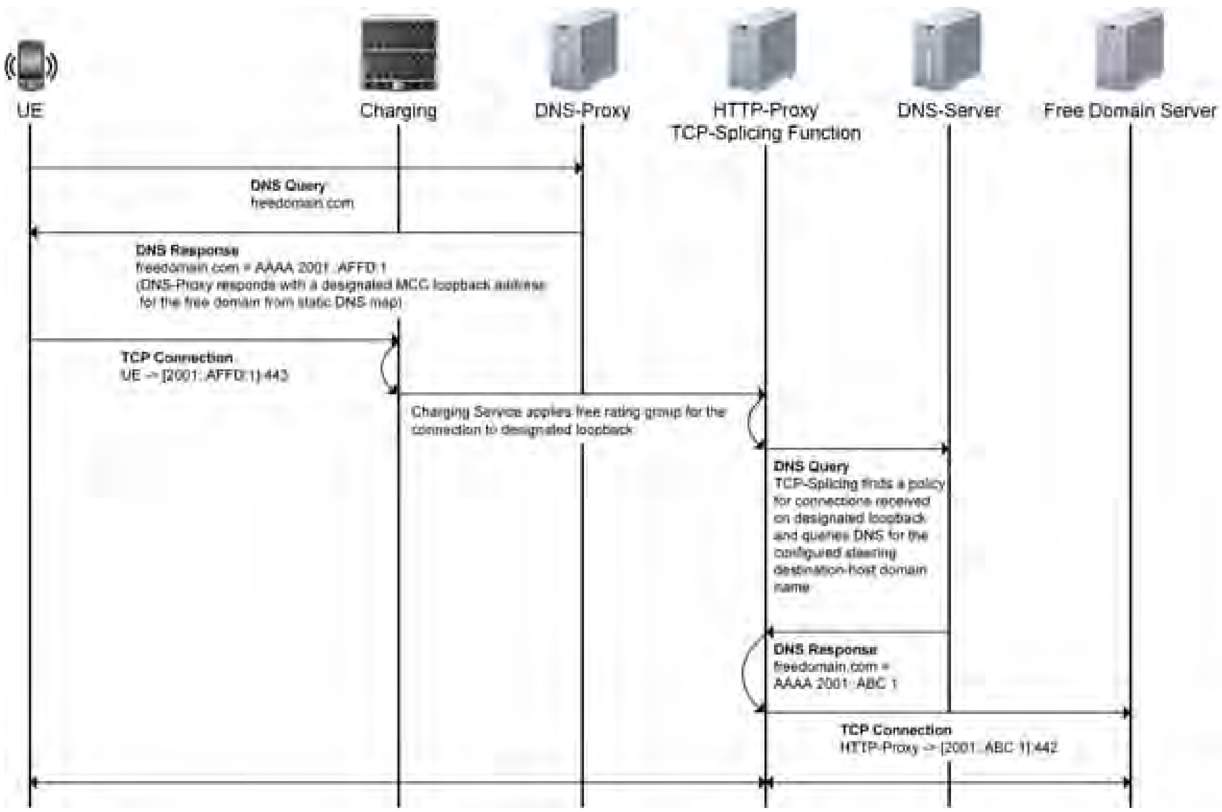
4 MCC uses workflow (page 9-1) to match the destination address to the service-rule in http-ip-flow and forwards the address to the http-proxy, after applying the free charging policy associated with the workflow.

5 The http-proxy matches the destination address with the tcp-splicing-policy within http-ip-flow and applies the policy containing the steering-group with *freedomain.com* in its destination-host.

6 MCC triggers a DNS lookup to obtain the real ip-address [2001::ABC1].

7 Upon a successful DNS response, MCC establishes a TCP connection to the server on port 443. Up to this point, and end-to-end communication channel was established between the UE and freedomain.com and data was spliced over the channel.

Figure 2-1 TCP Splicing Policy - Example Call Flow



Use the following steps to configure a TCP splicing policy. The configured service-rule must have a packet-filter and must not have an http-rule-group.

Note: *If ip-transparency is enabled in Configuring a Multi-protocol Proxy IP Flow (page 2-25), applying a tcp-splicing-policy causes a change in destination address and the request is rejected.*

---

1 Configure a name for the TCP splicing policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
tcp-splicing-policy <name>
```

2 Specify a unique priority for the tcp-splicing-policy between 1 - 255, where 1 is the highest. If multiple polices are configured in the http-ip-flow, the policy is evaluated in the order of priority.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
tcp-splicing-policy <name> priority 1
```

3 Specify the name of a service-rule associated with this tcp-splicing-policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
tcp-splicing-policy <name> service-rule <SR-name>
```

4 Specify the name of a configured traffic profile associated with this policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
tcp-splicing-policy <name> traffic-control-profile <TP-name>
```

Configuring a Multi-protocol Proxy Toll Free Profile

This section describes how to configure the toll free data profile.

Access	config
Limitations	None

Operators can configure a toll-free-profile to support Toll Free Data (TFD). TFD is a feature in which a third-party (content provider) pays for a subscriber's data usage when the subscriber is downloading or using the third-party company's Apps, Websites, and services. If a toll-free-data profile is enabled, then TFD is turned on for that proxy instance.

The Gateway support for Toll Free Data (International Roaming) feature enables the Operator to determine the roaming zone of a PDN subscriber session in the gateway for the TFD feature. The roaming zone information is relayed to the proxy module. The proxy module determines if the TFD feature must be supported for the subscriber session, based on the subscriber's current roaming zone. The TFD feature is not supported for outbound international roaming sessions. If the "international roaming" state is not known, TFD is not supported.

The PCRF determines the roaming zone for subscribers using the SGW IP address received in the IP address AVP in the Gx CCR message. The PCRF includes roaming zone information in the cpsRoamZone AVP in the Gx CCA message. If the cpsRoamZone custom AVP isn't received in the Gx CCA message, the gateway sets the roaming zone to unknown and the TFD feature is not supported for the sessions.

Use the following show command to determine the roaming zone:

```
an3000-mcm-slot7cpu1#show subscriber pdn-session <pdn-session-id>
vsa-roam-zone  International
```

Subscriber Profiles (SPs) are objects containing mobile user specific information. SPs are used to store Toll Free Data (TFD) manifests and are also used to store future per-subscriber parameters such as user preferences, content feature eligibility, optimization opt-outs, content-filtering subscription status, etc.

SPs contain:

- subscriber key (for example, session ID or MSISDN)
- a list of TFD manifests (each with manifest ID + expiration time + URL regexes)

For more information, see the *Affirmed Networks System Administration Guide*.

Use the following steps to configure a toll free profile:

1. Configure a name for the toll free profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
toll-free-profile <name>
```

2. Set the admin-state to enabled (or disabled).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
toll-free-profile <name> admin-state
```

3. Configure a toll free data certificate server.

   Options include:

   - **`ca-certificate-list`** – Toll free data CA certificate list. The CA certificate must first be placed in */public/content/tls/cacerts* before it can be configured.
   - **`certificate-file-extension`** – Certificate file extension.
   - **`certificate-url`** – Certificate URL prefix - string, starting with http://
   - **`crl-url`** – Certificate revocation list (CRL) URL - string, starting with http://

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
toll-free-profile <name> certificate-server
```

   When toll-free-data manifest is received before Certificate Revocation List (CRL) is available to the MCC, the manifest is processed. After parsing and signature verification, the manifest is installed into the subscriber service profile.

   The manifest, using revoked content provider certificate, is only removed when the UE triggers a manifest refresh (re-downloads the manifest again).

   - **`loopback-ip`** – Loopback IP to use to connect to certificate server.
   - **`network-context`** – Network context to use connect to certificate server.

4. Enable or disable certificate verification properties. MCC supports a certificate-verification option to temporarily disable certificate verification when the TFD CA certificate is expired or near expiration. Setting this option to disabled allows TFD to process. Certificate verification is bypassed if certificates are expired and this configuration option is enabled.

   MCC checks all configured CA certificates every 15 minutes for expiry. If any of those certificates expire, MCC raises the **CntnCACertExpired** alarm. The **CntnCACertExpired** alarm clears when a valid CA certificate is installed with the same name or is removed from the CA certificate configuration.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
toll-free-profile <name> certificate-verification enabled
```

5 Enable the admin state to configure the certificate revocation list (CRL) refresh interval.

Options include

- **unit** – Specify CRL Refresh Interval Unit:min/hr.
- **value** – Specify the CRL-Refresh Time Interval Value.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
toll-free-profile name crl-refresh-interval admin-state enabled
```

6 Configure the toll free data record.

Options include:

- **file-name** – File name for streaming TFD record locally.
- **file-type** – File type for streaming TFD record locally.
- **tfd-template** – TFD data record template.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
toll-free-profile <name> data-record-configuration
```

7 Configure toll free data manifest properties as bytes or kbytes. The maximum supported manifest size is 256 KB. Options include:'

- **manifest-max-size** – Maximum allowed size of TFD manifest. This setting is configurable.
- **manifest-max-size-unit** –Manifest max size units: byte/kbyte.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
toll-free-profile <name> manifest manifest-max-size 256
manifest-max-size-unit kbyte
```

8 Configure toll free header names.

Options include:

- **tf-access** – Toll free access.
- **tf-callback-url** – Toll free callback URL.
- **tf-eligible** – Toll free eligible.
- **tf-manifest-id** – Toll free manifest ID.
- **tf-manifest-indication** – Toll free manifest indication.
- **tf-sid** – Toll free sponsor ID.
- **tf-status** – Toll free status.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
toll-free-profile <name> tf-header-names
```

Local Toll Free Data Records

This section lists the Toll Free Data Record (TFDR) fields and data types. TFDRs are used to keep track of users' billing information for each toll free data transaction. The data is presented in Abstract Syntax Notation 1 (ASN.1) format.

Table 2-2 lists the name, data type and description for each field contained in TFDRs.

Table 2-2 Toll Free Data Record Fields

Name	Data Type	Description
CDR Entity	Unsigned64	Indicates the record sequence number.
Recording Entity	String	Indicates the Proxy identity.
Client Identity/MSISDN	String	Indicates the Mobile Station (MS) ISDN number (MSISDN) of the User Equipment (UE) that originated the request. This field is empty if the MSISDN is not available.
Destination URL	String	Indicates the URL from which the content was retrieved.
Content Size from Terminal	32 Bit Unsigned Integer	Indicates the volume of data received from the terminal, in bytes.
Content Size to Terminal	32 Bit Unsigned Integer	Indicates the volume of data sent to the terminal, in bytes.
Content Size from Server	32 Bit Unsigned Integer	Indicates the volume of data received from the origin server, in bytes.
Content Size to Server	32 Bit Unsigned Integer	Indicates the volume of data sent to the origin server, in bytes.
HTTP Method	String	Indicates the HTTP method used for the Toll Free Data (TFD) transaction. This field is empty if the HTTP method is not available.
Session ID	String	Identifies the MCC session ID for the TFD transaction.
Charging ID	String	Identifies the MCC charging ID for the TFD transaction.
Network Access Type	String	Indicates the network access type used for the TFD transaction.
APN	String	Indicates the Access Point Name (APN) used to make the request.
SGSN IP Address	String	Indicates the IP address of the Serving GPRS Support Node (SGSN).
Source IP Address	String	Indicates the IP address allocated to the terminal.

Table 2-2 Toll Free Data Record Fields

Name	Data Type	Description
Destination IP Address	String	Indicates the IP address derived from the destination URL.
IMSI	String	Indicates the International Mobile Subscriber Identity (IMSI) of the User UE that originated the request.
Negotiated QoS Profile	String	Indicates the 3GPP negotiated Quality of Service (QoS) profile.
Return Code	String	Indicates the HTTP response code.
HTTP Request Timestamp	String	Indicates the time when the HTTP request was sent to the origin server. The time is accurate to the nanosecond.
HTTP Response Timestamp	String	Indicates the time when the HTTP response was received from the origin server. The time is accurate to the nanosecond.
Virtual GW Name	String	Indicates the virtual name of the MCC gateway.
Virtual GW External IP Address	String	Indicates the external IP address of the virtual MCC gateway.
Event Timestamp	String	Indicates the event time of the recorded TFD transaction. The time is accurate to the second.
Radio Access Type	String (Integer presented in String format)	Indicates the Radio Access Technology (RAT) type currently used by the Mobile Station, when available. The following values are defined: ■ 0 = Reserved ■ 1 = UTRAN ■ 2 = GERAN ■ 3 = WLAN ■ 4 = GAN ■ 5 = HSPA-Evolution ■ 6 = EUTRAN ■ 101 = IEEE 802.16e ■ 102 = 3GPP2-eHRPD ■ 103 = 3GPP2-HRPD ■ 104 = 3GPP2-1xRTT ■ 105 = 3GPP-EPS
IMEI	String	Indicates the International Mobile Equipment Identity (IMEI) of the UE that originated the request.
Roaming Information	String	Indicates the roaming information of the TFD transaction.

Table 2-2 Toll Free Data Record Fields

Name	Data Type	Description
Subscriber Plan	String	Indicates the subscriber plan that was active during the processing of this request.
Sponsor ID	String	Indicates a unique identifier for the content provider.
Application ID	String	Indicates a unique identifier for the content provider's application.
TFD Charging Type	String	Indicates whether the UE or the content provider (CP) is charged for the transaction. The following values are defined: ■ CP ■ UE
TFD Charging Reason	String	Indicates the reason for charging either the UE or the CP, such as "UE and manifest toll free".
TFD Failure Reason	String	Indicates the reason for the failure during the TFD transaction.
Operator Specific Attribute 1	String	Indicates additional TFD parameters as defined by the network operator.

Toll Free Data Record ASN.1 Definition

The following is a complete ASN.1 definition of the Toll Free Data Records.

```
AFFIRMEDNETWORKS
DEFINITIONS IMPLICIT TAGS ::=
BEGIN
HttpDataRecord ::= CHOICE
{
   tfdDataRecord            [0] IMPLICIT TfdDataRecord
}
TfdDataRecord ::= [0] SEQUENCE
{
    cdrEntity                    [0]     INTEGER
   ,timeStamp                    [8]     GeneralizedTime
   ,recordingEntity              [1]     IA5String OPTIONAL
   ,msisdn                       [3]     IA5String OPTIONAL
   ,url                          [5]     IA5String OPTIONAL
   ,contentSizeFromTerminal      [7]     INTEGER OPTIONAL
   ,contentSizeToTerminal        [6]     INTEGER OPTIONAL
   ,contentSizeFromServer        [117]   INTEGER OPTIONAL
   ,contentSizeToServer          [118]   INTEGER OPTIONAL
```

```
    ,httpMethod                 [21]    IA5String OPTIONAL
    ,sessionId                  [33]    IA5String OPTIONAL
    ,chargingId                 [32]    IA5String OPTIONAL
    ,networkAccessType          [31]    IA5String OPTIONAL
    ,apn                        [52]    IA5String OPTIONAL
    ,sgsnIPAddress              [28]    IA5String OPTIONAL
    ,sourceIPAddress            [26]    IA5String OPTIONAL
    ,destinationIPAddress       [25]    IA5String OPTIONAL
    ,imsi                       [36]    IA5String OPTIONAL
    ,negotiatedQoS              [62]    IA5String OPTIONAL
    ,returnCode                 [18]    IA5String OPTIONAL
    ,httpRequestTime            [42]    IA5String OPTIONAL
    ,httpResponseTime           [41]    IA5String OPTIONAL
    ,virtualGWName              [77]    IA5String OPTIONAL
    ,virtualGWExternalIpAddress [78]    IA5String OPTIONAL
    ,radioAccessType            [81]    IA5String OPTIONAL
    ,imei                       [82]    IA5String OPTIONAL
    ,roamingInformation         [83]    IA5String OPTIONAL
    ,subscriberPlan             [131]   IA5String OPTIONAL

    ,sponsorId                  [159]   IA5String OPTIONAL
    ,applicationId              [160]   IA5String OPTIONAL
    ,tfdChargeType              [161]   IA5String OPTIONAL
    ,tfdChargeReason            [162]   IA5String OPTIONAL
    ,tfdFailureReason           [163]   IA5String OPTIONAL
END
```

AVP Definitions for cpsRoamZone

AVP Name	Vendor ID	AVP Code	AVP Type	Possible Values	Description	Multi-Value	Mandatory /Optional
cpsRoamZone	9 (cisco)	131512	UTF8String	On-Net Off-Net International	Roaming Zone	N	O

Configuring a Multi-protocol Proxy Traffic Control Profile

This section describes how to configure the profile for handling of splicing traffic.

Access	config
Limitations	The following limitations apply to savings-estimation: ■ In case of CSM failure, the savings data for that transaction is lost. ■ The MCC relies on the session bitrate to obtain data. This is an approximate method used to calculate savings. ■ Byte savings is updated only at the end of a quic rate-limiting session and not for each UdpProxyFlow. ■ Since the data inside a packet is encrypted, there is a small possibility of measuring non-video content for savings estimation. The following limitations apply to rate-limiting: ■ Rate limiting is performed on a domain basis rather than a server IP address basis. This means that different videos to the same domain with the same user are grouped into a single session and the system applies the configured rate limiting. ■ If a single user is watching multiple videos from the same domain, the delivered rate may be less than intended. ■ If an SNI is not specified in the client hello, processing reverts to using the server IP address. Then, all flows without an SNI with that server IP address are added to the same video session. ■ If a video flow matches a policy with a specified manifest bitrate, that rate is used for as the target rate for that video session. ■ A maximum of 64 rate-limited flows is added to a video rate-limiting session. Additional flows matching that video session are bypassed.
Guidelines	The following guidelines apply to ABR shaping for multi-stream HTTPS connections to the same server. ■ This feature is used only when rate-limiting is enabled for a subscriber and video. ■ Flows can be aggregated into 'sessions' to apply rate limiting across all flows. Disable this setting when multiple users share the same connection (tethering). ■ When flow-aggregation-state is disabled, MCC uses the existing behavior. ■ MCC identifies flows that are associated with a common domain, regardless of server FQDN, as candidates for aggregated rate limiting. If the domain is unavailable, MCC reverts to the server IP address. ■ Streams to a user from different servers within the same domain are eligible for aggregation. ■ MCC applies a rate-limiting bandwidth algorithm that is ‘fair’ to all streams, preventing bandwidth ‘starvation’ and ‘hogging’ for each flow. ■ Dynamic fluctuations in flow bandwidth is re-calculated per stream.

Configure a profile to handle non-http traffic on Port 80. The http-proxy in transparent mode does not break non-http protocols that traverse TCP Port 80. In addition, it does not break addressable http-proxy calls to an external proxy that traverses TCP Port 80. Configure the action (allow or block) to apply to non-http traffic.Configure the loopback-ip-list and network-context-override for non-http connections. Configure the instance for this non-http-profile.

MCC support ABR shaping for multi-stream HTTPS connections to the same server. This feature enables Operators to apply HTTPS rate limiting (shaping) to all flows concurrently, as a group, for a single logical video.

Browsers often open multiple flows for a single video in order to download content faster. As a result, each of those flows receives the same configured bandwidth, diminishing the utility of rate limiting. MCC enables Operators to optionally aggregate those flows and apply the limited rate collectively to all flows.

non-http profile

The non-http profile is used to handle non-http traffic on Port 80. The http-proxy in transparent mode does not break non-http protocols that traverse TCP Port 80. In addition, it does not break addressable http-proxy calls to an external proxy that traverses TCP Port 80.

Use the following steps to configure the traffic control profile:

1. Configure a name for the traffic control profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile <name>
```

2. Configure the maximum data to store when processing downstream data, units KB.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile <name> downstream-buffering-limit 128
```

3. Configure loopback IP list for splicing connections.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile <name> loopback-ip-list
```

4. Configure network-context override for the traffic-control-profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile <name> network-context-override
```

5 Configure rate-limiting.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile <name> rate-limiting
```

Options include:

- **admin-state** – Enable/Disable Rate Limiting
- **bitrate** – Configure Rate Limiting bitrate.
  - **bitrate-units** – Configure the bitrate-units. Options include: bps, kbps, mbps, and gbps.
- **burst-size** – Configure Rate Limiting burst size.
  - **burst-size-units** – Configure the burst-size-units. Options include: bytes, kbytes, mbytes, and gbytes. When NOABR is present in the acwhttppolicy Vendor-Specific Attribute (VSA) received from the PCRF, ABR video rate limiting and encrypted video rate limiting is bypassed. This mechanism is used to allow a subscriber to opt-out from video rate limiting.
- **flow-aggregation-state** – Enable or disable flow aggregation.

6 Enable/disable TCP Rate Limited KPI Savings. This feature enables the MCC to estimate optimization byte savings for rate-limited QUIC video traffic and report the savings in logs, statistics, and HTDRs.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile <name> savings-estimation
```

Options include:

- **sampling-interval** – Specify the sampling interval for savings estimation 1.. 43200(72).
- **sampling-interval-unit** – Specify the sampling-interval-unit: min/hr.

7 Configure the segment threshold in bytes for bypassing rate-limiting.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile <name> segment-threshold
```

8 Configure the steering destinations to use for this profile. The steering-group allows the UDP flow to be steered to the destination configured within itself, when the profile is selected. In packet processing, after UDP policies are evaluated, the associated traffic-control-profile is selected. The steering-group within the profile can have multiple steering-destinations. A round-robin approach is used between destinations. In case of a single steering-destination, the current UDP flow is steered to that destination.

If the destination is a host-name, DNS lookup is invoked and the flow is diverted to the resolved IP address. This action causes MCC to issue the Profile - Steered to Host statistic or Profile - Steered to IP statistic.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile <name> steering-group
```

9 Set toll-free-data to enabled (or disabled) to indicate whether toll free data would apply for this profile. See Configuring an Encryption Profile (page 2-54).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile <name> toll-free-data enabled
```

10 Configure the maximum data to store when processing upstream data, units KB.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
traffic-control-profile <name> upstream-buffering-limit
```

Configuring Multi-protocol Proxy UDP

This section describes how to configure UDP proxying parameters.

Access	config
Limitations	None

This feature enables Operators to selectively apply rate limiting to Quick UDP Internet Connection (QUIC) protocol flows. This feature complies with Google's Quic Version Q039. Using this feature, UDP traffic with a configured destination network port is sent to the UDP proxy.

This feature supports the following:

- Quic network domain detection
- Quic service rules
- Transparent and non-transparent modes
- Quic video rate limiting
- Quic statistics and KPIs
- Quic flow data records

Affirmed Networks supports Google QUIC versions Q039, Q043, Q044, and Q046.

Use the following steps to configure UDP parameters:

1 Configure UDP Splicing idle timeout in seconds.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp
idle-timeout
```

2 Set quic-rate-limiting to enabled (or disabled).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp
quic-rate-limiting enabled
```

3 Configure QUIC reset timeout in seconds.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp
quic-reset-timeout
```

Configuring a Multi-protocol Proxy UDP Policy

This section describes how to configure the udp-policy and the name of a service-rule associated with this policy.

Access	config
Limitations	None

Use the following steps to configure a UDP policy:

1. Configure a name for the UDP policy (quic policy name).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp-policy
<name>
```

2. Configure the congestion control type: head-drop/tail-drop. Options include: head-drop and tail_drop.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp-policy
<name> congestion-control head-drop
```

3. Enter a unique priority for the quic-policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp-policy
<name> priority 5
```

4. Enter the traffic control profile associated with this policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp-policy
<name> traffic-control-profile <name>
```

5. Enter the name of a service rule associated with this policy.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy udp-policy
<name> service-rule <name>
```

Configuring Multi-protocol Proxy Video Pacing

This section describes how to configure video pacing.

Access	config
Limitations	None

Set the admin status to enabled.

```
an3000-mcm-slot8cpu1(config)# services multi-protocol-proxy
video-pacing admin-state enabled
```

Configuring Multi-protocol Proxy Web Image Optimization

This section describes how to configure web image optimization. This feature implements a Web optimization technique in which image compression and transcoding produce a large gain in bandwidth savings for the Operator. The web-image-optimization must be associated with an http-request-policy.

Access	config
Limitations	None

This feature enables Operators to convert gif files to jpeg and png files and png files to jpeg files. This feature also enables Operators to set image conversion parameters such as metadata, quality, percentage, and size. For example, configure the predicted size savings percentage threshold that can be optimized between 0-100 (default 33). For example, configure the target JPEG quality between 0 (coarsest) and 100 (finest quantization). The default is 75. See Configuring a Multi-protocol Proxy Server Connections (page 2-24) for additional details.

The following image files account for 99% of HTML image requests:

- *PNG and GIF* – The file format is lossless and reduction in file size is accomplished by rescaling the image prior to re-encoding.
- *JPEG* – The file format is lossy. In addition to rescaling, reduction in file size is accomplished by adjusting the encoding quality factor setting. Subjective quality impact is most likely to occur from using a quality setting that is too aggressive.

Use the following steps to enable web image optimization and configure web image optimization profiles.

1. Set the admin status to enabled.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization admin-state enabled
```

2. Enter a name for the web image optimization profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name>
```

3. Allow conversion of GIF to JPEG.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> convert-gif-jpeg
```

4. Allow conversion of GIF to PNG.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> convert-gif-to-png
```

5 Allow conversion of PNG to JPEG.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> convert-png-to-jpeg
```

6 Allow interlace image.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> interface-image
```

7 Allow requantization of palettized image.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> quantize-image
```

8 Allow removal of image metadata.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> remove-metadata
```

9 Allow resizing of image by reducing the width and height attributes.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> resize-image
```

10 Allow 4:2:0 chroma subsampling of JPEG.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> sample-jpeg-to-420
```

11 Configure the predicted size savings percentage threshold that can be optimized.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> size-saving-threshold
```

12 Configure source maximum size (in kbytes) that can be optimized.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> source-max-size
```

13 Configure source minimum size (in kbytes) that can be optimized.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> source-min-size
```

14 Configure target JPEG quality: 0 - coarsest, 100 - finest quantization.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> target-jpeg-quality
```

15 Configure target maximum size (in kbytes).

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
web-image-optimization profile <name> target-max-size
```

Configuring an Encryption Profile

Configure encryption profile.The encryption-profile is used in HTTP Header Enrichment (HHE) to determine the algorithm and keys to be used for header encryption.

Access	config
Limitations	None

The Advanced Encryption Standard (AES) profile under service-construct stores encryption algorithm information and keys depending on the algorithm configured by the Operator. AES allows the Operator to configure a KEY in the CLI. MCC generates a random IV and includes this as an extra header in each encrypted header. MCC supports a key in hex format (without spaces).

MCC supports the AES-128/192/256-CBC encryption algorithm and base64 and hex encoding.

For each profile, enter:

- key – Key for AES encryption. Input HEX format without space
- IV – Static Initialization Vector for AES encryption. Input a unique IV in HEX format without space equal 16 bytes of length.

Use the following steps to configure the encryption profile:

1 Set the admin state to enabled (or disabled) for this encryption profile.

```
an3000-mcm-slot7cpu1(config)# service-construct encryption-profile
<name> admin-state enabled
```

2 Configure the encryption profile as follows:

a Encrypted by Advanced Encryption Standard 128-bit Cipher Block Chaining.

```
an3000-mcm-slot7cpu1(config)# service-construct encryption-profile
<name> aes-128-cbc
```

b Encrypted by Advanced Encryption Standard 192-bit Cipher Block Chaining.

```
an3000-mcm-slot7cpu1(config)# service-construct encryption-profile
<name> aes-192-cbc
```

c Encrypted by Advanced Encryption Standard 256-bit Cipher Block Chaining.

```
an3000-mcm-slot7cpu1(config)# service-construct encryption-profile
<name> aes-256-cbc
```

d Encrypted by Rivest Cipher 4 Standard 128-bit Stream Cipher encryption.

```
an3000-mcm-slot7cpu1(config)# service-construct encryption-profile
<name> rc4md5-128
```

3 Select an encoding format for encrypted information. Options include: base64 and hex.

```
an3000-mcm-slot7cpu1(config)# service-construct encryption-profile
<name> encoding hex
```

Configuring HTTP Header Enrichment (HHE)

This section describes how to configure HTTP header enrichment.

Access	config
Limitations	None

The function of the HHEE algorithm is to encrypt or hash the header value in HTTP request headers. This ensures that sensitive user information (such as IMEI, IMSI, MSISDN, UE IP Address etc) is securely communicated to the public network.

The Operator provides a KEY from the CLI and the MCC encrypts the key and verifies that the receiver decrypts the header value properly. MCC supports HEX (default) and base64 encoding.

The http-header-enrichment object defines the HTTP headers/cookies that an http-policy adds, removes, or replaces in the HTTP request or response header. Multiple headers for the same value are allowed. These headers are used by origin servers and other nodes in the network to determine the treatment of the packet and for tracking purposes. Operators can specify the string of the cookie and the value (imsi, msisdn, ue-ip-addr or value) to use in the cookie. The http-header-enrichment object supports first-request, which applies the http-header-enrichment only for the first request on a client connection. This object is referred to by the service-rule object. The http-header-enrichment profile is applied to the profile in Configuring a Multi-protocol Proxy HTTPS Profile (page 2-20).

Long Term Mobile ID (LTMI) is a variant of Anonymous Customer Reference. It supports the insertion of an encrypted subscriber ID in the http-request header. LTMI enables the subscriber ID to be communicated to the origin server in a secure manner.

HTTP proxy supports embedding variables in HTTP headers. MCC HTTP-Proxy header enrichment functionality supports the following variables:

- IMSI
- MSISDN
- HTTPVERSION

The feature enables support for the following use cases:

- Ut (VoLTE) interface support for adding *X-3GPP-Asserted-Identity* header with embedded *MSISDN* variable. The Ut interface is defined in *3GPP TS 24.623*.
- Adding and updating *Via* header with embedded *HTTPVERSION* variable to HTTP requests and responses. The *Via* header is defined in *IETF RFC7230*.

The http-header-enrichment profile supports the addition of a string macro to include variables for add-header header-string value and rewrite-header header-string append-value. MCC uses "`$()`" in the string macro to indicate a variable. Any "`$`" or "`()`" that does not conform to this format is considered a normal character in the string instead of a variable. The string-macro may contain as many variables as needed. The string length limitation is 255.

This feature also enables Operators to add or rewrite a *Via* header. If a *Via* header already exists and rewrite is configured, MCC rewrites the header with the original information and appends the configuration with "`,`".

```
an3000-mcm-slot7cpu1(config)# service-construct http-header-enrichment
HHE request add-header header-string X-IMEI value $(IMEI)
Possible completions:
encrypted   Include encrypted custom value in the header by the
algorithm configured in encryption profile
md5         Include MD5 hashed custom value in the header
<cr>
an3000-mcm-slot7cpu1(config)# service-construct http-header-enrichment
hhee request add-header header-string X-AES-MSISDN msisdn
Possible completions:
dynamic-encryption   Include Anonymous Customer Reference(ACR) dynamic
encrypted MSISDN in the header
encrypted            Include encrypted MSISDN in the header by the
algorithm configured in encryption profile
md5                  Include MD5 encrypted MSISDN in the header
static-encryption    Include Anonymous Customer Reference(ACR) static
encrypted MSISDN in the header
  <cr>
```

Use the following steps to configure HTTP header enrichment:

1 Enter a name for the encryption profile.

```
an3000-mcm-slot7cpu1(config)# service-construct http-header-enrichment
<name>
```

2 Configure an encryption profile used to encrypt request header value.

```
an3000-mcm-slot7cpu1(config)# service-construct http-header-enrichment
<name> encryption-profile <name>
```

3 Apply HTTP header enrichment add-header and add-cookie to the first request on a client connection.

```
an3000-mcm-slot7cpu1(config)# service-construct http-header-enrichment
<name> encryption-profile <name> first-request
```

4 Configure HTTP header enrichment support for request packets. .

```
an3000-mcm-slot7cpu1(config)# service-construct http-header-enrichment
name first-request request
```

Options include:

- **`add-cookie`** – Add a new cookie to request packets and specify the string of the cookie:
  - User-Identity-Forward-IMSI – Add subscriber IMSI to cookie.
  - User-Identity-Forward-MSISDN – Add subscriber MSISDN to cookie.
  - cookie-string – Add a custom cookie string
- **`add-header`** – Add a new header field to request/response packets. MCC adds a new header only if the same header does not already exist. Add a new header name/string:
  - X-Forwarded-For – Add X-Forwarded-For header name
  - X-Forwarded-Proto – Add X-Forwarded-Proto header name
  - X-IMSI – Add X-IMSI header name
  - X-MSISDN – Add X-MSISDN header name
  - header-string – Add a custom header name. See Header String Values (page 2-58).
- **`remove-header`** – Remove a header field from request packets.
- **`rewrite-header`** – Rewrite a header field or value from request packets.

5 Configure HTTP header enrichment support for response packets.

```
an3000-mcm-slot7cpu1(config)# service-construct http-header-enrichment
name first-request response
```

Options include:

- **`add-header`** – Add a new header field to request/response packets. MCC adds a new header only if the same header does not already exist. Add a new header name/string:
  - X-Forwarded-For – Add X-Forwarded-For header name.
  - X-Forwarded-Proto – Add X-Forwarded-Proto header name.
  - X-IMSI – Add X-IMSI header name.
  - X-MSISDN – Add X-MSISDN header name.
  - header-string – Add a custom header name. See Header String Values (page 2-58).
- **`remove-header`** – Remove a header field from request/response packets.
  - **`rewrite-header`** – Rewrite a header field or value from request/response packets. The http-header-enrichment request/response feature supports the following objects in the rewrite-header header-string:
  - **`append-value`** – Append a value to an existing header. string (any character except Ctrls and seperators, may include variable such as IMSI, MSISDN, HTTPVERSION, example: $(MSISDN)).
  - **`replace-header`** – Replace header string with a new string (HTTP header field name string).
  - **`replace-value`** – Replace header value with a new value (HTTP token - any character except Ctrls and seperators).
- **`rewrite-header`** – Rewrite a header field or value from response packets.

Header String Values

MCC supports the following header-string values:

- **`charging-characteristic`** – Include charging-characteristic in the header.
- **`encrypted`** – Include encrypted custom value in the header by the algorithm configured in encryption profile
- **`http-protocol`** – Include http-request protocol (http/https) in the header.
- **`imei`** – Include IMEI in the header.
- **`imsi`** – Include IMSI in the header.
- **`mapped-apn-name`** – Include mapped APN name in the header.

- **md5** – Include MD5 hashed custom value in the header
- **msisdn** – The system provides the ability to encrypt the Mobile Subscriber Integrated Services Digital Network-Number (MSISDN). See header-string msisdn (page 2-60) for the supported encryption types.
- **original-apn-name** – Include original APN name in the header.
- **time-zone** – Includes UE time zone information in HTTP request header. This feature adds HTTP header enrichment options for configuring the UE time zone and daylight savings. The header value follows the 3GPP standard and uses a 4-digit hex string. MCC updates the MCD table with the UE time zone information when the session is first created or anytime the UE time zone changes.
- **original-url** – Include original URL in the header. MCC inserts the original URL in the request prior to sending the request to the server.
- **rat-type** – Include Radio Access Type (2G, 3G or 4G) in the header.
- **reseller-id** – The MCC supports the following three text string VSAs used to generate LTMI and the reseller ID header: subID, miDial, and resellerID. The VSAs are retrieved from the PCRF when a call is established and delivered to the http-proxy when an http-request is processed.
- **serving-gw-ip-addr** – Include Serving Gateway IPv4 Address in the header.
- **serving-gw-ipv6-addr** – Include Serving Gateway IPv6 Address in the header.
- **serving-network-plmnid** – Include Serving Network PLMNID in the header.
- **subscriber-id** – Insert the encoded and encrypted subscriber ID (LTMI) into the http-header.
  - **ltmi-dynamic-trusted** – Include Dynamic Trusted ID in the header
  - **ltmi-external** – Include External ID in the header
  - **ltmi-static-trusted** – Include Static Trusted ID in the header
  - **plain-text** – Include plain text ID in the header
- **ue-ip-addr** – Include UE IP Address in the header.
- **ue-ip-addr-port** – Include UE IP Address and Port number in the header.
- **ue-ipv4-mapped-ipv6-addr** – Include original UE IP address in the HTTP header; IPv4 address is mapped into IPv6 address following the specified IPv6 address prefix.
- **ue-port** – Include UE Port number in the header.
- **user-location-info** – Include User Location Information in the header.

- **`value`** – Value to include for the header. May include variable such as IMSI, MSISDN, HTTPVERSION, UEIPADDR, ORIGINALAPN, IMEI, ULICGI, RATTYPE, GWIPADDR, PLMNID, CHARGINGCHAR, TIMESTAMP, SUBSCRIBERID, VMNAME, example: $(MSISDN):
  - **`encrypted`** – Include encrypted custom value in the header by the algorithm configured in encryption profile
  - **`ltmi-dynamic-trusted`** –Encrypt data using Dynamic Trusted LTMI algorithm in the header
  - **`ltmi-external`** – Encrypt data using External LTMI algorithm in the header
  - **`ltmi-static-trusted`** – Encrypt data using Static Trusted LTMI algorithm in the header
  - **`md5`** – Include MD5 hashed custom value in the header

header-string msisdn

The system provides the ability to encrypt the Mobile Subscriber Integrated Services Digital Network-Number (MSISDN). The MSISDN number uniquely identifies a subscription in a GSM or UMTS mobile network. MCC supports the following pre-defined format value: *mobile country code + mobile network code + ACR type* and the *date* in ISO8601 format. For example:

mobile country code

ACR type and date in ISO8601 format

mobile network code

234015STATRSV2017-09-11T14:55:31ZVYtFxFlYEupSrp3108iemQx9NpXj8q+
3YCHxbbOokcARlwzzkej2XFrUJNASo6aT/DMGH5dPwNBSRpUdg5TJV5Erb
DVS9ojLiPq7Ztm/HhrI9hELlW2dHiLrN0e5KgWsub6jRTPLt+p/7CPWE9zSSsI9
fQzsJgDYXVfdMX8Cqp+12/a/CLnlmXkWjmIOu4EP0o2nkwamtFYqHl3

encrypted MSISDN

MCC supports the following encryption types:

- **`static-encryption`** – For certain URLs, static encryption is done on the MSISDN. In order to make the encryption result deterministic for the static encryption, the encryption must have “NO PADDING”. This ensures that the same sequence of bytes is produced each time. Use this setting for trusted partners.

- **`static-acr`** – Supports a predefined format value + encrypted value of MSISDN.

- **dynamic-encryption** (default) – For non-partner websites, dynamic encryption occurs on the “timestamp of PDP context creation + MSISDN”. Dynamic encryption is created for one pdp-context. MCC used dynamic encryption for the duration of this pdp-context. After disconnection, MCC creates a new pdp-context and new encryption with a different value.
- **dynamic-acr** – Supports a predefined format value + encrypted value of timestamp and MSISDN.

Configuring a Static DNS Map

Use the instructions in this section to configure a static-dns-map for the http-proxy service. MCC supports up to 128 static-dns-maps and each static-dns-map supports up to 1024 host names in a fully qualified format. Each host supports up to 5 ipv4 and 5 ipv6 IP addresses.

Access	config
Limitations	None

Use the following steps to configure a static DNS map:

1. Create a name for the static DNS map.

```
an3000-mcm-slot7cpu1(config)# service-construct static-dns-map <name>
```

2. Configure a list of hosts and possible IP address. For example: foo.bar.com

```
an3000-mcm-slot7cpu1(config)# service-construct static-dns-map <name>
host
```

3. Associate this static map with a network context

```
an3000-mcm-slot7cpu1(config)# service-construct static-dns-map <name>
network-context <name>
```

4. Specify how the returned host address list will be constructed. Options include: static and round-robin.

```
an3000-mcm-slot7cpu1(config)# service-construct static-dns-map <name>
returned-addr-order static
```

Loading Unoptimized Content Using Shortcut Keys

Users are able to load unoptimized content using shortcut keys from Web browsers. The shortcut keys trigger a no-cache HTTP request header to disable the following HTTP optimization features:

- dns-preemption
- minification
- web-image-optimization

Table 2-3 Shortcut Keys for Web Browsers

Shortcut Key	Function	Supported Browser
F5	Refreshes a page with If-Modified-Since header, but no no-cache header	All browsers
<Ctrl>F5	Refreshes a page with one or both of the HTTP headers to ignore cached response: ■ Cache-Control: no-cache ■ Pragma: no-cache	■ Internet Explorer10 ■ Chrome ■ Firefox
<Shift>F5	Refreshes a page with one or both of the HTTP headers to ignore cached response: ■ Cache-Control: no-cache ■ Pragma: no-cache	Chrome
<Shift>Reload	Refreshes a page with one or both of the HTTP headers to ignore cached response: Cache-Control: no-cache Pragma: no-cache	Safari

Multi-protocol Proxy Show Commands

HTTP proxy supports the following show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*.

```
an3000-mcm-slot7cpu1# show services multi-protocol-proxy
Possible completions:
ltmi-key            ltmi-key stats
service-data-flow   Each service-data-flow that is evaluated on each
multi-protocol-proxy during flow classification, in order of
evaluation
statistics Statistics
|                   Output modifiers
<cr>
```

To view current and past congestion levels:
show subscriber proxy-session detail

To view a summary of past congestion levels:
show subscriber proxy-session summary

To view the current congestion level of the session:

```
an3000-mcm-slot7cpu1# show subscriber proxy-session detail 163847
Tcp Measurement - Current Congestion Level           Moderate
```

To view the last 10 transitions of congestion levels with their corresponding timestamps:

```
an3000-mcm-slot7cpu1# show subscriber proxy-session detail 163847
Tcp Measurement - Congestion Level History           "Fri Aug 18
14:59:11 2017:  Severe-->    Fri Aug 18 15:05:20: Moderate"
```

This show command displays the congestion transition from Severe to Moderate with the associated timestamp. In this example, session ID 163847 is in Severe congestion at the `14:59:11 2017` timestamp. However, the network situation improves and the session moves into the Moderate congestion level at the `15:05:20` timestamp. MCC supports a maximum of 10 simultaneous entries. For subsequent entries, MCC replaces the first set of entries with new entries.

Chapter 3

TCP Optimization

This chapter describes an overview of TCP optimization and how to configure a TCP optimization profile.

The tcp-measurement feature reports the TCP parameters and provides a status of the downstream GN radio link by calculating the downstream client throughput to determine whether it may be in a congested state. In this case, downstream client throughput refers to the available theoretical bandwidth on the downstream air link.

In addition, this feature supports TCP performance statistics for every connection made to the proxy for all user sessions. See *Acuitas Performance Statistics*.

Note: *The retrieval of TCP parameters and the collection of data records and HTDRs may impact performance. MCC uses a rolling average and this may result in a slight deviation from the actual average value. However, this method saves memory space and computation to calculate and store the exact values every second. As a result, this method reduces performance implications.*

Note: *TCP measurements may be inaccurate for very low latencies.*

MCC reports the average, maximum, and minimum throughput in the last 1 minute of each downlink connection for each subscriber.

The congestion detection algorithm is based on observing the TCP RTT, MSS, CWND, and downstream throughput. The **tcp-measurement** object is disabled by default. If the Operator enables **tcp-measurement**, the MCC calculates the TCP performance parameters for the sampling frequency set in the CLI for both uplink and downlink connections as follows:

MCC measures the following parameters for congestions measurement and reporting:

- RTT: round trip time of a connection (ms)
- Downstream MSS: Maximum segment size of the Client to Proxy connection (client connection) in the downstream direction (bytes)
- Downstream CWND: The congestion window of the client connection in the downstream direction
- Downstream throughput: The throughput of the client connection in the downstream direction (Kbps)

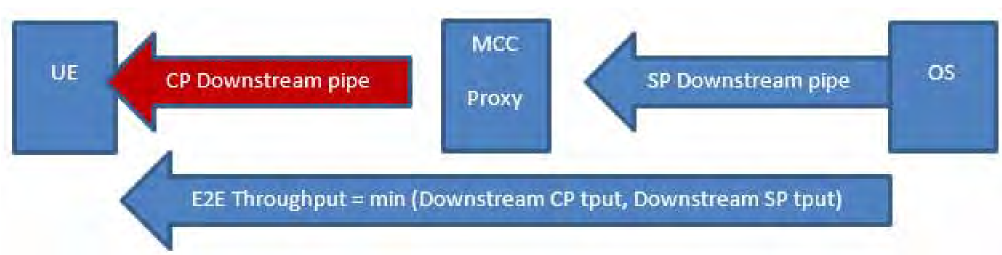
  In this case, downstream client connection throughput is indicated as follows:

In a typical network deployment, UE devices connect to the MCC Proxy from the Gn or Gi-Gateway access interface. These connections traverse through high latency, high packet loss radio networks. The MCC proxy subsequently uses a Gi, or Internet side connection to communicate with Origin Servers.

In relation to tcp-measurement statistics, MCC measures the characteristics of the Gn/access-side connection and not the Gi/Internet-side connection. The objective is to estimate the available downstream throughput of client connections between the MCC proxy and the UE.

The following figure shows the different aspects of throughput measurements. In this case, *CP* refers to *Client-Proxy* and is indicated by the two endpoints involved in the measurement. *E2E* refers to *end–to–end*, indicating that the connection is viewed as a complete connection from server to client (that is, the user equipment). *SP* refers to *Server-Proxy.*

Figure 3-1 Congestion/Throughput Measurement



The MCC proxy tcp-measurement feature reports information in the CP downstream pipe (shown in the red arrow in Figure 3-1). Client downstream throughput indicates the size of the pipe in the downstream direction between the MCC proxy and UE. This is *not* always equivalent to the end-to-end throughput (which may be limited by other factors such as server side connection speed, proxy latency, server speed, etc.). The objective is to determine and report to the Operator, the state of congestion in the radio network. This measurement does not take into account the bottleneck caused by external factors to the client-side connection such as proxy and server limitations.

TCP Measurement Examples

The example in Figure 3-1 on page 3-2 describes how this feature is utilized. Considering the following two examples, the decision to use TCP measurement on the Gn side alone accurately reflects the condition of the radio network.

Origin Server Transmission Rate

Typically, origin servers send data at a very fast rate, therefore, the server is not a bottleneck in this calculations. For example:

- The server sends data at 5 Mbps (the SP downstream pipe in Figure 3-1 on page 3-2)
- Client can handle 10 Mbps of data (the CP downstream pipe in Figure 3-1 on page 3-2)

In this case, the end-to-end throughput is calculated as 5 Mbps because it is always the minimum (server pipe, client pipe). This does not necessarily indicate that the client is congested. Therefore, if end-to-end throughput is used for the measurement, it incorrectly reports that there is a problem on the client-side network.

Rate Limiting and Video Pacing

This section describes an example connection with rate limiting and video pacing features enabled in the MCC proxy. For example:

- The server sends data at 100 Mbps (the SP downstream pipe in Figure 3-1 on page 3-2)
- Client can handle 10 Mbps of data (the CP downstream pipe in Figure 3-1 on page 3-2)

In this case, it is expected that the end-to-end throughput reports the same as what the client downstream pipe can handle, as the server is sending data at 100 Mbps. However, if video data is flowing through a connection with rate limiting or video pacing applied, the MCC proxy applies server-side pacing, which throttles the rate at which the MCC proxy reads from the server. In addition, the MCC proxy also paces the rate at which data is sent to the client. In this case, the end-to-end throughput incorrectly reflects that the network is congested.

Considering these two factors, the decision to use the measurement on the Gn side alone most accurately reflects the condition of the radio network.

Using this method, the Operator can detect a potential problem with the network and detect congestion based on the trend of the available client downstream throughput. If there is a consistent degradation over time, it is assumed that the network is in some form of congestion.

Configuring Global TCP Parameters

This section describes how to configure global TCP parameters.

Access	config
Limitations	None
Guidelines	The following ip-protocols tcp configuration options are not supported if user-space-networking (see page 2-3) is enabled: *rmem-min, wmem-min, frto, tcp-early-retrans, tcp-mem-high-threshold, tcp-mem-low-threshold, and tcp-mem-pressure-threshold.*

Use the following steps to configure global TCP parameters:

1. Configure F-RTO options. Set to disable-frto or enable-frto. Enable forward retransmission timeout recovery to detect and avoid unnecessary TCP data retransmissions.

   ```
   an3000-mcm-slot7cpu1(config)# ip-protocols tcp frto enable-frto
   ```

2. Configure the auto-tuning send/receive buffer ranges. Buffer sizes start with 1MB and increases as needed, up to 32 MB. The range is 4096 to 33554432 bytes. TCP dynamically adjusts the size of the send/receive buffer within the configured range, depending on available memory.

   a. **`auto-tuning-recv-buffer-size`** – Configure the auto-tuning range for TCP receive buffer size. Set the minimum, default, and maximum buffer size in bytes for each socket. Options include:

      – **`rmem-default`** – Configure the default size of the receive buffer for each socket (Bytes) 4096 .. 33554432 (1048576)

      – **`rmem-max`** – Configure the maximum size of the receive buffer for each socket (Bytes) 4096 .. 33554432 (33554432)

      – **`rmem-min`** – Configure the minimum size of the receive buffer for each socket (Bytes) 4096 .. 33554432 (65536)

   b. **`auto-tuning-send-buffer-size`** – Configure the auto-tuning range for TCP send buffer size. Set the minimum, default, and maximum buffer size in bytes for each socket. Options include:

      – **`wmem-default`** – Configure the default size of the send buffer for each socket (Bytes)4 096 .. 33554432(1048576)

      – **`wmem-max`** – Configure the maximum size of the send buffer for each socket (Bytes) 4096 .. 33554432(33554432)

      – **`wmem-min`** – Configure the minimum size of the send buffer for each socket (Bytes)4 096 .. 33554432(65536)

3 Set the TCP max-syn-backlog threshold high enough to avoid TCP SYN Flood attack prevention (TCP SYN Cookies) under a normal system load. Options include: 32768. 16384 .. 1048576 (32768)

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp frto enable-frto
max-syn-backlog 32768
```

Use **`show ip-protocols tcp statistics cluster-level`** to monitor if SYN Cookies are sent.

4 Enable TCP early retransmission and tail-loss-probe functionality.

5 Configure the tcp early retransmit and tail loss probes options. Options include: disable-er, enable-er, delayed-er, delayed-er-and-tlp, and enable-tlp-only.

```
an3000-mcm-slot7cpu1(config)#ip-protocols tcp tcp-early-retrans
delayed-er-and-tlp
```

6 Configure TCP ECN options: disable-ecn, enable-with-request-ecn-on-outgoing, and enable-without-request-ecn-on-outgoing. Disable TCP ECN to avoid compatibility issues with servers which do not support ECN

```
an3000-mcm-slot7cpu1(config)#ip-protocols tcp tcp-ecn disabled
```

7 Do not set memory thresholds unless TCP pressure threshold is exceeded during normal system operation.

Monitor the **TCP Memory - Peak In Use** statistic using the **`show ip-protocols tcp statistics node-level`** command.

```
an3000-mcm-slot7cpu1(config)#ip-protocols tcp tcp-mem-
```

a **`tcp-mem-high-threshold`** – Configure the high threshold of tcp stack memory (KB).

b **`tcp-mem-low-threshold`** – Configure the low threshold of tcp stack memory (KB).

c **`tcp-mem-pressure-threshold`** – Configure the pressure threshold of tcp stack memory (KB).

Managing the Kernel TCP configuration

This section describes how to manage the kernel TCP configuration. MCC supports the following TCP optimization enhancement options:

- TCP FRTO (FRTO and FRTO response)
- TCP ECN
- TCP memory threshold (low, pressure, high)

Use the following steps to manage the kernel TCP configuration:

1 Configure IP protocols TCP.

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp
```

2 Configure the auto tuning range for the TCP receive buffer size.

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp
auto-tuning-recv-buffer-size
```

Options include:

- `rmem-default` – Configure default size of the receive buffer for each socket(Bytes)4096 .. 33554432(1048576)
- `rmem-max` – Configure maximum size of the receive buffer for each socket(Bytes) 4096 .. 33554432(33554432)
- `rmem-min` – Configure minimum size of the receive buffer for each socket(Bytes) 4096 .. 33554432(65536)

3 Configure the auto tuning range for the TCP send buffer size.

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp
auto-tuning-send-buffer-size
```

Options include:

- `wmem-default` – Configure default size of the send buffer for each socket(Bytes) 4096 .. 33554432(1048576)
- `wmem-max` – Configure maximum size of the send buffer for each socket(Bytes) 4096 .. 33554432(33554432)
- `wmem-min` – Configure minimum size of the send buffer for each socket(Bytes) 4096 .. 33554432(65536)

4 Configure the TCP F-RTO options. F-RTO is an enhanced recovery algorithm for TCP retransmission timeouts (RTOs). It is used in wireless environments where packet loss is typically due to random radio interference rather than intermediate router congestion. **F-RTO may negatively impact with the packet counting of the SACK-enabled TCP flow.**

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp frto
```

Options include:

- **`disable-frto`** – Disable this functionality.
- **`basic-frto`** – Enable the basic version F-RTO algorithm.
- **`sack-enhanced-frto`** – Enable SACK-enhanced F-RTO if flow uses SACK. The basic version can also be used when SACK is in use.

5 Configure TCP ipv4 max syn backlog.

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp max-syn-backlog
```

6 Configure the tcp Early Retransmit and Tail Loss Probes options: delayed-er, delayed-er-and-tlp, disable-er, enable-er, enable-tlp-only.

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp tcp-early-retrans
```

7 Configure TCP ECN options. Enable RFC 2884 Explicit Congestion Notification. When enabled, connectivity to some destinations may be affected due to older, misbehaving routers along the path, causing connections to be dropped.

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp tcp-ecn
```

Options include:

- **`disable-ecn`** – Neither initiate nor accept ECN.
- **`enable-with-request-ecn-outgoing`** – ECN when requested by incoming connections and also request ECN on outgoing connection attempts.
- **`enable-without-request-ecn-on-outgoing`** – ECN when requested by incoming connections, but do not request ECN on outgoing connections.

8 Configure the high threshold of tcp stack memory (KB). Configure TCP memory options (measured in units of the system page size) to track its memory usage. MCC calculates the defaults at boot time from the amount of available memory. This value is the maximum number of pages, globally, that TCP allocates and this value overrides any other limits imposed by the kernel.

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp tcp-mem-high-threshold
```

9 Configure the low threshold of tcp stack memory (KB). Configure TCP memory options (measured in units of the system page size) to track its memory usage. MCC calculates the defaults at boot time from the amount of available memory. TCP doesn't regulate its memory allocation when the number of pages it has allocated globally is below this number.

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp tcp-low-high-threshold
```

10 Configure the pressure threshold of tcp stack memory (KB). Configure TCP memory options (measured in units of the system page size) to track its memory usage. MCC calculates the defaults at boot time from the amount of available memory. When the amount of memory allocated by TCP exceeds this number of pages, TCP moderates its memory consumption. This memory pressure state is lifted after the number of allocated pages falls below the low mark.

```
an3000-mcm-slot7cpu1(config)# ip-protocols tcp
tcp-mem-pressure-threshold
```

Configuring the TCP Optimization Profile

This section describes how to configure the TCP optimization profile.

Access	config
Limitations	None
Guidelines	■ Configure separate TCP optimization profiles for access (client-side) and core (server-side) network connections. When multiple access networks are used (for example 2G, 3G, 4G, fixed-line), configure separate access profiles for each access network. ■ Configure a single TCP optimization profile for server-side connections. ■ When user-space-networking is set to enabled or disabled, certain CCAs are available. See Congestion Control Settings (page 3-13). See Configuring Multi-protocol Proxy Services (page 2-3).

The TCP optimization functionality supports enabling TCP optimization for TCP-based protocols such as HTTP, HTTPS, SSH, SCP, and SFTP. The TCP or HTTP proxy splits a TCP connection into a pair of UE-to-proxy (client) and proxy-to-server (server) TCP connections. The TCP proxy splices data between the pair of client and server connections. Splitting the TCP connection provides a significant reduction of latency (reducing round trip time (RTT)) resulting in an improved throughput and user download/upload data experience.

TCP optimization is supported for TCP-based protocols configured for interception by HTTP proxy and TCP proxy. These include HTTP and terminated or originated HTTPS connections. These also include spliced TCP connections (served by HTTP proxy TCP splicing feature) for protocols such as SSH, SCP, SFTP and pass-through HTTPS. Client connections, both terminated and spliced, are optimized using the configured client's tcp-optimization profile. Similarly, server connections are optimized using the configured server connections tcp-optimization-profile. The client and server tcp-optimization-profiles are configured on a per-HTTP proxy instance basis. In addition, terminated HTTP/HTTPS connections can be tcp-optimized using configured HTTP request policies.

TCP proxy supports a typical proxy mode where connections to the server originate from the MCC loopback. It also supports ip-transparent mode where server connections originate from the UE IP address.

The TCP optimization feature allows tuning of the TCP/IP stack parameters to improve the throughput and user experience in the mobile network. This feature enables Operators to configure TCP parameters such as congestion control algorithm (CCA), mss, window size, and send and receive buffer sizes. The changes can be independently applied to client and server TCP connections.

Multiple TCP optimization profiles can be configured and applied to different TCP/HTTP proxy service instances. Different proxy service instances (and thus TCP optimization profiles) can be selected by the Gateway Session Analyzer based on session attributes such as APN or RAT-Type.

To manage the TCP kernel configuration and configure options for TCP optimization tuning, see Managing the Kernel TCP configuration (page 3-6).

Congestion Control Algorithms

The following CCAs are supported for the tcp-optimization-profile for both client and server connections: bic, cdg, chd, cubic, dctcp, hd, highspeed, htcp, hybla, illinois, lp, newreno, reno, scalable, vegas, veno, westwood, yeah. Also see Congestion Control Settings (page 3-13) for guidelines on the user-space-networking setting.

Hamilton Delay

The Hamilton Delay (HD) congestion control algorithm is an implementation of the Hamilton Institute's delay-based congestion control which aims to keep network queuing delays below a particular threshold (queue_threshold).

HD probabilistically reduces the congestion window (cwnd) based on its estimate of the network queuing delay. The probability of reducing cwnd is zero or less at hd_qmin, rising to a maximum at queue_threshold, and then back to zero at the maximum queuing delay.

Loss-based congestion control algorithms such as NewReno, probe for network capacity by filling queues until there is a packet loss. HD competes with loss-based congestion control algorithms by allowing its probability of reducing cwnd to drop from a maximum at queue_threshold to zero at the maximum queuing delay. This method works well when the bottleneck link is highly multiplexed.

CAIA Hamilton Delay

CAIA Hamilton Delay (CHD) enhances the HD algorithm implemented in cc_hd(4). It provides tolerance to non-congestion related packet loss and improvements to coexistence with traditional loss-based TCP flows, especially when the bottleneck link is lightly multiplexed.

Similar to HD, the CHD algorithm aims to keep network queuing delays below a particular threshold (queue_threshold) and decides to reduce the congestion window (cwnd) probabilistically based on its estimate of the network queuing delay.

CHD differs from HD in the following aspects:

- The probability of cwnd reduction due to congestion is calculated once per round-trip instead of each time an acknowledgement is received (as done by cc_hd(4)).
- Packet losses that occur while the queuing delay is less than queue_threshold do not cause cwnd to be reduced.
- CHD uses a shadow window to help regain lost transmission opportunities when competing with loss-based TCP flows.

CAIA-Delay Gradient

CAIA-DelayGradient (CDG) is a hybrid congestion control algorithm which reacts to both packet loss and inferred queuing delay. It attempts to operate as a delay-based algorithm where possible, but utilizes heuristics to detect loss-based TCP cross traffic. CDG is therefore incrementally deployable and suitable for use on shared networks.

During delay-based operation, CDG uses a delay-gradient based probabilistic back-off mechanism, and also tries to infer non-congestion related packet losses and avoid backing off when they occur. During loss-based operation, CDG essentially reverts to cc_newreno(4)-like behavior.

CDG switches to loss-based operation when it detects that a configurable number of consecutive delay-based back-offs have had no measurable effect.

It periodically attempts to return to delay-based operation, but switches back to loss-based operation as required.

Use the following steps to configure the TCP optimization profile:

1 Set the TCP congestion control algorithm depending on the type of access network.

Note: *If you attempt to configure an unsupported CCA, the system defaults to* ***cubic****.*

Note: *If user-space-networking is enabled (see* *Configuring Multi-protocol Proxy Services on page 2-3**), see the guidelines in* *Congestion Control Settings (page 3-13)**.*

- For low to medium speed networks (2G, 3G) with occasional data loss (congestion), use **`reno`** or **`westwood`**; these algorithms are designed to avoid congestion.
- For higher speed networks (4G, fixed), use a more aggressive algorithm such as **`cubic`**; this setting finds the optimal send window sooner (cubic search).

- For higher speed wired server-side networks, use a more aggressive TCP CCA algorithm such as **`cubic`**; this setting finds an optimal send window sooner (cubic search).
- For networks experiencing increased network delay after TCP optimization is enabled, use **`yeah`**; this setting uses both packet loss and network delay to detect congestion.
- The CCA parameter must be tuned, depending on the conditions in the access network.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp congestion-control
```

2 The TCP/IP stack enables the Operator to configure the number of duplicate acknowledgment necessary to trigger fast-retransmit algorithm. A low duplicate ack threshold causes unnecessary extra retransmissions on congested networks. Affirmed Networks recommends setting this value to 3.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp fast-retransmit-dupacks 3
```

3 The TCP/IP stack allows the Operator to define the initial congestion window for each subscriber. This feature prevents the congestion control algorithm from activating prematurely for wireless links with higher latency.

- On the client side – Set a large initial send window to optimize small web page transfers (part of TCP slow start avoidance). Affirmed Networks recommends a value of 32.
- On the server side – Set a large initial send window (16-64) to optimize file uploads (TCP slow start avoidance).

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp initial-congestion-window 32
```

4 Configure initial TCP receive window in units of mss. Advertise a large initial window size to avoid TCP slow start for upstream data transfers.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp initial-receive-window
```

5 Enable (or disable) tcp server-side buffering. This feature intelligently manages the TCP buffers so that data flow from the Internet to the Radio network causes minimal increase in traffic on the Internet side. TCP-Proxy maintains a data throughput rate from the Internet to be approximately equal to the speed at which data is sent over the Radio network to the UE, to create balance and minimize traffic differential.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp intelligent-buffering
```

6 The TCP/IP stack allows the Operator to enable or disable the Nagle algorithm for each subscriber independently. Enabling this feature reduces the overhead in the wireless link and allows combining small packets at the source.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp nagle-algorithm disabled
```

7 MSS for a connection is negotiated at the time of establishing a TCP session. The TCP/IP stack allows the Operator to set different MSS values for different users based on workflow configuration.

- *On the client (access) side* – Configure MSS to the highest value to avoid fragmentation on the access network when access tunneling is in use. For example, use 1420 to accommodate GTP tunneling. This setting impacts upstream traffic only; for downstream traffic, configure mss-adjust on access network-context or access interfaces.
- *On the server side* – Configure MSS to match client-side value. This setting impacts downstream traffic only; for upstream traffic, configure mss-adjust on server-side network-context or server-side interfaces.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp receive-mss 1420
```

8 Enable or disable auto-tuning of TCP window. Enable TCP buffer auto-tuning to allow the TCP stack to dynamically tune buffer sizes based on the current connection speed. This feature is tunable for each subscriber TCP flow. Set to enabled to use auto-tuning. Set to disabled to manually configure the recv-buffer-size or send-buffer-size.

Set the buffer size on the access side to a large value (16MB-32MB) to ensure that enough data is buffered and available when the TCP send window opens, otherwise use auto-tuning.

With auto-tuning, TCP constantly adjusts buffer sizes based on the current data transfer rates and RTT. Affirmed Networks recommends using auto-tuning instead of manually setting the buffer size. If you are manually setting the buffer size, set the maximum buffer sizes lower than the access side (6-10MB); setting this value too high causes increased traffic volume on server-side connections.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp recv-buffer-auto-tuning enabled
```

9 The system allows the Operator to define send and receive buffers for each TCP connection for each subscriber. This option is configurable only if recv-buffer-auto-tuning is set to disabled. Affirmed networks recommends using 8388608.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp recv-buffer-size
```

10 Configure the TCP send buffer size in bytes. This option is configurable only if recv-buffer-auto-tuning is set to disabled. Affirmed networks recommends using 8388608.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp send-buffer-size
```

11 The TCP/IP stack for HTTP Proxy provides the capability to enable or disable TCP behavior defined in RFC2861. If this parameter is not configured for a subscriber, the congestion window times out after an idle period, preventing slow start algorithm for persistent connections.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp slow-start disabled
```

12 Configure TCP window size in bytes.

- On the client side – Affirmed Networks recommends not setting this value. Use auto-tuning instead.
- On the server side – Advertise a large initial window size to avoid TCP slow start for downstream data transfers.

```
an3000-mcm-slot7cpu1(config)# service-construct
tcp-optimization-profile <name> tcp window-size
```

Congestion Control Settings

The following congestion-control settings are available by platform based on the user-space-networking status (enabled/disabled). When the user attempts to configure an unsupported congestion-control setting, the system uses the cubic setting instead.

Congestion Control setting	ISM user-space-networking enabled	ASM/ISM user-space-networking disabled
bic	Unsupported/cubic	Unsupported/cubic
cdg	Supported	Unsupported/cubic
chd	Supported	Unsupported/cubic
cubic	Supported	Supported
dctcp	Supported	Unsupported/cubic
hd	Supported	Unsupported/cubic
highspeed	Unsupported/cubic	Supported
htcp	Supported	Supported
hybla	Unsupported/cubic	Supported
illinois	Unsupported/cubic	Supported
lp	Unsupported/cubic	Supported
newreno	Supported	Unsupported/cubic
reno	Supported	Supported
scalable	Unsupported/cubic	Supported
vegas	Supported	Supported

Congestion Control setting	ISM user-space-networking enabled	ASM/ISM user-space-networking disabled
veno	Unsupported/cubic	Supported
westwood	Unsupported/cubic	Supported
yeah	Unsupported/cubic	Supported

Configuring TCP Congestion-Triggered Optimization

This section describes how to configure TCP congestion-triggered optimization.

Note: *To use this feature, you must enable the admin-state for tcp-measurement.*

Use the following steps to configure TCP congestion-triggered optimization:

Configure the congestion-level threshold values in the instance.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<INSTANCE-NAME> tcp-measurement congestion-level
```

- `mild` – Mild congestion-level throughput in kbps. 10,000 kbps (10 Mbps)
- `moderate` – Moderate congestion-level throughput in kbps. 5,000 kbps (5 Mbps)
- `severe` – Severe congestion-level throughput in kbps. 2,000 kbps (2 Mbps)

1 Create a session-rule-group.

```
an3000-mcm-slot7cpu1(config)# service-construct session-rule-group
<SESSION-RG-NAME> rule <RULE-NAME> congestion-level mild
```

- Options include: mild, moderate, and severe.

2 Add the session-rule-group to the service-rule.

```
an3000-mcm-slot7cpu1(config)# service-construct service-rule <SR-NAME>
session-rule-group <SESSION-RG-NAME>
```

3 Associate the service rule to the http-request policy, http-response-policy, and tcp-splicing-policy as follows:

- HTTP-Request Policy:

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-request-policy <HTTP-REQ-POL-NAME> service-rule <SR-NAME>
```

- HTTP-Response Policy:

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-response-policy <HTTP-RESP-POL-NAME> service-rule <SR-NAME>
```

- TCP-Splicing Policy:

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
tcp-splicing-policy <TCP-SPLICING-POL-NAME> service-rule <SR-NAME>
```

4 Associate the http-request-policy to the ip-flow:

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-ip-flow <IP-FLOW-NAME> http-request-policy <HTTP-REQ-POL-NAME>
```

5 Associate the http-response-policy to the ip-flow:

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-ip-flow <IP-FLOW-NAME> http-response-policy <HTTP-RESP-POL-NAME>
```

6 Associate the tcp-splicing-policy to the http-ip-flow:

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-ip-flow <IP-FLOW-NAME> tcp-splicing-policy
<TCP-SPLICING-POL-NAME>
```

Note: *Congestion Triggered Rule is applied only for Offline Serving Policy.*

---

7 Use the following command to view the current and past congestion level.

`an3000-mcm-slot7cpu1#` **`show subscriber proxy-session detail`** command.

8 Use the following command to view the current congestion level of the session.

```
an3000-mcm-slot7cpu1# show subscriber proxy-session detail 163847
Tcp Measurement - Current Congestion Level          Moderate
```

9 Use the following command to view the last 10 transitions of congestion levels with their corresponding timestamps.

```
an3000-mcm-slot7cpu1# show subscriber proxy-session detail 163847
Tcp Measurement - Congestion Level History          "Fri Aug 18
14:59:11 2017:  Severe-->    Fri Aug 18 15:05:20: Moderate"
```

This show command displays the congestion transition from Severe to Moderate with the associated timestamp. In this example, session ID 163847 is in Severe congestion at the `14:59:11 2017` timestamp. However, the network situation improves and the session moves into the Moderate congestion level at the `15:05:20` timestamp. MCC supports a maximum of 10 simultaneous entries. For subsequent entries, MCC replaces the first set of entries with new entries.

TCP Congestion-triggered Optimization Use Case

TCP congestion-triggered optimization enables Operators to configure optimization policies for users experiencing congestion. Optimization is based on a subscriber's measured downstream throughput. Operators trigger optimization policies based on a subscriber's access network congestion level: mild, moderate, or severe. This feature enables Operators to apply differentiated optimization polices based on current network conditions.

Operators configure a service-rule with a session-rule-group and a congestion-level rule construct. Operators can apply these different congestion levels to various VAS services to create differentiated levels of services depending on the calculated congestion levels with their correspond policies; such as:

- Web Image Optimization
- HTTP Compression and Minification
- ABR and HTTPS Rate Limiting
- Video Pacing

For example, if an Operator sets the severe congestion level threshold to 500 kbps, all subscribers with an average 1-min downstream throughput less than this value are considered severely congested.

MCC calculates the saturated downlink throughput based on Maximum Segment Size (MSS), Congestion Window (CWND), and Round Trip Time (RTT).

The following use cases apply to this feature:

- Applying high optimization for severely congested users
- Applying medium optimization for moderately congested users
- Applying low optimizations for mildly congested users
- Applying no optimization for users with no congestion
- Removing optimization after user is out of congestion

Configuring Round-trip Time Offsets

This section describes how to configure a multi-protocol-proxy RTT location offset profile and parameters.

This feature adds a configuration option for Round-trip Time (RTT) offsets when RTT is used in the congestion detection's throughput formula. RTT offsets can be configured in data centers for both *nearby* and *distant* subscribers to avoid misidentifying *distant* subscribers as congested.

This feature reports the length of time that the client connection remains in each congestion level in the HTDR/EDR and also reevaluates policies for flows where the congestion levels change.

This feature adds the following new HTDR/EDR fields in tcpClientConnectionInfo:

- CongestionLevelNoneTime
- CongestionLevelMildTime
- CongestionLevelModerateTime
- CongestionLevelSevereTime

Use the following steps to configure RTT location offset:

1 Configure the RTT location offset profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
rtt-location-offset-profile RTT-PROFILE

Possible completions:

compromise-rtt-offset RTT location offset for subscribers without a
known TAC

location-offset RTT location offset for subscribers with a known TAC
```

2 Configure the location offset.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
rtt-location-offset-profile RTT-PROFILE location-offset RTT-LOCATION1

an3000-mcm-slot7cpu1(config-location-offset-RTT-LOCATION1)# ?

Possible completions:

rtt-location-offset   RTT location offset(ms)

tac-high-byte         high byte of TAC
```

3 Configure TCP measurement.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
USTCP tcp-measurement

Possible completions:

admin-state Enable/Disable TCP Measurement

congestion-level Congestion Level for applying Optimizations based on
Client Side TCP measurement

rtt-location-offset-profile   RTT Location Offset Profile for this
instance

sampling-interval Client Side TCP measurement Sampling Interval (1/ms)
```

IP Protocols TCP Show Commands

This section provides an overview of the tcp statistics and tcp-optimization show commands.

show ip-protocols tcp statistics

```
an3000-mcm-slot7cpu1# show ip-protocols tcp statistics
Possible completions:
  cluster-level   Cluster Level
  node-level      Node Level
  |               Output modifiers
  <cr>
```

HTDR/EDR

The following protobuf fields support congestion detection and record the following HTDR values:

- Minimum RTT (micro-seconds)
- Average RTT (micro-seconds)
- Maximum RTT (micro-seconds)
- Minimum Downstream MSS (bytes)
- Average Downstream MSS (bytes)
- Maximum Downstream MSS (bytes)
- Minimum Downstream CWIND
- Average Downstream CWIND
- Maximum Downstream CWIND
- Minimum Downstream Throughput (Kbps)
- Average Downstream Throughput (Kbps)
- Maximum Downstream Throughput (Kbps)

The following protobuf fields support encrypted connection congestion detection and record the following EDR values for the uplink:

- tcpClientConnectionInformation
- minRTT
- avgRTT
- maxRTT
- minDownstreamMSS
- avgDownstreamMSS
- maxDownstreamMSS
- minDownstreamCWND
- avgDownstreamCWND
- maxDownstreamCWND
- minDownstreamTPUT
- avgDownstreamTPUT
- maxDownstreamTPUT

Chapter 4

Configuring a DNS Proxy and Caching Service

This section describes the DNS proxy and caching service.

Access	config
Limitations	■ The DNS proxy builds the DNS reverse cache by looking only at DNS packets delivered over UDP. DNS over TCP is not supported. ■ The DNS proxy is passively observing the DNS responses triggered by UE DNS requests to build the DNS reverse cache. The DNS proxy does not have a mechanism to trigger DNS resolution of names of interest. There is no mechanism to trigger refresh of existing resolved names. ■ Every DNS reverse cache instance is local to the workflow instance.

Integrated within the MCC workflow, the DNS caching service is used to reduce the amount of DNS message traffic towards an Operator's DNS servers. When enabled, the DNS caching service caches DNS response traffic, allowing subsequent DNS lookup requests for the same domains to be served locally from the DNS cache. Separate caches are maintained per network-context, and the DNS caching service observes the time-to-live (TTL) contained in each DNS response to age out individual cache entries.

For any HTTP, HTTPS, and QUIC flows originating from the UE for IPv4 and IPv6, the TTL or hop limit value is decremented by one, respectively. This applies to transparent IP and non-transparent IP traffic traversing the proxy, and to traffic bypassing the proxy, regardless of whether the User Space IP stack or the Linux IP stack is used.

Additional cache-control configuration parameters allow the Operator to:

- Enable/disable the entire DNS cache
- Enable/disable the caching of DNS response data on a per-DNS server basis
- Override the TTL values that are used to age DNS cache entries

The DNS caching service also supports a configurable static host mapping table, allowing Operators to associate up to five IP addresses to a domain name. This configuration table is loaded into the DNS cache, and any request for a domain that is statically mapped is served directly by the MCC without any further exchange with an Operator's DNS servers. Each static mapping table can also be associated with a network-context, allowing different IP addresses to be returned, based on the network-context in which a UE resides.

Configuring a DNS Proxy Service

This section describes how to configure a DNS proxy service.

Access	config
Limitations	None

Use the following steps to configure a DNS proxy:

1. Enter the following command to configure a DNS proxy.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name>
```

2. Enable the admin-state.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name> admin-state
enable
```

3. Configure DNS IP flow. See Configuring a DNS IP Flow (page 4-4).
4. Configure DNS policy. See Configuring a DNS Policy (page 4-5).
5. Configure DNS reverse cache. See Configuring DNS Reverse Cache (page 4-9).
6. Configure the inactive blocking domain age time for DNS tunnel detection.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name>
dns-tunnel-age-time
```

7. Configure DNS proxy service instance.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name> instance
```

8. Enter the maximum number of cache entries for each DNS cache.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name>
max-cache-entries
```

9. Enter the TTL override in seconds for caching negative DNS responses.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name>
negative-response-ttl-override
```

10. Enter the time in seconds to retain pending DNS request cache entries.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name>
pending-request-timeout
```

11. Enter the TTL override in seconds for caching positive DNS responses.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name>
positive-response-ttl-override
```

12. Enter the TTL to return in any response for a static DNS entry. See Configuring a Static DNS Map (page 4-11).

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name>
static-map-entry-ttl
```

13 Enter the TTL to return in any response for a static DNS entry.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name>
static-map-entry-ttl
```

14 Configure DNS tunnel detection profile. See Configuring a Tunnel Detection Profile (page 4-11).

```
an3000-mcm-slot7cpu1(config)# services dns-proxy <name>
tunnel-detection-profile
```

Configuring a DNS IP Flow

This section describes how to configure a DNS IP flow.

Access	config
Limitations	None

Use the following steps to configure a DNS IP flow:

1 Enter a name for the DNS IP flow.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-ip-flow <name>
```

2 Enable the admin-state for the DNS IP flow.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-ip-flow <name>
admin-state enabled
```

3 Enter the DNS policies associated with this DNS IP flow.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-ip-flow <name>
dns-policy <name>
```

4 Enter a unique priority of the DNS IP flow.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-ip-flow <name>
priority
```

5 Enter the service rule name associated with this DNS IP flow.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-ip-flow <name>
service-rule <SR-name>
```

MCC supports bypassing the DPI module for the traffic by detecting the corresponding host name in the DNS query/response and blacklisting the host names.

To achieve bypassing of DPI based on blacklisting of domain-names:

a Configure a `services dns-proxy instance`. Configure the associated `dns-ip-flow service-rule` with:

- External activation through PCRF (`service-activation = external-activation`) (*Optional*)
- A `domain-rule-group` used by workflow to classify certain connections and take specific actions for certain domains

See the *Affirmed Networks Gateways Administration Guide* for details on the service-construct service-rule service-activation setting.

The DNS proxy-populated **dns-reverse-cache** matches an incoming IP address with the hostname. If a domain is blacklisted with a higher priority **service-rule**, the domain bypasses the DPI module. All other traffic goes through configured **service-rule** and is subjected to DPI (if enabled).

6. Enter the DNS tunnel detection profile associated with this DNS proxy.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-ip-flow <name>
tunnel-detection-profile <TDP-name>
```

Configuring a DNS Policy

This section describes how to configure a DNS policy. The dns-proxy supports the DNS whitelisting capability. This feature allows Operators to allow only DNS queries to trusted DNS servers. This feature prevents any DNS fraud where the user may have configured the IP address of a rogue DNS server.

Access	config
Limitations	None
Guidelines	See DNS Request Action Guidelines (page 4-7).

Use the following steps to configure a DNS policy:

1. Enter the following commands to configure a DNS policy.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-policy <name>
```

2. Configure the DNS request action for this policy. Operators can define the trusted DNS servers in the network and enforce the following actions: allow, block, redirect. The action dictates how the DNS server handles DNS requests. Configuration guidelines for each action are outlined in DNS Request Action Guidelines (page 4-7).

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-policy <name>
action allow
```

3. Configure the DNS response cache control action for this policy.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-policy <name>
cache-control cache-all
```

4. Set a unique priority for this dns-policy and dns-ip-flow. Associate the service-rule and dns-policy to this dns ip-flow.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-policy <name>
priority 100
```

5. Select a configured ip-list name.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-policy <name>
redirect-ip-list
```

6 Configure TTL override for DNS response packets. Use a low priority DNS policy to capture a ttl-override or other specific action in the policy. Set a high priority (wide open) dns-policy to allow normal DNS traffic flow through the DNS proxy. See TTL Override (page 4-8) for more information.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-policy <name>
ttl-override 0
service-rule SR-TTL-OVERRIDE (can configure multiple service-rule
here)
!
service-construct service-rule SR-TTL-OVERRIDE
domain-rule-group                          DRG-safe-search-TTL
!
service-construct domain-rule-group DRG-safe-search-TTL
 rule r1
  domain-name regex ".*\.google\..*"
 !
 rule r2
  domain-name regex ".*\.bing\..*"
 !
 rule r3
  domain-name regex "(.*\.|)youtube(|i\.googleapis|-nocookie)\..*"
```

7 Configure one service-rule.

```
services dns-proxy dns-policy <name>
service-rule <service-rule-name>
```

DNS Request Action Guidelines

Action	DNS Server Response	Configuration Guidelines
`allow`	Allow DNS queries to Operator-hosted DNS servers (designated DNS servers in the network) Allow DNS queries to other trusted DNS servers (for example Google DNS)	**1** Configure a set of service-construct packet-filter objects for all Operator-hosted and other trusted DNS servers. **2** Configure services dns-proxy dns-ip-flow and associate it with the packet-filter for the trusted DNS servers. **3** Optionally, associate the dns-ip-flow with a dns-policy with action set to **`allow`** (default). The DNS-proxy based on the configuration above matches the DNS query with one of the packet-filters and the dns-ip-flow. Since the dns-ip-flow has a dns-policy with action set to **`allow`**, the dns packet is forwarded unaltered to the trusted DNS server.
`block`	Block DNS queries to untrusted DNS servers	**1** Configure a catch-all service-construct packet-filter object for all other (untrusted) DNS servers. **2** Configure services dns-proxy dns-ip-flow with a lower priority and associate it with the packet-filter for the untrusted DNS servers. **3** Associate the dns-ip-flow with a dns-policy with action set to **`block`**.
`redirect`	Redirect DNS queries to untrusted DNS servers to a set of designated DNS servers	**1** Configure a catch-all service-construct packet-filter object for all other (untrusted) DNS servers. **2** Configure services dns-proxy dns-ip-flow” with a lower priority and associate it with the packet-filter for the untrusted DNS servers. **3** Associate the dns-ip-flow with a dns-policy with action set to **`redirect`**. **4** Set the dns-policy redirect-ip-list to refer to a service-construct ip-list containing a list of IP addresses of trusted DNS servers.

TTL Override

This object supports a service-rule based TTL override in the DNS policy. For each configuration update in the DNS policy, the DNS module analyzes all sessions and updates the service-rule index.

- Cache-control and action in the DNS policy is applied based on service-rule matching for upstream traffic.
- TTL override in the DNS policy is applied based on service-rule matching for downstream traffic.

If there is no dns-policy matching the upstream traffic, MCC drops the packet (current behavior).

HTTP proxy DNS resolver

If the subscriber is using the HTTP proxy as an explicit proxy to divert requests, the DNS lookup to the actual original server occurs in the HTTP proxy. If the request hostname matches the safe-search configuration, the HTTP proxy DNS resolver sends the DNS query to lookup the forced safe-search domain and establishes a server connection to that server.

CRN-based safe search

The service-rule option under **services content-filter instance safe-search** enables users to set that service-rule to **external-activated** which is triggered by the CRN. MCC applies the external-activated service-rule to the DNS proxy and HTTP proxy safe search.

Note: *DNS policy configuration changes are not applied to existing sessions.*

Configuring DNS Reverse Cache

This section describes how to configure DNS reverse cache.

Access	config
Limitations	None

DNS reverse cache is a database that provides the ability to lookup an IP address (IPv4 or IPv6) and return a list of host names that resolve to that IP address. This database is generated with information extracted from DNS responses by the DNS proxy. DNS reverse cache is cached from all DNS responses.

DNS reverse cache is used by a service-construct domain-rule-group. See the *Affirmed Networks Gateway Administration Guide*.

Use the following steps to configure DNS reverse cache:

1. Enter the following commands to configure DNS reverse cache.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-reverse-cache
```

2. Enable the admin-state.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-reverse-cache
admin-state enable
```

3. Set the cache full alarm threshold between 0-100. The default is 90. A value of 0 disables the threshold.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-reverse-cache
cache-full-alarm-threshold 90
```

4. Enable domain rule group filtering. The default is enabled.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-reverse-cache
domain-rule-group-filtering enabled
```

5. Set the maximum number of reverse cache entries for each DNS cache.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-reverse-cache
max-cache-entries 10000
```

6. Configure the TTL override. Options include add, multiply, and set.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy dns-reverse-cache
ttl-override add
```

Configuring Global Blocking Domain Clean Time

This section describes how to configure inactive blocking domain age time for DNS tunnel detection.

Access	config
Limitations	None

Use the following steps to configure domain clean time:

1 Enter the following command to configure domain clean time.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy domainCleantime
```

Options include:

- **dns-tunnel-age-time** – Inactive blocking domain age time for DNS tunnel detection
- **dns-tunnel-age-time-units** – dns-tunnel-age-time-units: seconds, minutes, hours, days

Configuring a DNS Proxy Instance

This section describes how to configure a DNS proxy instance.

Access	config
Limitations	None

Use the following steps to configure a DNS proxy instance:

1 Enter the following command to configure a DNS proxy instance.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy instance <name>
```

2 Enable the DNS proxy admin-state.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy instance <name>
admin-state enable
```

3 Associate a DNS IP flow with this proxy instance.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy instance <name>
ip-flows
```

Configuring a Static DNS Map

This section describes how to configure a static DNS map.

Access	config
Limitations	None

Enter the following command to configure a name for the static DNS map.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy static-dns-map <name>
```

Configuring a Tunnel Detection Profile

This section describes how to configure a DNS tunnel detection profile.

Access	config
Limitations	None

Use the following steps to configure a DNS tunnel detection profile:

1. Enter the following command to configure a DNS tunnel detection profile.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy
tunnel-detection-profile <name>
```

2. Enable the admin state.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy
tunnel-detection-profile <name> admin-state enable
```

3. Configure the time to turn off detail monitoring after the total hit is under threshold and there is no blocked domain, unit seconds.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy
tunnel-detection-profile <name> clean-time 80
```

4. Configure the number of recent domains to monitor.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy
tunnel-detection-profile <name> domain-cache-size 120
```

5. Configure the penalty time for blocked domain, unit seconds.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy
tunnel-detection-profile <name> domain-penalty-time 120
```

6. Configure the refresh rate for monitor interval, unit seconds.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy
tunnel-detection-profile <name> refresh-rate 10
```

7. Configure the threshold for hitting rate of one singe domain in detail monitoring mode.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy
tunnel-detection-profile <name> single-domain-threshold 30
```

8. Configure the threshold for overall DNS traffic rate to turn on detail monitoring mode.

```
an3000-mcm-slot7cpu1(config)# services dns-proxy
tunnel-detection-profile <name> totoal-dns-hit-threshold 250
```

DNS Proxy Show Commands

The dns-proxy supports the following show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*.

```
an3000-mcm-slot7cpu1# show services dns-proxy

Possible completions:

cache                            DNS Cache Display

cache-entry                      DNS Cache Entry Display

cache-stats                      DNS Proxy Cache Statistics

module-stats                     DNS Proxy Module Statistics

tunnel-blocked-domain           Domains blocked by DNS tunnel detection

tunnel-blocked-domain-history    Domains blocked by DNS tunnel
detection history - domains once detected as blocked but inactive now

|                                Output modifiers

<cr>
```

Chapter 5

Configuring Content Filtering

This section describes the content filtering services.

Access	config
Limitations	None
Guidelines	To enable a content-filter, the http-proxy must be enabled.

The content-filter configures the filtering capability for a subscriber workflow (page 9-1). The user's HTTP transactions can be filtered using the following filter types:

- blacklist-file
- whitelist-file
- static-category
- service-rule
- online-category-group
- malware-category-group
- phishing-category
- iwf-category
- filter-hash-list

For each of these filter types, Operators specify the corresponding filtering action (allow, block, redirect, terminate) for a matching HTTP request.

The filter types online-category-group, malware-category-group, phishing-categorization, and iwf-categorization require configuration of the content-categorization. The content-categorization functionality uses third-party databases to categorize user requests. User requests are categorized to one or more of the well-known online-category-group, malware-category-group, phishing-categorization, or the iwf-categorization. Affirmed Networks updates the databases periodically, as often as every hour.

Operators can also filter web search engine results for the Bing search engine using the safe-search object.

Content filter instances are used to configure a set of filtering rules for a group of subscriber sessions. For example, a content filter instance can be created for all subscriber sessions from a given APN.

Configure a content-filter using a service-rule, static-category, online-category-group, filter-hash-list, or malware-category-group. Operators can enable or disable the global-content-filter in instance.

Configuring Content Filter Services

This section describes how to configure content filter services.

Note: *If a global filter (see* *page 5-14**) is not configured, HTTP traffic is filtered at the instance level (see* *page 5-17**).*

Access	config
Limitations	None
Guidelines	■ The file must be uploaded through Acuitas and located on the MCC when the content-filter is configured. Operators can also use **`fault-management clear`** to scp the file to directory */public/content/filters/*. ■ If you make changes to the file, you must import the file from the CLI (using the **`content-filter import`** debug command) to rebuild the database for the new file to take effect. Contact the TAC for assistance. ■ When file-type is set to domain, the file should have a Fully Qualified Domain Name (FQDN) for each line for HTTP request HOST exact-match: – The protocol schema is ignored and anything before "//" and including "//" is skipped. Only the domain name following "//" is used. – If line contains only the domain (without the forward slash), only the domain is added to the database for HOST exact-match. – If line contains a URL (with a forward slash), only the domain name is added in the database for HOST exact-match. – If line contains a star "*", the line is ignored. ■ When file-type is set to url, the file should have a Fully Qualified Domain Name (FQDN) for each line for HTTP request HOST or URL exact-match: – The protocol schema is ignored and anything before "//" and including "//" is skipped. Only the URL following "//" is used. – If line contains only the domain (without the forward slash), the entry is added to the database for HOST exact-match. – If line contains a URL (with a forward slash), the entry is added in the database for URL exact-match. – If line contains a star "*", the line is ignored.

Use the following steps to configure content filter services:

1. Enter the following command to configure the content filter.

   ```
   an3000-mcm-slot7cpu1(config)# services content-filter
   ```

2. Enable the admin state.

   ```
   an3000-mcm-slot7cpu1(config)# services content-filter admin-state
   enabled
   ```

3. Configure a content filter static category. See Configuring Content Categorization (page 5-5).

4. Configure the maximum filter-database-size for different filter types to filter web traffic based on hash-list databases. The hash-list databases are used for web filtering in certain markets such as Turkey, where Safe-Internet filtering database is provided by Turkish authorities.

   MCC supports importing of the hash-list databases and performs periodic or on-demand updates.

   The databases are used to filter web traffic, including HTTP and HTTPS transactions.

   ```
   an3000-mcm-slot7cpu1(config)# services content-filter
   filter-database-size
   ```

   a. **filter-hash-list** – Adjust the filter hash list based filter total entry size (requires slot reboot on CSM).

   b. **static-filter** – Adjust the static filter file, including blacklist, whitelist, and static-category, based filter total entry size (requires slot reboot on CSM).

   c. **url-list** – Adjust the URL list based filter total entry size (requires slot reboot on CSM and SSM).

   In HTTP Transaction Event Data Record, the following filter-information fields containing the following subfields support this feature:

   – filterName

   – filterType

   – actionApplied

   – filterInstanceName

   – onlineCategory

   d. **static-filter** – Adjust static filter file, including blacklist, whitelist, static-category, based filter total entry size (requires slot reboot on CSM).

   e. **url-list** – Adjust URL list based filter total entry size (requires slot reboot on CSM and SSM).

   ```
   an3000-mcm-slot7cpu1(config)# service-construct filter-hash-list
   <name> importFrom [import-from-local-file | import-from-server]
   filename <name> file-download-profile <name>
   ```

> ***Note:*** *A web browser can add the www prefix to the input URL (for example, it can change imdb.com to www.imdb.com). To avoid a mismatch, add both imdb.com and www.imdb.com to the filter-hash-list.*

5 Configure a global content filter service. See Configuring Global Content Filter Services (page 5-14).

6 Configure a content filter instance. See Configuring a Content Filter Instance (page 5-17)

7 Configure a content filter static category. See Configuring a Static Category (page 5-20).

Configuring Content Categorization

This section describes how to configure content categorization using online category databases.

Access	config
Limitations	None
Guidelines	Extended storage is required for content categorization. If you do not currently have extended storage, you must add extended storage and reboot both MCMs (either separately through timed MCM reboots or together through a cluster reboot). If content categorization is enabled and extended storage is unavailable, the system displays the **anPlaFileDOwnloadFail** warning-level alarm with the following error message: `Extended_Storage_unavailable_and_necessary_for_content-categorization` The alarm clears when extended storage becomes available or if content categorization is disabled.

Use the following steps to configure content categorization:

1 Configure categorization of web transactions using a third-party database. The categorization database is sourced from an online Internet server (pitcher1/pitcher2.rulespace.com). The database updates are downloaded periodically (as frequently as every hour and configured using the database-interval-parameter), using HTTPS protocol. The downloaded database changes are applied in-service and the changes take effect within seconds after each download.

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization
```

2 Enable the admin state for the content filter service.

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization admin-state enabled
```

3 Configure a customer identifier string.

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization custom-id <string>
```

4 Configure the database update interval in hour(s).

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization database-update-interval 5
```

5 Enable/disable IWF child abuse categorization. With content categorization configured, user requests can be categorized to a well-known Internet Watch Foundation category. The iwf-categorization contains web sites hosting illegal child abuse images and content anywhere in the world. For the iwf-categorization, the Operator can define a filter with the action block, terminate, redirect, or allow.

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization iwf-categorization enabled
```

6 Enable/disable malware categorization. When enabled, the content filter categorizes requests to the malware-category-group: malware-all, malware-domain, malware-object, malware-spyware, malware-victim.

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization malware-categorization enabled
```

7 Configure a content filter using a list of malware categories. For each malware-category-group, the Operator can define a filter with the action block, terminate, redirect, or allow. Also see Malware Category Group (page 5-13).

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization malware-category-group <name> malware-category
```

User requests can be categorized to one or more well-known malware categories:

- malware-attack-botnet
- malware-domain
- malware-object
- malware-spyware
- potential-malware-victim

8 Configure a network context to use when connecting to third-party servers to update the category database.

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization network-context <NC-name>
```

9 Operators can group a set of well-known online categories using the online-category-group. For each online-category-group, Operators can define a filter with action, block, terminate, redirect, or allow. See Online Category Group.

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization online-category-group <name> online-category
```

10 Enable/disable phishing categorization.

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization phishing-categorization enabled
```

11 Enable/disable real-time categorization of web pages containing dynamic or unknown content.

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization real-time-categorization enabled
```

12 Enable/disable url categorization.

```
an3000-mcm-slot7cpu1(config)# services content-filter
content-categorization url-categorization enabled
```

Online Category Group

The following table lists the online-category-group descriptions.

Table 5-1 Online Categories

Name	Sites that...
abortion	Provide information or arguments in favor of or against abortion; describe abortion procedures; offer help in obtaining or avoiding abortion; provide testimonials on the physical, social, mental, moral, or emotional effects of abortion.
advertising	Provide Internet advertising services.
alcohol	Promote or sell alcoholic beverages; supply recipes or paraphernalia to make alcoholic beverages; glorify, tout, or otherwise encourage alcohol consumption or intoxication.
alcohol-kids	The category is created in compliance with the BBFC guidelines. Sites which include the promotion, glamorization or encouragement of drinking alcohol to excess that is aimed at children or that may have a significant appeal to children
anonymizer	Offer anonymous access to websites through a PHP or CGI proxy, allowing users to gain access to websites blocked by corporate and school proxies as well as parental control filtering solutions.
art-and-museums	Include art galleries, artists, and museums. Includes all types of visual arts such as performing arts, theater, painting, drawing, sculpture, photography, natural history, science, and children's museums.
art-nudes	Contain non-pornographic, tasteful, and artful display of the naked body.
automated-web-application	Allow a computer to automatically open an http connection for reasons such as checking for operating system or application updates.
automotive	Relate to motor vehicles, dealers, sales and clubs.
bikini	Offer the sale of bikinis, mico-kinis, mono-kinis, and thongs marketed as beachwear rather than swimwear. Also sites that feature galleries and/or videos of models in bikinis.
blogging	Contain blogs.

Table 5-1 Online Categories

Name	Sites that...
`business`	Are sponsored by or devoted to individual businesses not covered by any other categories.
`cash-gambling`	Involve the wagering and exchange of money online.
`chat`	Offer users the ability to chat online.
`content-delivery-service`	Are content delivery/distribution networks (CDN) designed to accelerate the delivery of content rich Internet pages or large files to end users.
`criminal-skills`	Provide instruction for threatening or violating the security of property or the privacy of people; and how to avoid complying with legally mandated duties and obligations.
`cults`	Feature well-known, organized new religious movements and sects regarded as exploitative or unorthodox.
`cyberbullying`	Sites or pages where people post targeted, deliberate and slanderous or offensive content about other people with the INTENT to torment, threaten, humiliate or defame them. Content is often sexual, malicious or hostile in nature and is submitted via interactive digital technology. Examples include: Malicious gossip; inappropriate photos posted without consent
`drugs`	Promote, offer, sell, supply, encourage or otherwise advocate the recreational or illegal use, cultivation, manufacture, or distribution of drugs, pharmaceuticals, intoxicating plants or chemicals and their related paraphernalia.
`dynamic`	Contain both appropriate and inappropriate user-generated content like social networking or blogging sites. Also, sites in which the page content changes based how the user interacts with it (for example, an Internet search).
`education`	Represent schools or other educational facilities, faculty, or alumni groups.
`energy`	Represent companies involved with the production and distribution of energy.
`enterprise-webmail`	Provide online e-mail services.
`entertainment`	Relate to personal entertainment as well as the entertainment industry.
`filesharing`	Make files available for other users to download over the Internet or private networks.
`finance-and-investing`	Provide the opportunity to establish, plan, research, or manage personal finances and investments.
`food-and-restaurants`	Provide information about food, recipes, specialty food shops, catering, food delivery, and restaurants.
`forums-and-messageboards`	Provide a web application enabling users to participate in the discussion of numerous topics, often in conjunction with online communities.
`freeware-and-shareware`	Offer online software download.

Table 5-1 Online Categories

Name	Sites that...
`gambling`	Allow users to place bets or participate in a betting pool (including lotteries) online; obtain information, assistance or recommendations for placing bets; receive instructions, assistance or training on participating in games of chance.
`gaming`	Sites related to computer or video games, game downloads, and online game sites.
`glamour`	Emphasize, promote, or provide information on how to achieve physical attractiveness, allure, charm, beauty, or style with respect to personal appearance.
`gore`	Display graphic violence and/or the infliction of pain or injuries. Gross violence towards humans or animals, such as scenes of dismemberment, torture, massive blood and gore, sadism, and other types of excessive violence.
`government`	Sites sponsored by government branches or agencies.
`hacking`	Promote or provide the means to practice illegal or unauthorized acts of computer crime using technology or computer-programming skills.
`hate`	Advocate hostility or aggression toward individuals or groups on the basis of race, religion, gender, nationality, ethnic origin, or other involuntary characteristics; sites that denigrate others on the basis of those characteristics or justifies inequality on the basis of those characteristics; sites that purport to use scientific or other commonly accredited methods to justify said aggression, hostility, or denigration.
`health`	Provide information or advice on personal health or medical services, health insurance, procedures, or devices; sites that include information on diets, nutrition, therapies, and counseling services.
`hobby`	Provide information on private pastimes.
`hosting`	Provide individuals or organizations with online systems for storing images, video, or any content accessible via the Web.
`internet-telephony`	Enable users to make telephone calls via the Internet or obtain information or software for that purpose.
`job-search`	Provide assistance or tools to help to find employment.
`kids`	Provide a safe and interesting Internet experience for children under 12 years of age.
`law`	Offer legal content and services.
`lifestyles`	Contain general material relevant to sexual orientation.
`malware`	Total number of requests categorized as Malware Online Category. See Malware Category Group.
`mature-content`	Contain sexually explicit information that is not of a medical or scientific nature.
`military`	Are sponsored by military branches or agencies as well as official and personal sites related to military history, ideology, or specific branches of the military.
`mobile-entertainment`	Offer a range of add-ons for hand-held devices, such as ring tones, wallpapers, games, and videos.

Table 5-1 Online Categories

Name	Sites that...
`music`	Are related to the music industry.
`news`	Report, inform, or comment, on current events or contemporary issues of the day, including sports, weather, editorials, and human interest news.
`non-profit`	Are owned by non-profit organizations.
`obscene`	Contain obscene material.
`occult`	Promote or offer methods, means of instruction, or other resources to affect or influence real events through the use of spells, curses, magic powers, or supernatural beings.
`online-backup-and-storage`	Provide the ability to backup or upload data files to the Internet providing long term storage, access from other computers or locations, or the ability to share data files with other users.
`peer-to-peer`	Provide centralized peer-to-peer networks.
`personals-and-dating`	Promote or provide opportunity for establishing or continuing romantic relationships.
`pets`	Are related to the care, maintenance, purchase, rescue, or breeding of any animal for companionship and enjoyment.
`photography`	Are related to taking, sharing, printing, studying, or manipulating photos at all skill levels, whether casual, amateur or professional.
`placeholder`	Are owned by domain name registrars, domain brokers, or Internet advertising publishers. They usually display dynamically generated content with the intent to monetize on traffic through linked advertising listings.
`plagiarism`	Are primarily designed to allow students to cheat at school by taking the work or ideas of someone else and pass it off as their own.
`politics`	Are related to politicians, election campaigns, political organizations, and publications. Includes official homepages of politicians and political parties and personal sites about politics and grass-root movements.
`pornography`	Contain sexually explicit material for the purpose of arousing a sexual or prurient interest.
`portal`	Offer a broad array of resources and services, such as email, forums, search engines, and on-line shopping malls.
`real-estate`	Are involved in the real estate business.
`reference`	Serve as a general resource to the public and feature factual information produced by authoritative and trusted sources.
`religion`	Are about religion or any set of beliefs and practices that have the function of addressing the fundamental questions of human identity, ethics, death, and the existence of the Divine.

Table 5-1 Online Categories

Name	Sites that...
`remote-access`	Sites or pages where people post targeted, deliberate and slanderous or offensive content about other people with the INTENT to torment, threaten, humiliate or defame them. Content is often sexual, malicious or hostile in nature and is submitted via interactive digital technology. Examples include: Malicious gossip; inappropriate photos posted without consent
`science`	Provide research materials in the natural and life sciences.
`search`	Support searching the Internet, news groups, or indexes and directories.
`self-harm`	Describe or discuss ways in which to self harm including eating disorders and self-injury.
`sexual-education`	Provide educational information on reproduction and sexual development, sexually transmitted disease, contraception, safe sexual practices, sexuality, and sexual orientation
`shopping`	Provide the means to purchase products or services online.
`sports`	Promote or provide information about spectator sports.
`streaming-media`	Host streaming media like television, movies, video, radio, or other media.
`suicide`	Describe or promote suicide.
`technology-and-telecommunication`	Provide information pertaining to computers, the Internet, and telecommunication.
`tobacco`	Encourage, promote, offer for sale or otherwise encourage the consumption of tobacco.
`tobacco-kids`	This is created to be in compliance with BBFC guidelines. Sites include the promotion, glamorization or encouragement of smoking that is aimed at children or that may have a significant appeal to children.
`travel`	Promote or provide opportunity for travel planning in a general sense, particularly finding and making travel reservations.
`violence`	Advocate or provide instructions for causing physical harm to people or property through use of weapons, explosives, pranks, or other types of violence.
`virtual-community`	Offer a variety of tools and mechanisms to enable a group of people to communicate and interact via the Internet.
`weapons`	Describe or offer for sale weapons including guns, ammunition, firearm accessories, knives, and martial arts.
`webmail`	Provide free, web-based email services, accessible through any Internet browser.
`wedding`	Are related to the traditions, customs, planning, and products involved in a marriage, commitment ceremony, or civil unions.
Note: *The following categories are for sites formatted for small format browsers like WML and xHTML:*	

Table 5-1 Online Categories

Name	Sites that...
`wireless-business`	Are sponsored by or devoted to individual businesses not covered by any other categories.
`wireless-chat-sms`	Offer users the ability to chat online.
`wireless-dynamic`	Contain both appropriate and inappropriate user-generated content like social networking or blogging sites. Also, sites in which the page content changes based how the user interacts with it (for example, an Internet search).
`wireless-entertainment`	Offer a range of add-ons for hand-held devices, such as ring tones, wallpapers, games, and videos.
`wireless-finance`	Provide the opportunity to establish, plan, research, or manage personal finances.
`wireless-gambling`	Allow users to place bets or participate in a betting pool (including lotteries) online; obtain information, assistance or recommendations for placing bets; receive instructions, assistance or training on participating in games of chance.
`wireless-news`	Primarily report, inform, or comment, on current events or contemporary issues of the day. Includes sports, weather, editorials, and human interest news.
`wireless-personals-and-dating`	Promote or provide opportunity for establishing or continuing romantic relationships.
`wireless-pornography`	Contain sexually explicit material for the purpose of arousing a sexual or prurient interest.
`wireless-sports`	Promote or provide information about spectator sports.
`wireless-travel`	Promote or provide opportunity for travel planning in a general sense, particularly finding and making travel reservations.
`unknown`	Are unknown to the category database.

Malware Category Group

Configure a content-filter using a list of malware categories. For each malware-category-group, the Operator can define a filter with the action block, terminate, redirect, or allow. User requests can be categorized to one or more well-known malware categories:

- **`malware-advertising`** – Sites associated with online advertising networks. They may accept and deliver ads for commercial spyware and adware as well as those delivered through adware applications. Some may use semi-malicious attacks, deception, or security exploits to deliver advertising or collect tracking information.

- **`malware-attack-botnet`** – Sites associated with malware that causes a system to perform automated tasks over the Internet.

- **`malware-domain`** – Sites where the domain was found to either contain malware or take advantage of other exploits to deliver adware, spyware, or malware.

- **`malware-object`** – Sites that contain direct links to malware file downloads (such as . exe, .dll, .ocx, and others). These URLs are generally highly malicious.

- **`malware-spyware`** – Sites of domains associated with vendors and distributors of spyware, adware, greyware, and other potentially unwanted advertising software. Some may run exploits to facilitate the installation of this unwanted software.

- **`potential-malware-victim`** – Sites that click fraud and adware components use when “phoning home” that could signal a device is infected with spyware or adware. This category should be used only for reporting, not for blocking as the target URLs can be legitimate sites.

Configuring Global Content Filter Services

This section describes how to configure global content filter services.

Note: *If a global filter is not configured, HTTP traffic is filtered at the instance level (see* *page 5-17**).*

Access	config
Limitations	None
Guidelines	■ The file must be uploaded through Acuitas and located on the MCC when the content-filter is configured. Operators can also use fault-management clear to scp the file to directory /public/content/filters/. ■ If you make changes to the file, you must import the file from the CLI (using the content-filter import debug command) to rebuild the database for the new file to take effect. ■ When file-type is domain, the file should have a Fully Qualified Domain Name (FQDN) for each line for HTTP request HOST exact-match: – The protocol schema is ignored and anything before “//” and including “//” is skipped. Only the domain name following “//” is used. – If line contains only the domain (without the forward slash), only the domain is added to the database for HOST exact-match. – If line contains a URL (with a forward slash), only the domain name is added in the database for HOST exact-match. – If line contains a star “*”, the line is ignored. ■ When file-type is url, the file should have a Fully Qualified Domain Name (FQDN) for each line for HTTP request HOST or URL exact-match: – The protocol schema is ignored and anything before “//” and including “//” is skipped. Only the URL following “//” is used. – If line contains only the domain (without the forward slash), the entry is added to the database for HOST exact-match. – If line contains a URL (with a forward slash), the entry is added in the database for URL exact-match. – If line contains a star “*”, the line is ignored. ■ Quic guidelines: – MCC does not apply content-filtering if the UDP flow is not recognized as a QUIC flow. – Content-filtering is not applied to QUIC flows that do not contain the SNI. – Content-filtering is ineffective when connection migration with QUIC occurs because it is mid-flow and the SNI is not present.

Use the following steps to configure global content filter services:

1. Configure global content filter services.

   ```
   an3000-mcm-slot7cpu1(config)# services content-filter global
   ```

2. Configure the content filter blacklist file that contains a list of domains or URLs to block/terminate/redirect. For a global content filter, Operators can configure up to ten blacklist-files. After the Operator configures the global blacklist-file, every subscriber is subjected to that service. For example, the Operator may want to block all subscribers from accessing certain inappropriate sites. This list is uploaded as a file to the system and the global blacklist points to this file.

   ```
   an3000-mcm-slot7cpu1(config)# services content-filter global
   blacklist-file <name>
   ```

   Options include:

   - **file-type** – Blacklist file type; domain/URL
   - **http-action** –http service action on match; block, redirect, terminate
     - **block-notification-message –** Block HTTP service on match
     - **redirect**: HTTP Request Redirect Code - 301|302|307
     - **redirect-url** – Allows the configuration of the original requested URL as part of the redirect URL for content filter features. When an incoming HTTP request matches one of the configured filters, the MCC content filtering service applies the configured filter action. If an action is **redirect**, MCC responds to the UE with a configured redirection page URL. The redirection URL may contain the original URL (blocked URL) in the following format: **http://213.14.227.50/landpage?ms=${ORIGINAL_URL}**
     - **terminate** – Terminate HTTP service on match.
   - **https-action** – https service action on match:
     - block
     - terminate
   - **quic-action** – QUIC service action on match:
     - block
     - terminate

3. Configure a service rule for the content filter. The global level supports one service-rule. A service-rule must contain at least one http-rule-group and no packet-filter in order to be applied to the content-filter service. See QUIC SNI Server Certificate Information (page 5-16).

   ```
   an3000-mcm-slot7cpu1(config)# services content-filter global
   service-rule <name>
   ```

   Options include:

   - **http-action** – http service action on match: block, redirect, terminate.
   - **https-action** – https service action on match: block or terminate.

4 Configure the content filter whitelist file that contains a list of domains or URLs to block or redirect. For a global content-filter, the Operator can configure up to ten whitelist files. Whitelist URLs and domains are accessible to all subscribers, regardless of any other filtering rules configured in workflows.

```
an3000-mcm-slot7cpu1(config)# services content-filter global
whitelist-file <name>
```

Options include:

- **`file-type`** – Whitelist file type; domain/URL
- **`action`** – Block HTTP service on match or Redirect HTTP service on match

QUIC SNI Server Certificate Information

This feature uses QUIC SNI information for content filtering. This feature enables Operators to configure and apply content-filtering and content-categorization policies based on the service-rule configuration based on SNI and configurable through the domain-rule-group. This feature also supports UDP proxy support for domain-based content-filtering and content-categorization.

This feature provides the ability to block/allow QUIC traffic based on the configured filter service-rules. Content categorization filters are applied based on the SNI of the QUIC flow. In other words, only QUIC flows apply to categorization.

The feature is useful in restricting access to content based on the SNI. This can be done for QUIC flows using content-filtering with the domain rule group in the service-rule or content-categorization.

The protocol-filtering quic object evaluates filtering for quic flows. See Configuring a Content Filter Instance (page 5-17).

The service-rule filter quic-action object determines the action to take for UDP flows. If configured as terminate, MCC drops the packet silently. If configured as block, MCC sends an RST packet back to the client. See Configuring Global Content Filter Services (page 5-14).

Configuring a Content Filter Instance

This section describes how to configure a content filter service instance.

Access	config
Limitations	Safe search has the following limitations: ■ Not applicable for subscribers with a local DNS cache. ■ Only rewrite type A and AAAA query is supported. ANY is not supported. ■ Rewrite safe search hostname config is global and you cannot set different values across multiple content-filter instances ■ TCP fallback is not supported in DNS proxy ■ Multi-level safe search for the same search engine is not supported. For example, only force moderate safe search or force strict safe search is supported for the same search engine but not both. ■ By default, the Bing safe search engine applies a moderated safe search for the results when using enforced safe search. As a result, the search results may contain a few non-child safe sites.

Note: *If content-filter instance is configured in workflow data-profile (page 9-1), the instance-level filter is applied for associated workflow traffic after the global filter is checked and no match is found.*

Use the following steps to configure a content filter instance:

1 Create the content filter instance.

```
an3000-mcm-slot7cpu1(config)# services content-filter instance <name>
```

2 Enable the admin-state for the content filter instance.

```
an3000-mcm-slot7cpu1(config)# services content-filter instance <name>
admin-state enabled
```

3 Configure a content filter using a service-rule, static-category, online-category-group, malware-category-group, phising-category, iwf-category, or filter-hash-list.

```
an3000-mcm-slot7cpu1(config)# services content-filter instance <name>
filter
```

Block or redirect HTTP service on match for the following filters:. Options include:

- filter-hash-list, iwf-category, malware-category-group, online-category-group, phishing-category, service-rule, static-category (see Configuring a Static Category (page 5-20)).
  - **`redirect-code`** – HTTP Request Redirect Code - 301|302|307
  - **`redirect-interval`** – Redirect Interval in Seconds. Operators can restrict the frequency of redirection during a defined service-rule. For example if a service-rule specifies a redirect to a URL, MCC caps the redirection at one during the redirect-interval. Operators can configure a time (in seconds) for which the action is applied only once.
  - **`redirect-url`** – MCC uses a CStringMacro to enrich the redirect-url fields. Allows dynamic subscriber information to be enriched. The following variables are used as part of redirect URL: $(IMEI), $(MSISDN), $(IMSI), $(UEIPADDR), $(TIMESTAMP) – milliseconds since Epoch, $(RANDOM) – 16 random octets, $(ORIGINALURL), $(SHA256[]) - base64 encoded hash of information (different macros or plain text) inside the square brackets.

Note: *MCC allows only one level of variable nesting; another SHA256 can not be used inside the SHA256 parameter.*
Note: *If dynamic information is being enriched, Affirmed Networks recommends configuring another filter with higher priority to allow the redirected URL.Otherwise, MCC will continuously redirect the URL.*

  - **`quic-action`** – QUIC service action on match - allow|block|terminate. If configured as terminate, MCC drops the packet silently. If configured as block, MCC sends an RST packet back to the client.

4 Configure a list of Geographic Redundancy Groups to which this content-filter instance is associated.

```
an3000-mcm-slot7cpu1(config)# services content-filter instance <name>
geo-redundant-group <name>
```

5 Enable/disable global content filters in instance.

```
an3000-mcm-slot7cpu1(config)# services content-filter instance <name>
global-content-filter enabled
```

6 Configure a priority. A lower value indicates a higher priority. Affirmed Networks recommends configuring a unique priority to avoid ambiguity in selection.

```
an3000-mcm-slot7cpu1(config)# services content-filter instance <name>
priority (<unsignedInt, 1 .. 4096>):
```

7 Enable or disable this filter instance for different protocols.

```
an3000-mcm-slot7cpu1(config)# services content-filter instance <name>
protocol-filtering
```

Options include:

- **`http`** – Set to enabled or disabled for filtering of HTTP protocol service.
- **`https`** – Set to enabled or disabled for filtering of HTTPS protocol service.
- **`quic`** – Set to enabled or disabled for filtering of QUIC protocol service.

8 Enable safe search for bing, google, or youtube. In addition, configure the content-filter instance and dnn-proxy instance. Operators can enable/disable the bing, google, and youtube safe-search filtering capability, which removes sites that contain inappropriate content from search results. When enabled for a specific subscriber, the system enforces safe-search even if an attempt is made to disable it from the UE.

```
an3000-mcm-slot7cpu1(config)# services content-filter instance test
safe-search <name>
```

The system content-filter supports the capability to enforce SafeSearch per search engine and per subscriber.

Content filter enforces safe search by rewriting the upstream DNS query to look for safe search only CNAME for the given search engine and return the safe search IP address to the subscriber for a search engine on which safe search support is received and enabled for the specific subscriber.

Configuring a Static Category

This section describes how to configure a static category.

Operators can define a category source or static category for matching user traffic. The static category is initialized with a list of URLs or domains (host names) specified in an Operator defined file (file-name).

A static category file contains many entries (<500,000) in a domain or url format that are used to build a filter database. The Operator can configure static-category filters, each with the action (allow/block/terminate/redirect) to be applied.

Use the following steps to configure a static category:

1. Configure the content filter static category.

```
an3000-mcm-slot7cpu1(config)# services content-filter static-category
```

2. Configure the category file name.

```
an3000-mcm-slot7cpu1(config)# services content-filter static-category
file-name <name>
```

3. Configure the category file type.

```
an3000-mcm-slot7cpu1(config)# services content-filter static-category
file-type
```

   a. Configure the domain and url for the file type.

Content Filter Show Commands

Content filter supports the following show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*.

```
an3000-mcm-slot7cpu1# show services content-filter

Possible completions:

category-database-import-history   Content-filter
online-categorization database import history

category-database-summary        Content-filter online-categorization
database import summary

statistics                         Statistics

|                                  Output modifiers

<cr>
```

Chapter 6

Configuring a Content Caching Proxy

This section describes the HTTP caching proxy services and how to configure a content caching proxy.

Access	config
Limitations	None

HTTP traffic and HTTP-based video traffic is growing exponentially in the mobile market. The biggest gain (in bandwidth and user experience) is in caching the static content at the edge. The HTTP caching proxy is an intercepting forward proxy that acts like a server to the UE. The proxy acts like a client to the origin server by splitting the TCP connection between the end user and the origin server without the knowledge of the client. The proxy caches the content (both video and static Web pages) and serves subsequent requests for the same object from the cache directly.

Content Caching Overview

The system supports caching of both Web objects (text, images, scripts, files etc) and HTTP-based video objects. HTTP web proxy and video caching involves the retrieval and caching of Web and video content using the cache to minimize internet bandwidth consumption. Video caching includes HTTP progressive download and all versions of HTTP adaptive streaming. Proxy cache also manages the cached content (both in-memory and disk storage) efficiently moving content between memory and disk and purging stale content from the cache. In addition, the HTTP proxy cache also interfaces with the transcoding function to transcode/transrate content from one format to another.

Caching is an important function of proxies aimed at reducing latency (faster Web-page loading) and wide area bandwidth usage, and reducing the load on origin servers.

A *caching proxy server* accelerates service requests by retrieving content saved from a previous request made by the same client or other clients. Caching proxies store local copies of frequently requested resources, allowing large Operators to significantly reduce their upstream bandwidth usage and costs, while significantly increasing performance.

The caching proxy server receives a new HTTP request, verifies that the request is stored in the cache, and obtains the content if it's not stored in the cache. Most Internet browsers include a *local cache* mechanism tuned to the characteristics of Web pages. This local cache saves time and bandwidth when the user clicks **Back** to view the previously loaded Web page. Caching proxies, unlike browser caches, service many different users simultaneously.

The system uses content cache related parameters to determine the behavior of the Web and video cache. Content cache enables Operators to configure global parameters for cache services such as eviction policy, eviction threshold, and cache life of an object with no cache directives.

The system implements inter-cache protocol to exchange information on the cache stored on each CSM. This feature implements a fetch and deliver concept to deliver the content to the user from the cache instead of routing the packet to the origin server.

HTTP Web proxy and video caching involves the retrieval and caching of Web and video content using the cache to minimize internet bandwidth consumption. A caching proxy server accelerates service requests by retrieving content saved from a previous request made by the same client or other clients. Caching proxies store local copies of frequently requested resources, allowing large Operators to significantly reduce their upstream bandwidth usage and costs, while significantly increasing performance.

The system integrates video caching with Web proxy functionality. For HTTP-based video streaming, the function of video proxy/cache is very similar to that of HTTP; video objects are retrieved and streamed to the client in the same manner.

The system provides the ability to store cached content at the edge of the network for mobile delivery, resulting in reduced Internet bandwidth consumption and reduced latency. The system retrieves video and web content and stores that content in the cache in its original format for delivery to the mobile device. Also see Multi-protocol Proxy (page 2-1) and Configuring Multi-protocol Proxy Services (page 2-3) for more information.

The system supports the following Web proxy and video-caching protocols and features:

Table 6-1 Web Proxy and Video Caching Protocols and Features

Feature	Description
HTTP proxy protocols	■ RFC2616 compliant and support HTTP/1.1 clients and servers – The HTTP proxy function acts as the origin server when communicating with the client and behaves like a cache when communicating with the origin server. ■ The HTTP proxy accepts requests from HTTP/1.0 and HTTP/1.1 clients. *Note: The HTTP proxy does not support requests from HTTP/0.9 client* ■ Bad requests to the client are supported by either closing the connection or sending a status code for bad request to the client. ■ HTTP Proxy uses HTTP/1.1 when communicating with origin servers. ■ IPv4 clients and servers.
Configuration, logging, and statistics	■ Logging of all video streaming activity for a given user for analytics. ■ The system supports the following statistics: – Number of video objects served from the cache. – Number of video objects served by the origin server. – Number of video objects transcoded. – Byte count for videos streamed from cache. – Byte count for videos streamed from the server. – Statistics for each video container and code. – Statistics for streaming (HTTP, Dynamic and Live streaming). – Top accessed video objects and the number of hits. – Top accessed video sites. – Delay statistics for video transfer (start time and duration). – Statistics for measuring Quality of Service (QoS) for each user.

Table 6-1 Web Proxy and Video Caching Protocols and Features

Feature	Description
Cache control	■ Capability to selectively purge caches based on URLs. ■ Least Recently Used (LRU) for cache replacement. ■ Least Frequently Used (LFU) cache replacement. ■ Capability to configure cache fill and eviction policy for the service. ■ Capability to bypass proxy/caching for specific users based on subscriber profile policy rules.
Availability and fault handling	Graceful handling of disk failures on any CSM30. TCP connections time out, cache bypass for all connections anchored on the CSM30.
Streaming protocol	Caching and proxy functionality for HTTP-based progressive download. HTTP progressive download is supported by Adobe plug-in, Windows Media Player, and Real player. This requirement is applicable for file formats and containers supported by the different architectures.
Caching algorithms	Maximum cache size limit for each video for each caching algorithm.

Cache Manager

Caching is one of the main and critical functions of the software architecture with respect to functionality and performance. The system supports two levels of cache for HTTP proxies; RAM cache and disk cache. Bookkeeping for both is performed using metadata containers that contain information about the object and a pointer to the storage location.

For each ASM/CSM, a single instance of cache manager is responsible for:

- Purging for RAM and disk cache based on expiration and cache size (eviction)
- Swapping content between RAM cache and disk cache based on popularity of the content
- Creation and management of metadata containers and RAM cache
- Choosing video objects to transcode periodically and interface with transcode to convert the object to the appropriate format.

Both metadata and RAM cache are created in a shared memory location accessible by all instances of proxy processes. The cache manager creates the shared memory region and all the processes attach to that region.

The system is designed to support the HTTP proxy function in forward explicit mode, offering some advantages in redirecting client requests to the right node and supporting additional revenue-generating features for the operator. Forward proxy mode is supported using the existing APN configuration on the mobile GW and uses the file to determine the proxy for each accessed URL.

Explicit forward mode proxy enables Operators to perform the following functions based on the subscriber's profile and configuration:

- Parental control
- Dedicate proxy nodes for different domains/URLs
- Request/Response modification

The system hosts proxy and cache functions in each Content Services Module (CSM). Each CSM can be viewed as an independent cache proxy with its own storage. The Subscriber Services Module (SSM) module hosts the load distribution function that terminates the TCP sessions and distributes the user sessions among all caches (CSM).

The system implements inter-cache protocol to exchange information on the cache stored on each CSM. This feature implements a fetch and deliver concept to deliver the content to the user from the cache, instead of routing the packet to the origin server.

Configuring a Content Cache Service

This section describes how to configure the content cache service. The system implements a transparent forward proxy (referred to as Intercepting Proxy) that allows a set of clients connected to the PGW to access any website in the Internet. The system is designed to support caching in intercepting mode.

Note: *Enable the http-proxy admin-state (**Chapter 2, Multi-protocol Proxy**) prior to enabling the content-cache admin-state.*

Access	config
Limitations	None

Use the following steps to configure the content cache service:

1. Configure the content cache service.

```
an3000-mcm-slot7cpu1(config)# services content-cache
```

2. Enable/disable the admin status for the content cache.

```
an3000-mcm-slot7cpu1(config)# services content-cache admin-state
enabled
```

3. Cache Dailymotion videos.

```
an3000-mcm-slot7cpu1(config)# services content-cache cache-dailymotion
```

4. Cache Metacafe videos.

```
an3000-mcm-slot7cpu1(config)# services content-cache cache-metacafe
```

5. Cache Youtube videos.

```
an3000-mcm-slot7cpu1(config)# services content-cache cache-youtube
```

6. Set the global compression level between 0 and 9.

```
an3000-mcm-slot7cpu1(config)# services content-cache compression-level
2
```

7. Configure cache lru or lfu eviction policy.

```
an3000-mcm-slot7cpu1(config)# services content-cache eviction-policy
lru
```

8. Evict content from disk when the configured percent of disk usage is between the configured maximum (max) or minimum (min) percentage value.

```
an3000-mcm-slot7cpu1(config)# services content-cache
eviction-threshold-disk min
```

9 Evict content from memory when the configured percent of memory usage is between the configured maximum (max) or minimum (min) percentage usage.

```
an3000-mcm-slot7cpu1(config)# services content-cache
eviction-threshold-memory min
```

10 Configure the content cache instance.

```
an3000-mcm-slot7cpu1(config)# services content-cache instance <name>
```

a Configure a service rule for the content cache instance.

```
an3000-mcm-slot7cpu1(config)# services content-cache instance
<name> service-rule <SR-name>
```

b Enable or disable serving from cache.

```
an3000-mcm-slot7cpu1(config)# services content-cache instance
<name> service-rule <SR-name> cache-bypass
```

c Enable or disable caching the response from the server.

```
an3000-mcm-slot7cpu1(config)# services content-cache instance
<name> service-rule <SR-name> cache-server-response
```

d Configure the priority for the content cache service rule.

```
an3000-mcm-slot7cpu1(config)# services content-cache instance
<name> service-rule <SR-name> priority
```

11 Configure the maximum cache lifetime in seconds.

```
an3000-mcm-slot7cpu1(config)# services content-cache max-cache-life
3600
```

12 Configure maximum object size (in bytes) that can be cached on disk.

```
an3000-mcm-slot7cpu1(config)# services content-cache
max-object-size-disk 2147483648
```

13 Configure maximum object size (in bytes) that can be cached in memory. The proxy checks the maximum size of data that can be cached in the disk to avoid running out of cache buffers.

```
an3000-mcm-slot7cpu1(config)# services content-cache
max-object-size-memory 8192
```

14 Configure minimum number of hits to cache object.

```
an3000-mcm-slot7cpu1(config)# services content-cache min-hit-cache 1
```

Content Cache Show Commands

Content cache supports the following show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*.

```
an3000-mcm-slot7cpu1# show services content-cache statistics
cluster-level detail 38
Possible completions:
bytes-evicted-from-disk           Total number of bytes evicted from
disk
bytes-evicted-from-memory         Total number of bytes evicted from
memory
bytes-purged-to-disk             Total number of bytes purged to disk
bytes-served-from-cache           Total number of bytes served from
cache
bytes-served-from-disk-cache      Total number of bytes served from
disk cache
bytes-served-from-memory-cache    Total number of bytes served from
memory cache
current-bytes-in-memory           Total number of current Bytes in
memory
current-bytes-on-disk            Total number of current Bytes on disk
current-cache-state               Current VM Nodes Caching
current-content-blocks            Total number of current content
blocks
current-content-buffer           Total number of 4KB current content
buffer
current-content-buffer-64k       Total number of 64KB current content
buffer
current-metadata                  Total number of current MetaData
current-mutex-blocks             Total number of current mutex blocks
current-object-in-memory          Total number of current Object in
memory
current-object-on-disk            Total number of current Object on
disk
current-video-objects-in-cache    Total number of current video
objects in cache
discontinuity-timestamp           Time counters were last cleared
objects-evicted-from-disk        Total number of objects evicted from
disk
objects-evicted-from-memory      Total number of objects evicted from
memory
objects-missed-from-cache        Total number of objects missed from
cache
objects-purged-to-disk            Total number of objects purged to
disk
objects-served-from-cache        Total number of objects served from
cache
```

```
objects-served-from-disk-cache     Total number of objects served from
disk cache
objects-served-from-memory-cache   Total number of objects served from
memory cache
total-content-buffer-4k            Total number of available 4KB
content buffers
total-content-buffer-64k           Total number of available 64KB
content buffers
video-bytes-served-from-cache      Total number of video bytes served
from cache
video-objects-served-from-cache    Total number of video objects
served from cache
|                                  Output modifiers
  <cr>
```

Chapter 7

HTTP Server

This section describes the HTTP server. The HTTP server function allows Operators to configure multiple HTTP servers and associate those servers with different services. Each HTTP server must have an instance name.

This section describes the following HTTP server functions:

- Configuring an Entitlement Server and Service Mappings (page 7-3)
- Configuring the HTTP Server Instance (page 7-5)
- Creating Server Policies for the HTTP Server (page 7-6)
- Configuring Secure HTTP Identify Protocol Services (page 7-7)
- Configuring the HTTP Server (page 7-14)
- Configuring a List of HTTP Servers (page 7-15)
- Configuring a Header Enrichment List (page 7-15)

Access	config
Limitations	None

HTTP server supports the following features:

- MMS STI (see MMS Services on page 8-1)
- Simple and Secure Entitlement
- Banner and Overlay Insertion server
- Secure HTTP Identity Protocol (SHIP)

The http-server function allows Operators to configure multiple HTTP servers and associate those servers with different services. For example, this feature enables Operators to enable and configure an http-server instance for the MMS STI service. Each http-server must have an instance name. The HTTP server supports the following options:

- STI profile associated with the server instance
- network-context to be used for the server instance
- ua-profile database associated with the server instance
- MMS STI profile to be used with the HTTP server instance (this is a configurable Universal Resource Indicator (URI) for the STI). The URI is an MMSC URL location in the STI HTTP POST message. The system matches the configured URI value with the received value prior to processing the STI packet.

Configuring an Entitlement Server and Service Mappings

This section describes how to create an entitlement server and configure entitlement service mapping.

The Entitlement server is used by subscriber mobile devices to obtain information as to which services a subscriber is entitled. An entitlement-server global configuration specifies the mapping between the URI and the internal VSA name. The entitlement server provides the ability for the UE to query and determine if a subscriber is entitled to certain services. The entitlement server is also used to obtain a signed MSISDN that is used with the registration server.

Note: *Affirmed Networks implements the entitlement server using a server plugin method. The plugin uses this mapping to determine whether the service exists in the Vendor Specific Attributes (VSAs) coming from the service-metadata.*

To upgrade the plugin debian outside of a normal software upgrade or online software upgrade, see the Affirmed Networks System Administration Guide. For example:

`image-management releasemodifier version plugin_2014-09-30`

To display the status of the current plugin version, see the Affirmed Networks System Administration Guide.
For example: **`show image-management release-modifier-status`**

The entitlement server supports two types of entitlement:

- *Simple (http)* – Uses GET requests with no security. The simple entitlement protocol flow is as follows:
  - Subscriber device sends HTTP GET request to entitlement server; the URL contains service name
  - Entitlement server responds with HTTP status code (success or disallowed) for the queried service
- *Secure (https)* – Uses POST method and the body is in JSON format. The secure entitlement protocol flow is as follows:
  - PGW obtains a list of services to which the subscriber is entitled. The Gx interface contains a set of Vendor Specific Attributes (VSAs)
  - Subscriber Device sends HTTPS POST request to entitlement server; the payload used JSON formatting and gzip compression
  - Entitlement server responds with subscriber status code (success or disallowed) for each queried service
  - Entitlement server responds with MSISDN and signature of the nonce+MSIDSN for the MSISDN query
  - HTTP response is sent in the gzipped JSON format

Use the following steps to configure an entitlement server and create service mappings:

1. Enter the following command to create the entitlement server.

```
an3000-mcm-slot7cpu1(config)# services http-server entitlement
entitlement-service <name>
```

2. Enter the entitlement vendor specific attribute value. This value is mandatory.

```
an3000-mcm-slot7cpu1(config)# services http-server entitlement
entitlement-service <name> entitlement-vsa-value
```

3. Enter a unique URI of the service. The uri should always begin with a forward slash.

```
an3000-mcm-slot7cpu1(config)# services http-server entitlement
entitlement-service <name> uri-path
```

Configuring the HTTP Server Instance

This section describes how to configure the HTTP server instance.

Access	config
Limitations	None

For an entitlement server, associate two policies with the instance: simple and secure entitlement. The policies reference a server-plugin script that is responsible for processing the request and generating a response and a service-rule to match the request to a policy. Associate this policy with a service-rule.

Use the following steps to configure the HTTP server instance:

1 Create the http server instance.

```
an3000-mcm-slot7cpu1(config)# services http-server instance <name>
```

2 Enable the admin-status for the http server.

```
an3000-mcm-slot7cpu1(config)# services http-server instance <name>
admin-state enabled
```

3 Configure the container for content insertion.

```
an3000-mcm-slot7cpu1(config)# services http-server instance <name>
content-insertion
```

Options include:

- `admin-state` – Enable/disable content-insertion.
- `content-insertion-profile` – Configure a list of content-insertion-profiles supported by this http-server service instance.
- `loopback-ip-list` – Configure loopback-ip-list used for content-insertion.

4 Configure the priority and server plug-in for this policy.

5 Configure the MMS STI profile to be used with this http server instance. MMS STI profile to be used with the HTTP server instance (this is a configurable Universal Resource Indicator (URI) for the STI). The URI is an MMSC URL location in the STI HTTP POST message. The system matches the configured URI value with the received value prior to processing the STI packet.

6 Configure the network-context to be used with this instance.

7 Configure the ua-profile database associated with this server instance.

The ua-profile object captures capability and preference information for wireless devices. This information can be used by content providers to produce content in an appropriate format for the specific device.

Note: *The ua-profile file must exist on the system prior to configuring the ua-profile options.*

---

a Configure a name for the ua-profile.

b Configure the user-agent-key used to indicate the association between the user-agent and ua-profile file. The user-agent-default-key is a mandatory parameter for ua-profile.

c Assign the ua-profile to a service rule and http-proxy instance.

d Configure the ua-profile applicable to the instance.

Creating Server Policies for the HTTP Server

This section describes how to configure a network context and server policies for the http server instance.

Access	config
Limitations	None

Use the following steps to configure the HTTP server instance:

1 Configure the HTTP server policy.

```
an3000-mcm-slot7cpu1(config)# services http-server http-server-policy
<name>
```

2 Enter a unique priority for the server policy.

```
an3000-mcm-slot7cpu1(config)# services http-server http-server-policy
<name> priority 1
```

3 Associate a server plug-in with this server policy.

```
an3000-mcm-slot7cpu1(config)# services http-server http-server-policy
<name> server-plugin <name>
```

4 Configure the service rule associated with this server policy.

```
an3000-mcm-slot7cpu1(config)# services http-server http-server-policy
<name> service-rule <name>
```

Configuring Secure HTTP Identify Protocol Services

This section describes how to configure Secure HTTP Identify Protocol (SHIP) services.

Access	config
Limitations	None

Traditionally, HTTP Header Enrichment (HHE) is used to pass information from a Network Operator to a Content Provider (CP). However, HHE is not practical when using an encrypted HTTPS connection. Therefore, the Secure HTTP Identify Protocol (SHIP) is used to convey information from the Network Operator to the Content Provider when using HTTPS.

The SHIP protocol consists of two forms:

- **Redirect based (Browser)** – This interface allows information exchange between the Network Operator when HTTPs connections are initiated from the UE's Internet browser. See Redirect-based Browser Call Flow (page 7-9).

- **Web API based (for Applications)** – This interface allows information exchange between the Network Operator, User Equipment (UE), and the CP by allowing an application running on a UE access to a WEB API from a SHIP server. See Web API-based Application Call Flow (page 7-10).

The following configuration describes the different examples of configuring a header with IMSI as its value.

- **Example 1**: Header X-CUSTOM-IMSI specifies the value directly. SHIP appends X-CUSTOM-IMSI=17567547897899 in the response.

- **Example 2**: Header X-IMSI specifies that IMSI is appended as the value of the Header in the response. X-IMSI=17567547897899.

```
an3000-mcm-slot7cpu (config)# service-construct header-enrichment-list
HELIST
header-name X-CUSTOM-IMSI
value 17567547897899
 !

header-name X-IMSI
imsi
 !
```

MCC supports the ability to differentiate URLs with the same domain name. To do this, multiple applications (inferred from different URL paths) offered by a single Content Provider (Inferred from the domain name) are treated as different content providers altogether.

MCC sorts the configured content providers (CPs) in decreasing order of length and matches each CP with the search string (domain name + URL path). MCC supports a maximum of 256 content providers.

For example, MCC treats the following three URLs (seen as multiple applications offered by a single CP) as three different content providers if they have the appropriate configuration:

- https://www.cp.com/app1
- https://www.cp.com/app2
- https://www.cp.com/app3

In this example, the URL path is different, while the domain name is the same. MCC allows the following services to be configured independently of the domain/URL:

- encryption-profile
- header-enrichment-list
- expiration-timeout
- http-redirect-code

For example:

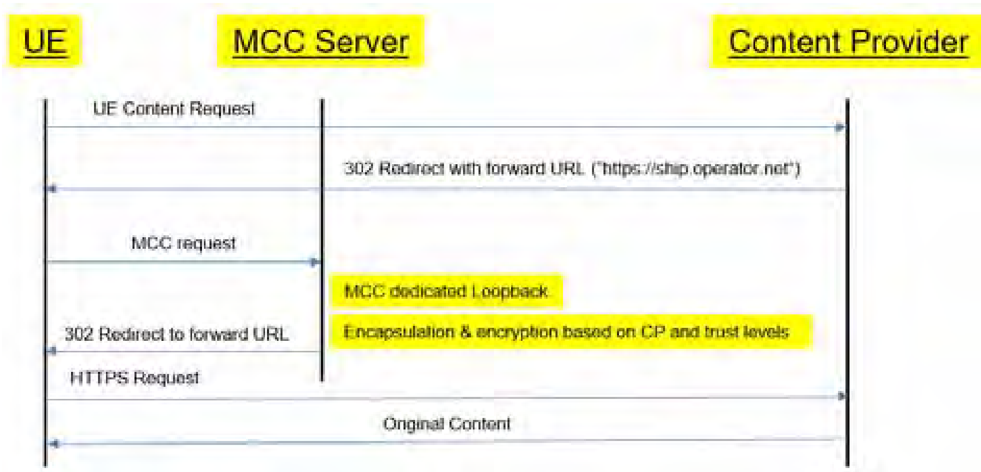
```
an3000-mcm-slot7cpu1(config)# services http-server ship
content-provider cp.com/appl

encryption-profile           ENCR-PROF1

header-enrichment-list    he1

expiration-timeout          720

http-redirect-code           302


an3000-mcm-slot7cpu1(config)# services http-server ship
content-provider cp.com/app2

encryption-profile          ENCR-PROF2

header-enrichment-list   he2

expiration-timeout         1440

http-redirect-code          303
```

```
an3000-mcm-slot7cpu1(config)# services http-server ship
content-provider cp.com/app3

encryption-profile          ENCR-PROF3

header-enrichment-list   he3

expiration-timeout         1440

http-redirect-code          301
```

Call Flows

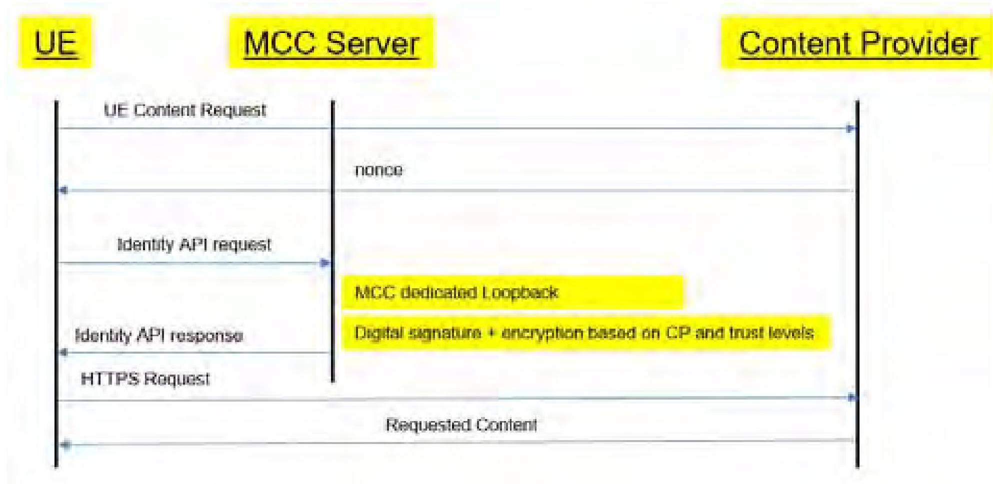
The following is the call flow for the redirect-based browser.

Figure 7-1 Redirect-based Browser Call Flow



The following is the call flow for the Web API-based application.

Figure 7-2 Web API-based Application Call Flow



Use the following steps to configure SHIP:

1 Configure an encryption profile with a user-defined encryption key. See Configuring an Encryption Profile (page 2-54) for details. For example:

```
an3000-mcm-slot7cpu1 (config)# service-construct encryption-profile
ENCR-PROF
admin-state enabled
aes-128-cbc key
$3$uRZlqo86T6gZgP64IV0SwAr6yraP9PCkoMkfFWrwzgf0Wevnk1kY1w==
encoding    hex
!
```

2 Configure the SHIP content provider, consent-id, encryption profile, expiration timeout, header enrichment list, and http redirect code. See Consent ID (page 7-13) for more information.

```
an3000-mcm-slot7cpu1(config)# services http-server ship ?
Possible completions:
content-provider   Configure SHIP Content Provider, Encryption and
header string
an3000-mcm-slot7cpu1(config)# services http-server ship
content-provider <cp.com>
Possible completions:
consent-id              Consent ID for the content provider - string
encryption-profile      Configure an encryption profile used for
content provider transactions
expiration-timeout      Request expire time for the content provider
header-enrichment-list   Configure a list of custom headers to be used
for content provider transactions
http-redirect-code      Content provider HTTP request success code
<cr>
```

3 Create two HTTP rule groups for Redirect and JSON SHIP. Use the following configuration to match the http-server policy:

```
an3000-mcm-slot7cpu1# show running-config service-construct
service-rule SR-REDIRECT-SNAP
service-construct service-rule SR-REDIRECT-SNAP
admin-state                                 enabled
priority                                    100
service-data-flow-id                        1
http-rule-group                             HRG-REDIRECT-SHIP
tcp-filter                                  all
service-activation                          always-on
install-default-bearer-packet-filters-on-ue disabled
include-service-data-flow-id-in-offline-sdc enabled
an3000-mcm-slot7cpu1# show running-config service-construct
service-rule SR-API-SNAP
service-construct service-rule SR-API-SNAP
admin-state                                 enabled
priority                                    100
service-data-flow-id                        1
http-rule-group      HRG-API-SHIP
tcp-filter                                  all
service-activation                          always-on
install-default-bearer-packet-filters-on-ue disabled
include-service-data-flow-id-in-offline-sdc enabled
Associate the service-rule with the http-server-policy:
an3000-mcm-slot7cpu1# show running-config services http-server
http-server-policy
services http-server http-server-policy HTTP-SERVER-POL-REDIRECT-SNAP
priority      1
server-plugin redirect_snap
service-rule SR-REDIRECT-SHIP
!
!

an3000-mcm-slot7cpu1# show running-config services http-server
http-server-policy
services http-server http-server-policy HTTP-SERVER-POL-API-SNAP
priority      2
server-plugin api_snap
service-rule SR-API-SHIP
!
```

4 Associate the http-server-policies with the http-server instance:

```
an3000-mcm-slot7cpu1# show running-config services http-server
instance HTTP-SERVER-INSTANCE
services http-server instance HTTP-SERVER-INSTANCE
admin-state enabled
http-server-policy HTTP-SERVER-POL-REDIRECT-SHIP
!
http-server-policy HTTP-SERVER-POL-API-SHIP
!
```

5 Associate the http-server instance with the **workflow data-profile**:

```
an3000-mcm-slot7cpu1# show running-config workflow data-profile
DATA-PROFILE
workflow data-profile DATA-PROFILE
max-connections                1024
force-dpi-detection            false
force-packet-drop              false
multi-protocol-proxy           HTTP-PROXY
content-filter                 CF-QUIC
content-cache                  default
http-server                    HTTP-SERVER-INSTANCE
priority-evaluation-mode       hierarchical
dpi-pre-detection-max-pkts     0
dpi-pre-detection-max-bytes    0
first-packet-over-volume-quota allow

an3000-mcm-slot7cpu1(config)# services http-server http-server-policy
SIMPLE_ENT server-plugin ?
Possible completions:
[simple_entitlement]  secure_entitlement  simple_entitlement
redirect_ship api_ship
```

Consent ID

This feature enables closer coordination between Content Providers (CP) and end users who have given permission for that coordination. For example, if a CP requests to share subscriber information, this feature will verify if the user has consented to share their information with a given CP. If consent was granted, the data is shared; if not, access to subscriber information is rejected.

This feature receives data from the PCRF. A new string named ‘ConsentID’ is added to the PCRF. That string is relayed to the Proxy for processing.

If this feature is invoked and the content provider’s configured consent-id matches a VSA string in the metadata for that user, then MCC processes the request. Otherwise, MCC returns an error code.

This feature filters SHIP requests as follows:

- If the configured content provider consent-id is ‘`*`’ (wildcard) (default), then the SHIP request is processed, regardless of the user strings supplied. The SHIP request could still fail due to other errors.
- If the configured content provider consent-id is supplied and matches a VSA string provided for the user (not case sensitive), then the SHIP request is processed. The SHIP request could still fail due to other errors.
- If the configured content provider consent-id is supplied but does not match a VSA string supplied for the user, then the SHIP request fails with the failure return code (1004).

Configuring the HTTP Server

This section describes how to configure the HTTP server.

Access	config
Limitations	None

Use the following steps to configure the HTTP server:

1 Create the HTTP server.

```
an3000-mcm-slot7cpu1(config)# service-construct http-server <name>
```

2 Enable the HTTP server.

```
an3000-mcm-slot7cpu1(config)# service-construct http-server <name>
admin-state enable
```

3 Configure the server connection idle timeout in seconds.

```
an3000-mcm-slot7cpu1(config)# service-construct http-server <name>
idle-timeout 15
```

4 Configure the loopback IP to use to connect to the http server.

```
an3000-mcm-slot7cpu1(config)# service-construct http-server <name>
loopback-ip
```

5 Configure the network context to be used to select the loopback-ip.

```
an3000-mcm-slot7cpu1(config)# service-construct http-server <name>
network-context <NC-name>
```

6 Configure the server IP address.

```
an3000-mcm-slot7cpu1(config)# service-construct http-server <name>
server-ip-address
```

7 Configure the server port.

```
an3000-mcm-slot7cpu1(config)# service-construct http-server <name>
server-port
```

8 Configure the server URI path. The server-ip-address and server-port values are added to the uri-path to create the URL for HTTP POST. If the uri-path is not configured, the request message uses the default URI format /imsi/$(IMSI). See the *Affirmed Networks ePDG Administration Guide* for more information.

```
an3000-mcm-slot7cpu1(config)# service-construct http-server <name>
uri-path
```

Configuring a List of HTTP Servers

This section describes how to configure a list of HTTP servers.

Access	config
Limitations	None

Use the following steps to create a list of HTTP servers:

1. Configure a name for the HTTP server list.

```
an3000-mcm-slot7cpu1(config)# service-construct http-server-list
<name>
```

2. Associate an HTTP server with the server list.

```
an3000-mcm-slot7cpu1(config)# service-construct http-server-list
<name> http-server <name>
```

Configuring a Header Enrichment List

This section describes how to configure a header enrichment list.

Access	config
Limitations	None

Use the following steps to configure a header enrichment list:

1. Create the header enrichment list.

```
an3000-mcm-slot7cpu1(config)# service-construct header-enrichment-list
<name>
```

2. Configure the header string.

```
an3000-mcm-slot7cpu1(config)# service-construct header-enrichment-list
<name> header-string <name>
```

3. Add header names.

```
an3000-mcm-slot7cpu1(config)# service-construct header-enrichment-list
<name> header-string <name> header-name
```

Options include:

- **charging-characteristic** – Include Charging Characteristic in the header
- **header-string***-
- **imei** – Include IMEI in the header
- **imsi** – Include IMSI in the header
- **mapped-apn-name** – Include mapped APN name in the header

- **msisdn** – Include MSISDN in the header
- **original-apn-name** – Include original APN name in the header
- **rat-type** – Include Radio Access Type (2G, 3G or 4G) in the header
- **reseller-id** – Include reseller ID in the header
- **serving-gw-ip-addr** – Include Serving Gateway IPv4 Address in the header
- **serving-gw-ipv6-addr** – Include Serving Gateway IPv6 Address in the header
- **serving-network-plmnid** – Include Serving Network PLMNID in the header
- **subscriber-id** – Include subscriber id in the header
- **time-zone** – Include time zone in the header

HTTP Server Show Commands

HTTP server supports the following show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*.

```
an3000-mcm-slot7cpu1# show services http-server statistics
cluster-level
Possible completions:
content-insertion   Content Insertion Statistics
server-plugin       Server Plugin Statistics
summary             Http server Summary Statistics
|                   Output modifiers

an3000-mcm-slot7cpu1# show services http-server statistics node-level
Possible completions:
content-insertion   Content Insertion Statistics
server-plugin       Server Plugin Statistics
summary             Http server Summary Statistics
|                   Output modifiers
```

MMS Services

This section describes the Multimedia Messaging Service (MMS). MMS provides messaging capabilities for the delivery of multimedia messages composed of text, audio, graphics, photographs, and other media type on a network. Use this object to configure a Multimedia Messaging Service and options.

Access	config
Limitations	None

The Affirmed MMS proxy implements the MM1 protocol as defined in the WAP standards (Open Mobile Alliance and 3GPP). The MMS proxy routes client MMS messages over the HTTP protocol or WAP1.2 stack as defined in the MM1 interface specification and allows clients to retrieve messages from the MMSC. The system supports the following MMS proxy functions: MM1 transcoding, MM1 routing based on MSISDN, and MMS message ID.

For an MMS transaction, MCC forces a URL rewrite on every request. The rewritten URL could be the same as the original URL as MCC replaces the host in the original request with the configured host value. Therefore, the rewritten URL for the transaction data record for MMS transactions could be the same as the original URL.

All MMS devices communicate with the network over the Wireless Application Protocol (WAP). WAP is a technical standard that enables mobile device users to access information over a wireless network. WAP supports the requirements of various services in heterogeneous mobile networks and allows several network configurations to coexist within the WAP environment. The three most common configurations in a WAP environment include WAP 1.x, WAP HTTP Proxy, and direct Access. This guide describes the following two modes:

- WAP 1.x Protocol Stack (page 8-8)
- WAP HTTP Proxy with Wireless Profiled TCP and HTTP (page 8-10)

Note: *In direct access mode, there is no intermediate gateway and the client uses the HTTP stack to communicate with the server directly. This mode is not used in the Affirmed implementation and therefore is not included in this guide.*

WAP Gateway

A WAP browser is a web browser for mobile devices such as mobile phones that use WAP.

WAP is designed to work with most wireless networks such as CDPD, CDMA, GSM, PDC, PHS, TDMA, FLEX, ReFLEX, iDEN, TETRA, DECT, DataTAC, Mobitex and GRPS.

WAP is composed of the following layers:

- Wireless Application Environment (WAE)
- Wireless Session Protocol (WSP)
- Wireless Transaction Protocol (WTP)
- Wireless Transport Layer Security (WTLS) (see page 8-13)
- Wireless Datagram Protocol (WDP)

A WAP gateway is similar to a proxy and acts like a bridge between mobile devices using the WAP protocol and the Internet. The WAP gateway translates pages into a form suitable for mobile devices.

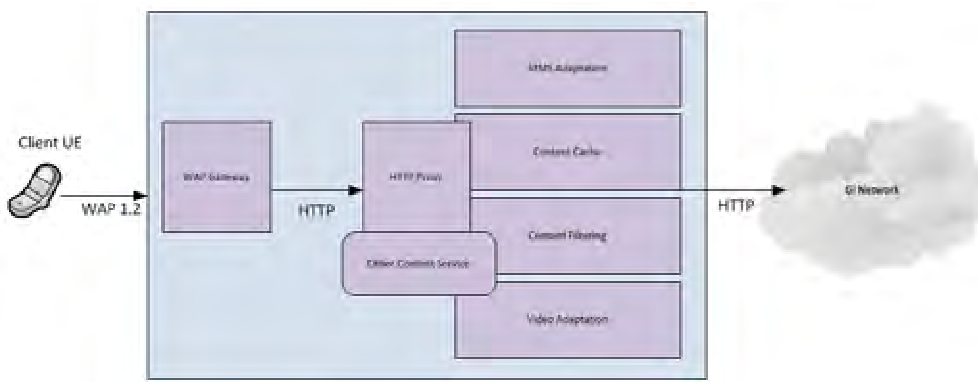
The system contains a WAP gateway designed to provide WAP clients with access and protocol translation between WAP 1.x and HTTP/1.1. The WAP gateway provides support for packet network access for older WAP 1.x devices until these devices are upgraded or phased out of the network. The WAP gateway implements the WAP 1.x stack toward the client and HTTP stack toward the content server and provides the translation layer between the two interfaces. It supports WAP 1.0 and WAP 1.2 clients with and without WTLS support. The system supports the protocols defined in the WAP standards (OMA).

The WAP gateway passes client traffic to the HTTP proxy service. The system is then able to process content using services such as MMS adaptation, content filtering, and content caching.

As shown in Figure 8-1, the WAP gateway processes and translates WAP requests to HTTP requests. HTTP responses are translated into WAP messages. Both WAP requests and responses are optimized for mobile networks using:

- Binary encoding of headers
- Caching of repeating headers and cookies
- Transport layer adaptive to high packet loss and congestion in the access radio network

Figure 8-1 WAP Architecture



WAP1.x Gateway

WAP Gateway is a service implemented in the Affirmed MCC. The service provides data access over WAP1.x protocols to mobile wireless clients.

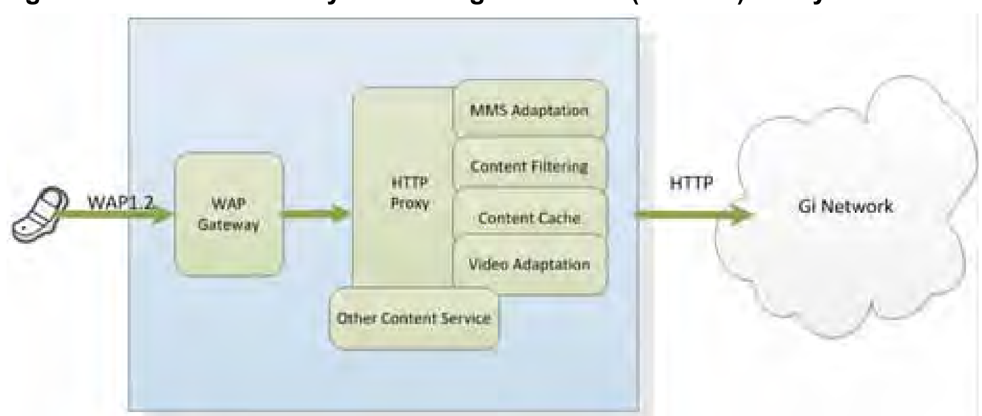
WAP-Gateway provides the following features:

- Efficient (saving wireless network bandwidth) encoding and decoding of messages sent over wireless network
- Protocol translation between WSP and HTTP/1.1
- Protocol translation between HTML content and WML
- Reliable message delivery
- Message segmentation and reassembly
- Connectionless (UDP port 9000) and connection-oriented (UDP port 9001) communication
- Secure communication

WAP Gateway works with the collocated HTTP Proxy to provide value-added services, such as MMS adaptation, content filtering, and content caching. When using the WAP Gateway, Operators configure the HTTP Proxy and optionally any other content services.

Figure 8-2 shows the WAP Gateway interfacing with HTTP (WAP2.0) Proxy.

Figure 8-2 WAP Gateway Interfacing with HTTP (WAP2.0) Proxy



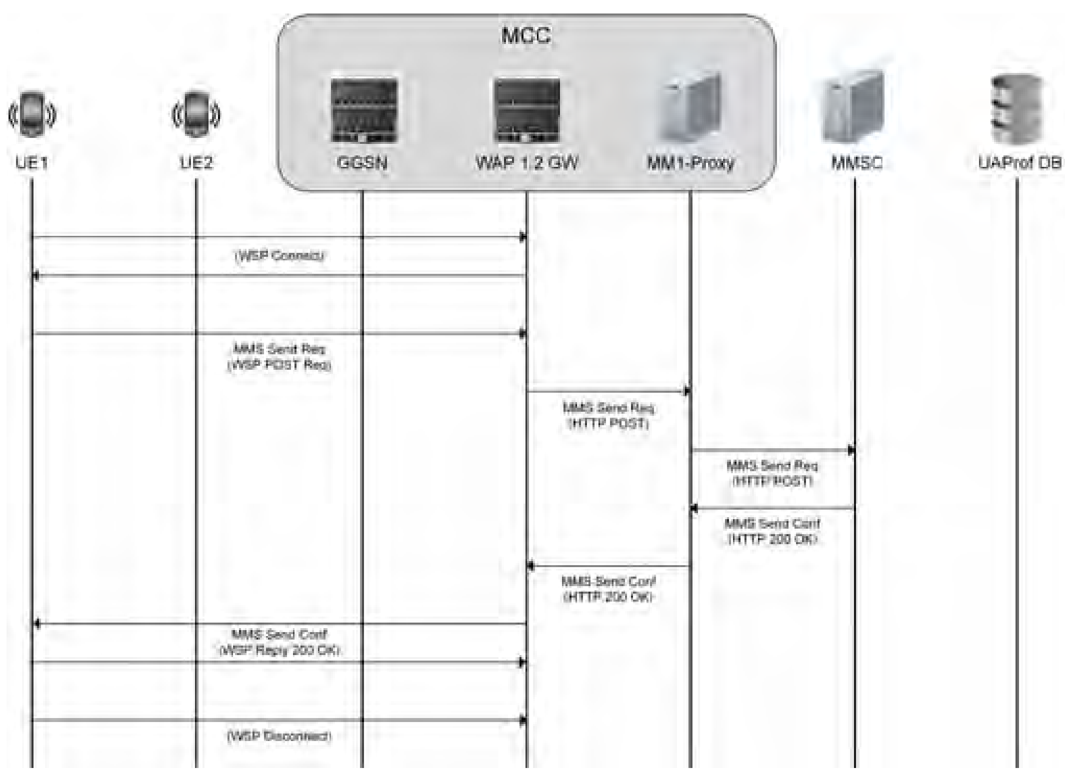
MMS MO Over WAP1.x

Figure 8-3 shows a typical flow for posting an MMS message by a mobile WAP1.x client. The message is posted via WAP1.x Gateway and MM1 proxy to the MMSC.

The sequence of events is as follows:

- The client first connects to WAP-Gateway using WSP Connect and WSP Connect Reply messages. When the connection is created, the connection capabilities (such as maximum message size) are negotiated. The HTTP headers and cookies common to all transactions made by the client are also passed on to the WAP-Gateway.
- The client sends an MMS m-send-req message encapsulated in WSP Post (and WTP Method Invoke) to WAP-Gateway. The WAP-Gateway, after successfully processing WSP Post, translates the message to the HTTP Post and forwards the message via MM1-Proxy to MMSC.
- The MMS m-send-conf message sent by MMSC via MM1-Proxy is received by WAP-Gateway. The HTTP headers of the message are converted to WSP equivalent and the MMSC response is sent to the client (delivered as WTP Result).
- If the WAP1.x client does not issue more transactions, it issues WSP Disconnect to terminate WAP1.x connection with WAP-Gateway.

Figure 8-3 MMS-MO Transaction Over WAP1.2



MMS MT Over WAP1.x

Figure 8-4 shows the typical flow of retrieval of the MMS message by WAP1.x client. The message is retrieved via WAP1.x Gateway and MM1 proxy from the MMSC.

The sequence of events is as follows:

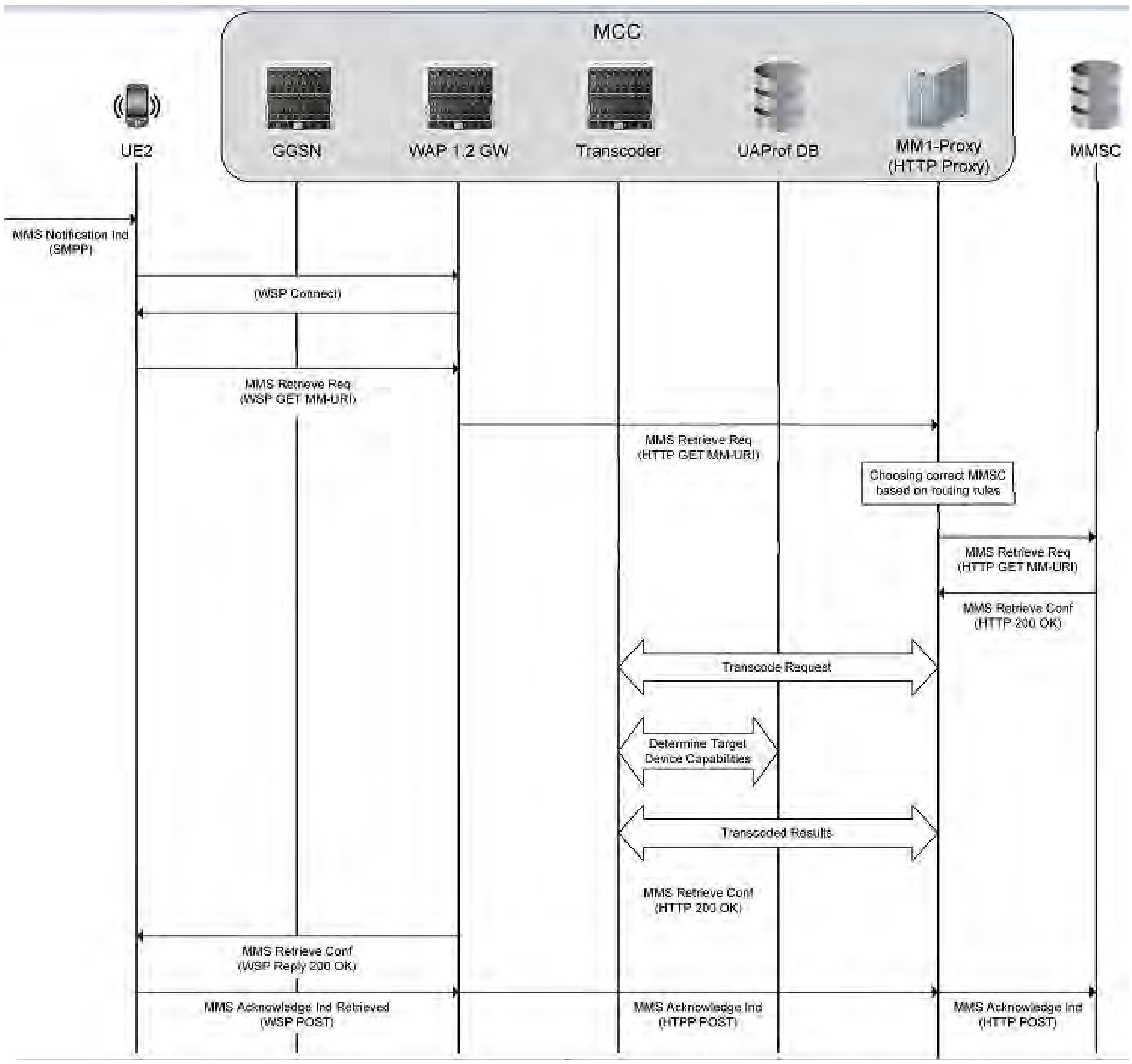
- The client is notified about a pending MMS message with an MMS Notification Indication message sent over SMPP protocol.
- If the client does not have an existing connection to the WAP-Gateway, it creates a new connection using a WSP Connect and WSP Connect Reply message exchange. When the connection is created, the connection capabilities such as maximum message size are negotiated. The HTTP headers and cookies common to all transactions made by the client are also passed on to the WAP-Gateway.

Note: *Due to WTP protocol limitations with standard segmentation and reassembly functions, the maximum message size supported by WAP-Gateway is 300 KB.*

- The client sends a WSP Get (over WTP Method Invoke) to the WAP-Gateway. The WAP-Gateway, after successfully processing the WSP Get, translates the message to an HTTP Get and forwards the message via the MM1-Proxy to the MMSC.
- The MMS m-retrieve-conf message sent by MMSC via the MM1-Proxy is received by the WAP-Gateway. The HTTP headers of the message are converted to a WSP equivalent and the MMSC response is sent to the client (delivered as WTP Result).
- The client posts an MMS m-acknowledge-ind (with WSP Post via WAP-Gateway, MM1-Proxy) to the MMSC to acknowledge successful retrieval of the MMS message.

If the WAP1.x client does not issue more transactions, it issues a WSP Disconnect to terminate the WAP1.x connection with the WAP-Gateway.

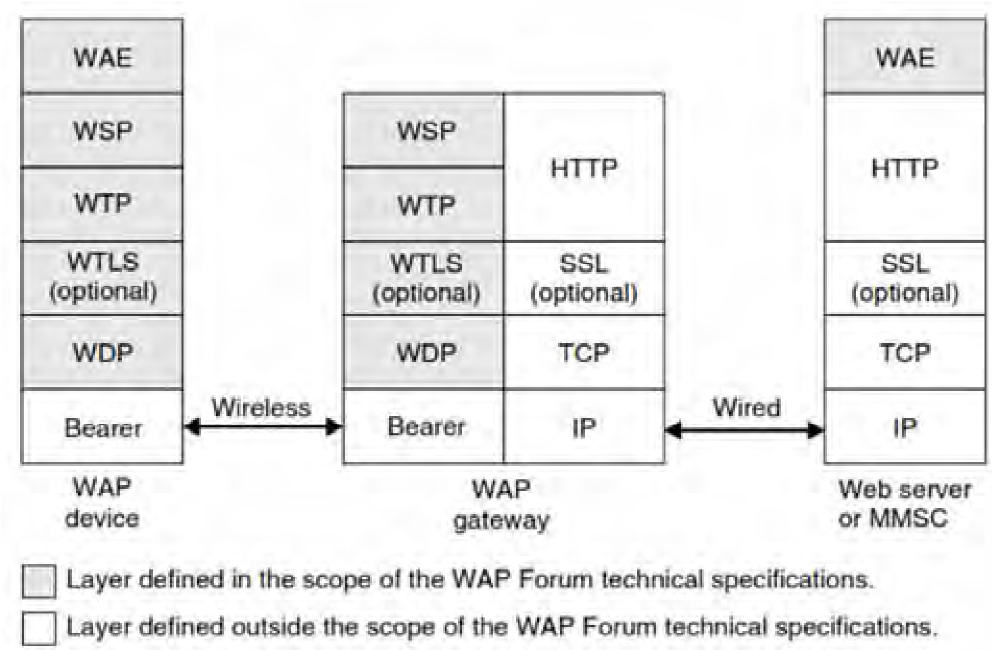
Figure 8-4 MMS-MT Call Flow Over WAP1.x



WAP 1.x Protocol Stack

Figure 8-5 shows the protocol stack of the configuration defined in the WAP specification suite 1.x. This configuration is also supported by the WAP specification suite 2.0 in addition to other configurations. In this configuration, the WAP device communicates with a remote server via an intermediary WAP gateway. The primary function of the WAP gateway is to optimize the transport of content between the remote server and the WAP device. For this purpose, the content delivered by the remote server is converted into a compact binary form by the WAP gateway prior to being transferred over the wireless link. The WAP gateway converts commands conveyed between datagram-based protocols (WSP, WTP, WTLS, and WDP) and protocols commonly used on the Internet (HTTP, SSL, and TCP).

Figure 8-5 WAP 1.x Protocol Stack



The *Wireless Application Environment (WAE)* is a general-purpose application environment in which Operators and service providers can build applications (for example, MMS clients) for a wide variety of wireless platforms.

The *Wireless Session Protocol (WSP)* provides features also available in HTTP (requests and corresponding responses). Additionally, WSP supports long-lived sessions and the possibility to suspend and resume previously established sessions. The WSP requests and corresponding responses are encoded in a binary form for transport efficiency.

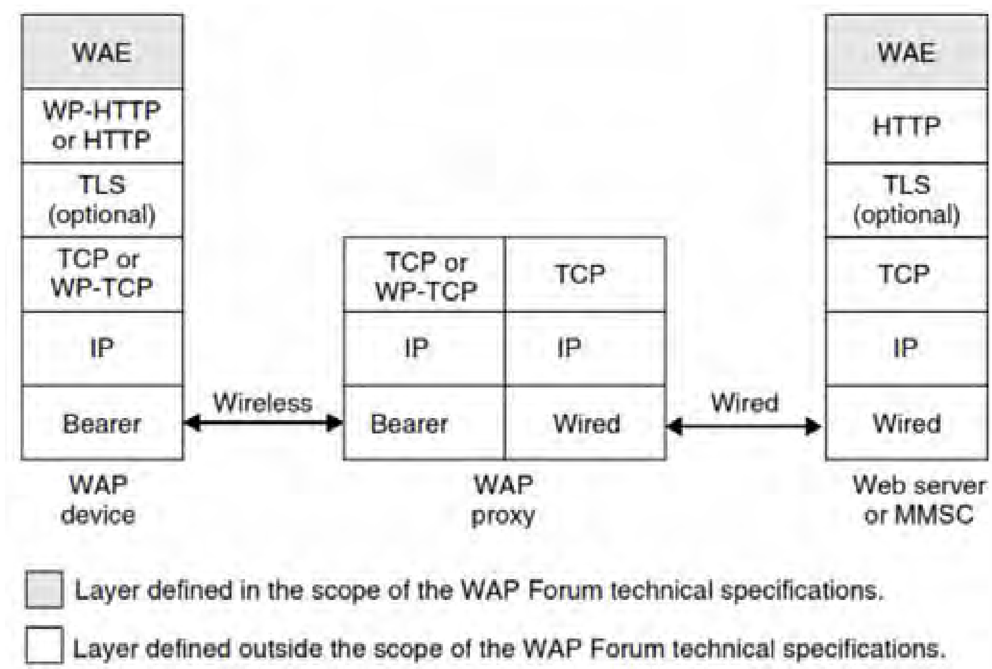
The *Wireless Transaction Protocol (WTP)* is a lightweight transaction-oriented protocol. The WTP improves the reliability over underlying datagram services by ensuring the acknowledgment and re-transmission of data grams. The WTP has no explicit connection set-up or connection release. WTP is a message-oriented protocol and therefore is appropriate in the implementation of mobile services such as browsing. Optionally, Segmentation And Reassembly (SAR) of packets composing a WTP protocol unit is also supported.

The optional *Wireless Transport Layer Security (WTLS)* provides privacy, data integrity, and authentication between applications communicating with the WAP technology. This includes the support of a secure transport service. The WTLS provides operations for the establishment and the release of secure connections. The *Wireless Data Protocol (WDP)* is a general datagram service based on underlying low-level bearers. The WDP offers a level of service equivalent to those offered by the Internet's *User Datagram Protocol (UDP)*. At the bearer level, the connection may be a circuit-switched connection (as found in GSM networks) or a packet-switched connection (as found in GPRS and UMTS networks). Alternatively, the transport of data at the bearer level may be performed over the SMS or over the Cell Broadcast Service.

WAP HTTP Proxy with Wireless Profiled TCP and HTTP

Figure 8-6 shows a configuration in which the WAP device communicates with web servers via an intermediary WAP proxy. The primary role of the proxy is to optimize the transport of content between the fixed Internet and the mobile network. The proxy also acts as a *Domain Name System (DNS)* for mobile devices. In this configuration, Internet protocols are preferred versus legacy WAP protocols to support IP-based protocols in an end-to-end configuration, from the Web server back to the WAP device. Figure 8-6 shows the protocol stack of this configuration as defined in the WAP specification suite 2.0.

Figure 8-6 WAP 2.0 Protocol Stack



The *Wireless Profiled HTTP (WP-HTTP)* is an HTTP profile specifically designed to cope with the limitations of wireless environments. The WP-HTTP is fully inter-operable with HTTP/1.1 and supports message compression.

The optional *Transport Layer Security (TLS)* ensures the secure transfer of content for WAP devices involved in the exchange of confidential information. The *Wireless Profiled TCP (WP-TCP)* offers a connection-oriented service. It is adapted to the limitations of wireless environments but remains inter-operable with existing *Transmission Control Protocol (TCP)* implementations.

WTLS Key Exchange

The Affirmed WAP-Gateway supports RSA anonymous (RSA_anon) mode for a key exchange suite. It is an RSA key exchange without authentication. The gateway sends its RSA public key. The client generates a secret value, encrypts it with the gateway's public key, and sends it to the gateway. The pre-master secret is the secret value appended with the server's public key.

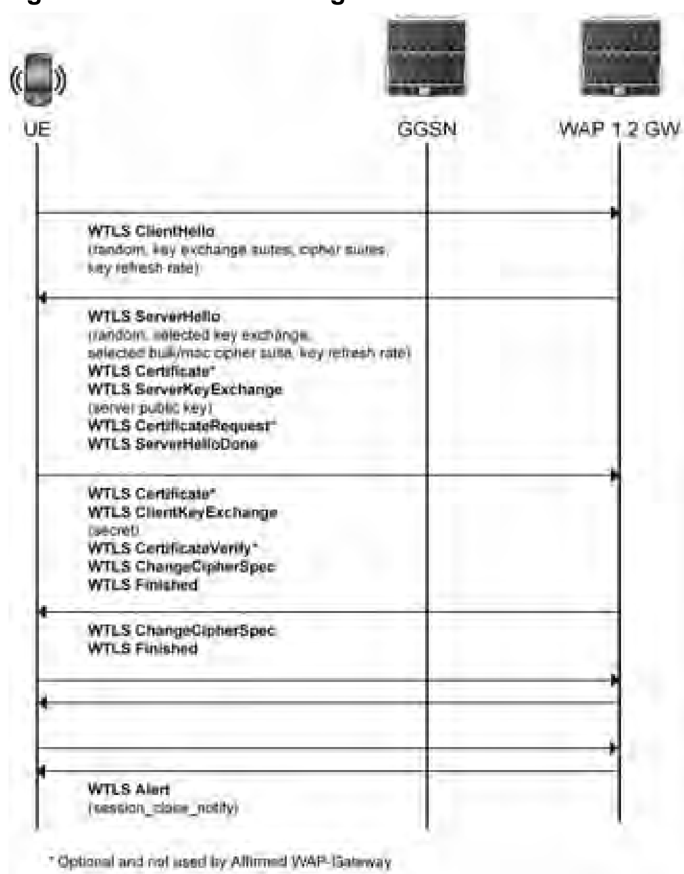
When WTLS is used, the Affirmed WAP-Gateway is configured with X.509 certificate and private key.

The Affirmed WAP-gateway supports a maximum key size of 2048 bits.

Other key exchange algorithms listed by the WTLS specification *are not* supported by the Affirmed WAP-gateway: NULL, shared-secret, Diffie-Hellman: DH_anon (with key sizes 512, 768, 2048), RSA_anon (with key sizes 512, 768), RSA (with key sizes 512, 768, 2048), Elliptic Curve DH: ECDH_anon (with key sizes 113, 131, 2048), ECDH-ECDSA (with key size 2048).

To ensure that keys are not compromised, the keys are periodically refreshed. The key refresh rate is negotiated between client and server when establishing a secure session. Figure 8-7 shows the WTLS message flows.

Figure 8-7 WTLS Message Flows



WTLS Authentication

The WAP1.x WTLS client can optionally authenticate WAP-Gateway using X.509 certificate. This procedure is used for non-anonymous secure session establishment.

The WAP-Gateway returns the WTLS server certificate in the initial handshake response and the certificate is validated by client.

If a WAP-Gateway needs to authenticate WAP1.x client, it sends CertificateRequest in the initial handshake response. The client responds with its Certificate and the CertificateVerify message. The gateway completes authentication by validating client certificate and returned signature in the client's CertificateVerify message. The Affirmed WAP-Gateway does not currently use this procedure, but instead uses the anonymous key exchange suite.

Confidentiality

Confidentiality is supported by encrypting transmitted user data. Received encrypted data from the client is decrypted by WTLS layer before passing it up the stack to WTP, WSP layers.

The Affirmed Networks WAP-Gateway supports the following encryption algorithms:

- DES-CBC (56)
- RC5-CBC (40, 56, 128)

The following algorithms listed by WTLS specification *are not* supported by the Affirmed Networks WAP-Gateway:

- NULL
- DES-CBC (40)
- 3DES
- IDEA
- RC5-CBC (64)
- IDEA (not supported due to patent issues)

WTLS Integrity

WTLS supports message integrity through the means of stamping each transmitted packet with a Message Authentication Code (MAC) - hash.

MAC on received message is validated and discarded in the event that MAC is incorrect.

The Affirmed Networks WAP-Gateway supports the following hashing algorithms:

- SHA1 (40, 80, 160)
- MD5 (40, 80, 128)

The SHA1-XOR-40 algorithm listed by the WTLS specification *is not* supported by the Affirmed WAP-Gateway.

Wireless Transport Layer Security

The Affirmed Networks WAP-Gateway supports the Wireless Transport Layer Security (WTLS) protocol. The WTLS protocol is derived from a popular and commonly deployed Transport Layer Security (TLS) protocol. Unlike TLS, which runs over connection-oriented transport, WTLS is adapted to datagram oriented transport (UDP). WTLS also provides for more efficient message encoding.

Similar to non-secure WAP1.x modes, WTLS operates in the following modes:

- Secure connectionless mode (UDP port 9202)
- Secure connection-oriented mode (UDP port 9203).

WTLS supports the following main features:

- Cryptographic algorithm negotiation
- Key exchange
- WAP-gateway, and optionally, client authentication
- Confidentiality (packet encryption)
- Integrity (packet signatures – hashes)
- Compression (not supported by Affirmed WAP-Gateway).

SMTP Proxy

The SMPT proxy is used to handle SMTP traffic. For example, this feature enables Operators to enable the smtp-proxy and create an instance name for each smtp proxy. This feature supports the following additional attributes:

- MM3 instance associated with the server instance.
- network-context to be used for the server instance.
- MMS MM3 profile to be used with the SMTP proxy instance.
- SMTP server to be used by the proxy.
- port number of the server.
- PC-Profile file to be used for transcoding (if any exist).

HTTP Post

HTTP post transmits/receives MMS messages even with incorrect MMS URL home page settings and routes messages to a specific target MMSC based on the MMS originator and destination address. The system extracts the destination address from the “To” field of the MMS message and uses the destination address for MMSC lookup as follows:

- The system looks up the MMSC based on the configured short-code: value, range, and regex.
- The system evaluates the MMSC with the lowest priority first and highest priority last.
- If multiple “To:”, “Cc:”, and “Bcc:” fields exist, the system uses the source address or “From” field to route the MMS message.
  - If the MMSC is not found by the “To:” field, the system uses the “From:” field to find MMSC based on the load-balance-suffix.
  - If the “From:” field is an invalid MSISDN (doesn’t have /TYPE=PLNM), the system uses the UE MSISDN.
  - If the MMSC is not found by the “To:” and “From” fields, the system uses the configured default MMSC in the mmsc-group.
  - If the connection to the MMSC fails, the system uses the configured backup MMSC (fall-back-mmsc).
- The system looks up the configured parameters of the determined MMSC and send an MMS message to the IP address and URL from the MMSC configuration file.

- When the HTTP POST request arrives, the system checks if the message is an MMS message.
  - If the message is not an MMS message, the system sends back `HTTP 403 (HTTP-FORBIDDEN)`.
  - If the message is an MMS message and mms message type = m-notifyresp-ind, m-acknowledge-ind, m-read-report-ind, the system extracts the mms message id to determine the destination MMSC. Message ids contain `@mms2.era.pl`, `@mms3.era.pl`, `@MMSCDA` suffixes. Configure Message ids form/suffix and their corresponding MMSC in a text file using the reqexp expression to allow changes if a new MMSC vendor is introduced to PTC.

The system forwards the HTTP GET request transparently to the MMSC if the URL matches. No changes are made to the URL or IP. When messages are downloaded from the MMSC, the system transcodes the MMS message according to the `uaprof.xml` file. Transcoding is enabled/disabled on a per MMSC basis. Messages from the MMSC_DA are transcoded using the STI interface and do not require transcoding again upon downloading.

In addition, configure the mmsc-group applicable to the instance.

Supported MIME Types

Video Input and Output

- video/mp4
- video/3gpp
- video/3gpp2
- video/quicktime
- video/mpeg
- application/vnd.rn-realmedia
- application/shockwave-flash
- video/flv
- video/avi
- video/x-ms-asf
- video/ogg (input only, copy to output)
- video/matroska (input only, copy to output)
- video/webm (input only, copy to output)

Acoustic Audio Input and Output

- audio/aiff
- audio/amr
- audio/amr-wb (input only, copy to output)
- audio/mpeg
- audio/mp3
- audio/ac3
- audio/au
- audio/x-pn-realaudio
- audio/mp4
- audio/3gpp
- audio/3gpp2
- audio/quicktime
- audio/aac
- audio/flac
- audio/wav
- audio/avi
- audio/x-ms-wma
- audio/ogg
- audio/matroska
- audio/webm

Synthesized Audio Input and Output

- application/vnd.smaf
- audio/midi
- audio/sp-midi
- audio/mobile-xmf
- audio/x-nokia-rng
- audio/iMelody

Image Input and Output

- image/jpeg
- image/pjpeg
- image/gif
- image/png
- image/jp2
- image/tiff
- image/bmp
- image/wbmp
- image/svg+xml (input only, copy to output)
- application/pdf (input only, copy to output)

Text Input and Output

- text/plain
- text/x-c
- text/html
- text/css
- text/x-javascript
- application/vnd.symbian.install (input only, copy to output)
- application/xml
- text/vnd.wap.wml
- text/x-vCard
- text/x-vNote
- text/x-vCalendar

Text Input and Output Character Set Types

- US-ASCII
- UTF-7
- UTF-8
- UTF-16
- UTF-16BE
- UTF-16LE
- UTF-32
- UTF-32BE
- UTF-32LE
- UCS-2
- UCS-2BE
- UCS-2LE
- UCS-4
- UCS-4BE
- UCS-4LE
- ISO-8859-1 ... ISO-8859-10
- WINDOWS-1250 ... WINDOWS-1258
- KOI8-R
- KOI8-U
- GBK
- GB18030
- GB2312
- BIG5
- ISO-2022-JP
- ISO-2022-JP-1
- ISO-2022-JP-2
- ISO-2022-CN
- ISO-2022-CN-EXT
- ISO-2022-KR
- SHIFT-JIS

Configuring Multimedia Message Service

This section describes how to configure the multimedia message service (MMS).

Access	config
Limitations	None

Use the following steps to configure the multimedia message service:

1 Configure the MMS service.

```
an3000-mcm-slot7cpu1(config)# service-construct mms
```

2 Configure the MMS destination address blacklist used to block certain destination addresses.

```
an3000-mcm-slot7cpu1(config)# service-construct mms blacklist <name>
```

Options include:

- **`destination-address`** – MMS address regex.
- **`ldap-in-use`** – LDAP query is in use.

3 Configure error response status and text in the event the multimedia message is rejected.

```
an3000-mcm-slot7cpu1(config)# service-construct mms error-response
<string>
```

4 Configure an MM3 interface. The MM3 interface allows the MMSC to exchange messages with external servers such as Email servers and SMSCs. The interface is typically based on IP-based Email protocols (such as SNMP).

```
an3000-mcm-slot7cpu1(config)# service-construct mms mm3-profile <name>
```

Options include:

- **`pc-profile-filename`** – Configure the PC-Profile filename.
- **`smtp-server-name`** – Configure SMTP Server domain name.
- **`smtp-server-port`** – Configure SMTP Server port.

5 Configure Multimedia Messaging Service Center (MMSC). Configure the maximum message size and MMSC IP address.

```
an3000-mcm-slot7cpu1(config)# service-construct mms mmsc <name>
```

MMS packets can be routed among a pool of configured Multimedia Messaging Service. Centers (MMSC), based on the following parameters:

- *address* – Configure MMSC IP address.
- *fall-back MMSC* – Used when the MMSC is chosen for an MM1 message but is not reachable from the MM1 Proxy.
- *host* – “Host” is used for MMS-MT (GET) routing.

- *load-balance-suffix range* – (round robin) Select the MMSC for an m-send-req MM1 message.
- *max-message-size* – Maximum message size allowed to be routed to the MMSC. The default is 3000 Kb.
- *max-recipients* – Maximum number of allowed destinations (To and Bcc field) and originator (From field).
- *message-id-suffix* – Configure the message ID suffix for this mmsc. Select the MMSC for m-notifyresp-ind, m-acknowledge-ind, m-read-report-ind MM1default MMSC in the group message.
- *priority* – Configure MMSC priority
- *short-code* – Configure MMS short code.
  - *short-code-range* – Configure short code range for this mmsc.
  - *short-code-regex* – Configure the short code regex for this mmsc.
  - *short-code-value* – Configure the short code value for this mmsc.
- **`transcode`** – Enabled/disabled transcoding for the MMSC
- *URL* – Instead of relying on UE’s URL to post a message, the MM1 Proxy uses the correct URL from the configured MMS table.

6 Configure a group of Multimedia Messaging Service Centers (MMSCs).

```
an3000-mcm-slot7cpu1(config)# service-construct mms mmsc-group <name>
```

Options include:

- **`mmsc`** – Configure the MMSCs for the group.
- **`reject-invalid-mms-requests`** – Reject GET requests if the host does not match MMSC host configuration.

7 Configure Standard Transcoder Interface profile. Affirmed Networks uses the Standard Transcoding Interface (STI) specification to resolve some of the integration and testing problems associated with deploying multimedia services across mobile devices. STI allows for simpler content adaptation by enabling the transcoding of different types of content files. For example, OMA MMS relies on STI to move rich content between the different mobile devices.

```
an3000-mcm-slot7cpu1(config)# service-construct mms sti
```

a Configure MMS URL location. The URI is an MMSC URL location in the STI HTTP POST message. The system matches the configured URI value with the received value prior to processing the STI packet.

```
an3000-mcm-slot7cpu1(config)# service-construct mms sti-profile
<name> uri
```

Service Construct Show Commands

MMS supports the following show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*.

```
an3000-mcm-slot7cpu1# show service-construct
Possible completions:
  code-mapping-lists        Code mapping list information
  control-service-profile   Configure control plane service profile
  device-database           Device database stats
  dhcp-mapping              Configure DHCP option mapping templates
  filter-hash-list          Filter-Hash-List stats
  interface                 Configure accounting authentication
authorization
                            interfaces
  radius-mapping            Configure RADIUS attribute mapping
templates
  url-list-statistics       URL-List stats
  workflow-data-stats       Workflow data statistics
  |                         Output modifiers
```

Chapter 9

Workflow Parameters

This section describes the workflow objects and parameters supported on the MCC. Workflow is an application running on the system that defines the treatment of a service flow across various services residing on the system. Workflow allows the Operator to customize a subscriber's packet treatment across various services. Workflow binds a single or set of subscribers to a workflow data profile and/or a workflow control profile. The workflow specifies what event triggers this binding (bearer setup). All CDRs, external communication, and statistics collection occur as if it's on the mapped-APN if configured.

Note: *See Acuitas Performance Statistics for details on statistics.*

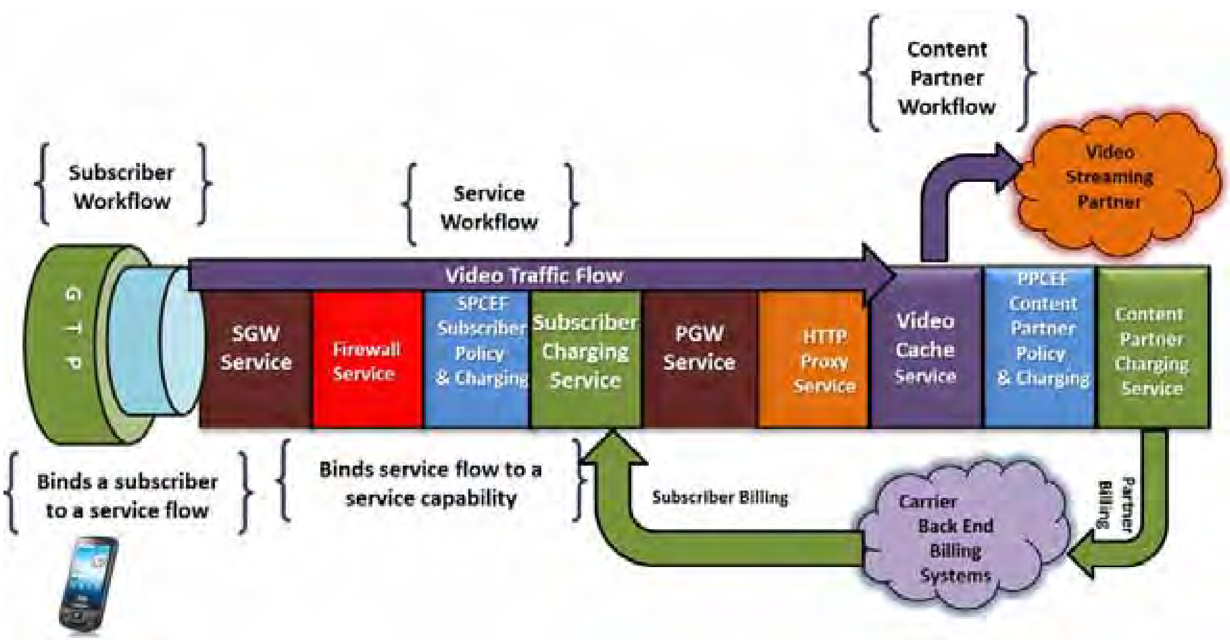
Access	config
Guidelines	None

Workflow enables Operators to customize the use of current services (Subscriber Management Gateways – SGW, PGW, GGSN, online and offline charging, accounting, policy, QoS, NAT, HTTP proxy, content filtering, TCP optimization, and caching) and add new services as they become available. Using workflow, Operators can customize each subscriber's packet treatment across various services on a per-flow basis.

As shown in Figure 9-1, workflow simplifies the creation of services in the Operator's network and provides the following carrier benefits:

- Reduces time to market for new services from service conception to service delivery, leading to rapid revenue generation for new services.
- Reduces capital expenditures by reducing network integration expenses.
- Reduces operating expenses by minimizing the need for professional services.
- Allows flexibility in service creation and development at Layers 4 through 7.
- Improves service-level monitoring by creating new service-monitoring processes.
- Simplifies device parameter and programming.
- Supports documented service flows and flow interactions.
- Adapts to existing services using workflow.

Figure 9-1 Workflow



Operators can use workflow to define combinations of treatment for user flows for the following functions, that together form the basis of a rich and dynamic, Operator-defined service:

Note: *Affirmed Networks currently supports the following functions. New functions are added to the same framework as they are supported. For the latest available features and functions, see the Release Notes.*

- MME
- PGW
- SGW
- GGSN
- QoS
- Charging (online, offline or hybrid online / offline)
- Content filtering
- HTTP proxy
- TCP / media delivery optimization
- Traffic steering

Workflow Framework

Workflow uses the subscriber analyzer to define the set of rules to select the subscriber sessions to apply a specific services policy. The subscriber analyzer uses the set of rules that can match signaling attributes for a subscriber session, such as IMSI, IMEI, UE device type, time of day, APN, SGSN, or MME IP address.

Each subscriber analyzer is tied to a workflow control profile that defines which external policy databases to consult (including internal or external policy manager) to control the subscriber policy and interpret results. The workflow control profile provides Operators with a powerful mechanism to manage subscriber policy for each individual subscriber using a combination of RADIUS or DIAMETER Authentication Servers, PCRF Servers, LDAP Servers (SPR), or Internal Policy Manager.

This approach provides Operators with the flexibility to control the exact mechanism used to manage the subscriber policy. For example, one workflow control profile may use a RADIUS authentication server to retrieve the name of the subscriber workflow policy, while another workflow control profile may use an external PCRF server, and yet another workflow control profile may look up the subscriber workflow policy from a locally defined database.

The Operator controls the subscriber analyzer filtering rules, such as IMSI and APN, and associated workflow control profiles through workflow configuration objects. In one scenario, the Operator may configure the subscriber policy to be driven entirely by an external PCRF selected using the workflow control profile. In the other scenario, the Operator may provision the system with the internal Policy Manager, or static workflow policies in the absence of a PCRF. The Operator can also choose a hybrid of the above two approaches, thus providing the Operator complete flexibility in defining and using the subscriber policies for managing different service behaviors.

The combination of subscriber analyzer and workflow control profile defines the service function rules for a particular subscriber in the workflow data-profile. The workflow data-profile defines a collection of rules for each service function to be examined and executed on a per-flow basis. Each rule defined in the workflow data-profile can either be pushed to the system by the policy server (internal or external) or predefined. Each rule can be defined based on per-flow filters such as L3/4 Packet Filter – 5 Tuples, L7 filters – HTTP URL / URIs, application being used, time of day, geographic location, etc. In addition, different rules may be effective for different functions for the same subscriber. For example, a charging function may be based on time of day while QoS may be based on a combination of HTTP URL and geographic location.

Workflow has been designed to define services that are both subscriber-aware and content-aware. In addition to providing services that are based on the subscriber parameters described above, services can also be based on content provider policies. For example, a service may not be charged (zero-rated) to a subscriber, but rather charged to the content provider for delivery of the service / content. For example, the charging service may be configured to charge for all HTTP traffic, while the traffic steering service may be configured to steer all uplink and downlink video traffic for a particular content site to either an internal or video caching service.

Workflow Application

Workflow binds a single or set of subscribers to a workflow data profile and/or a workflow control profile. Workflow specifies what event triggers this binding (bearer setup). All CDRs, external communication, and statistics collection occur as if it's on the mapped-APN if configured.

Workflow supports an apn mapping list that allows the Operator to specify criteria to which apn values can be mapped. For example, criteria such as PAP username can be used to select a specific APN to use for the specific subscriber.

Configuring a Workflow APN Mapping List

This section describes how to configure a workflow APN mapping list. The APN mapping list allows the user to specify criteria to which the APN values can be mapped. For example, criteria such as PAP username can be used to select a specific APN to use for the specific subscriber.

Access	config
Limitations	None

Tip: Affirmed Networks recommends not mapping an APN to another APN with the same name. Doing so may result in incorrect APN names in CDRs and AAA accounting records. Also see apn-reporting in the *Affirmed Networks Gateway Administration Guide.*

Note: *APN mapping is not supported on emergency APNs.*

Use the following steps to configure a workflow APN mapping list:

1. Create the workflow apn mapping list.

```
an3000-mcm-slot7cpu1(config)# workflow apn-mapping-list <name>
```

2. Specific an APN mapping entry that maps username or realm to APN name.

```
an3000-mcm-slot7cpu1(config)# workflow apn-mapping-list <name>
apn-mapping
```

3. Add a description for the APN mapping list.

```
an3000-mcm-slot7cpu1(config)# workflow apn-mapping-list <name>
user-description
```

Configuring a Workflow Control Profile

See the *Affirmed Networks Gateway Administration Guide* for details on configuring a workflow control profile.

Access	config
Limitations	None

The workflow control profile specifies the back-end interfaces to use for PCRF, authentication, accounting, tunneling, and charging systems. As an alternative, the profile allows Operators to configure subscriber profiles statically rather than using these interfaces. The profile also specifies the mechanism to dynamically obtain the name of a workflow data-profile that has been programmed on the system.

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